MN150013EN

PXM1000 – Dual-Port Ethernet Accessory Module
Multi-protocol Ethernet Communications with Wi-Fi
PXM1K-ETHMULTI and without WiFi PXM1K-ETHMULT-NW







Powering Business Worldwide

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Please read this manual carefully before installation, operation, and maintenance of the PXM1000 Dual-Port Ehernet Accessory Module.

The following symbols in this manual and on PXM 1000 meters are used to provide warning of danger or risk during the installation and operation of the meters.

Please note: The PXM1K-ETHMULT-NW is functionally identical to the PXM1K-ETHMULTI aside from having no WiFi capabilities or hardware, the remainder of this manual aside from references to WiFi apply to both modules.

Safety Precautions

All safety codes, safety standards, and/or regulations must be strictly observed in the installation, operation, and maintenance of this device.

⚠ WARNING

The warnings and cautions included as part of the procedural steps in this document are for personnel safety and protection of equipment from damage. An example of a typical warning call-out is shown above. This will help to ensure that personnel are alert to warnings that may appear throughout the document. In addition, cautions are all upper case and boldfaced as shown below.

⚠ WARNING

Completely read and understand the material presented in this document before attempting installation, operation, or application of the equipment. Only qualified persons should be permitted to perform any work associated with the equipment. The wiring, installation and application use instructions presented in this document must be followed precisely. Failure to do so could cause permanent equipment damage, bodily injury, or death.

⚠ WARNING

Do not attempt to install or perform maintenance on equipment while it is energized. Death, severe personal injury, or substantial property damage can result from contact with energized equipment. Always verify that no voltage is present before proceeding with the task, and always follow generally accepted safety procedures.

Eaton is not liable for the misapplication or misinstallation of its products.

To avoid complications, please read this manual carefully before installation and operation of the PXM1000 Dual-Port Ehernet Accessory Module.

1. Introduction to Ethernet

The PXM1K-ETHMULTI module provides a dual Ethernet and WiFi communication channel for PXM1000 series power meters. Users will be able to use both Ethernet ports and WiFi simultaneously with different networks and data acquisition systems.

This communications module provides users with an industry leading 100ms and 40ms response rate via Modbus TCP that allows users to attain real time updates to key parameters such as voltage, current and power from the PXM1000 Series meter.

Introduction to Ethernet

Ethernet was originally developed by Xerox and then further developed by DEC and Intel. This networking technology uses Carrier Sense Multiple Access with Collision Detection (CDSM/CD) protocol and provides transmission speeds up to 100Mbps.

Ethernet is a not a network but more of a standard. It is the most current communication standard Local Area Network(LAN). This standard defines the type of cable that is used and the method of Signal Processing. The PXM1K-ETHMULTI module supports two Ethernet channels.

IPv6

The PXM1K-ETHMULTI module also supports IPv6 which is the latest version for the internet protocol. The protocol uses 128-bit addressing in comparison to IPv4 which uses 32-bit addressing. The difference for addressing allows for more devices to be connected using IPv6 as opposed to the IPv4 protocol. The protocol is more efficient and provides more secure routing over the internet.

2. Functional Description of the Ethernet Module

The PXM1K-ETHMULTI module supports a wide range of communication protocols. Some of the more commonly used protocols are briefly explained below.

This module supports the Modbus-TCP protocol. When connected to the PXM1000 series meter, it is a server device that can only respond to queries. The default value for the Modbus Port is 502. The user defined range is 2000~5999.

The PXM1K-ETHMULTI grants users the ability to send emails based on a time interval or when there is a triggered event using the SMTP protocol. It can send mail from encrypted servers and servers that use different SMTP ports.

The PXM1K-ETHMULTI protocol supports HTTPS protocol. It is used as an HTTPS server and where the default value of the protocol port is 443. Using the HTTPS protocol, the PXM1K-ETHMULTI can send post requests to both HTTP and HTTPS servers.

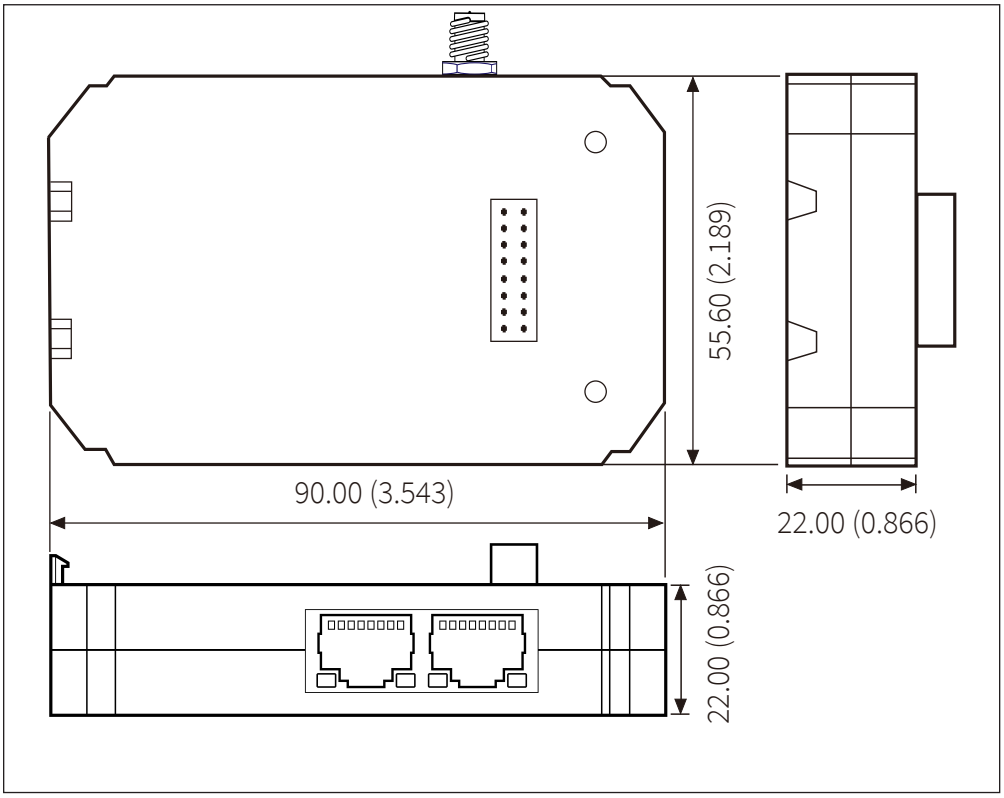
The following are all the protocols supported by the PXM1K-ETHMULTI module:

- Modbus TCP
- IPv6
- RSTP
- BACnet-IP
- SNMP V3
- SMTP
- NTP
- HTTP
- HTTPs
- FTP
- sFTP
- WiFi WPA, WPA2 Enterprise
- Ethernet IP

3. Appearance and Dimensions

The dimensions in the following diagram are in millimeters.

Dimensions listed in the brackets are inches.



4. Installation Method

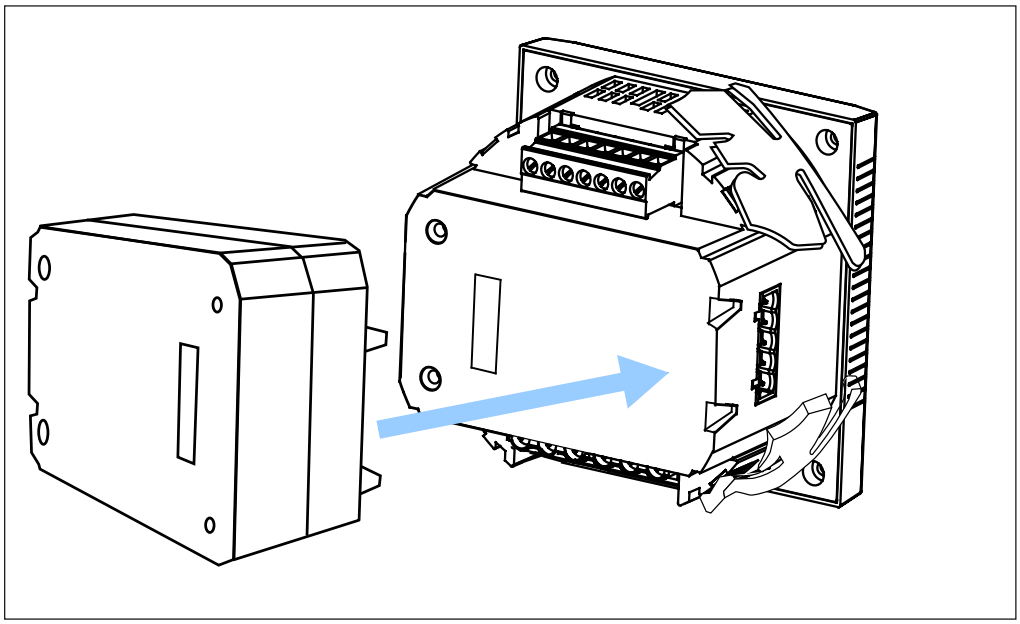
The PXM1K-ETHMULTI module is linked to the PXM1000 series meter by a communication plug. Other extended modules such as the IO modules can be linked to the PXM1000 series meter through the PXM1K-ETHMULTI.

1. Remove cover from the back of the PXM1000 series meter which will expose the socket.
2. Insert the installation clips to the grooves in the PXM1000 series meter and then press the PXM1K-ETHMULTI module lightly to establish a linking between meter and module.
3. Tighten the installation screws.

⚠WARNING

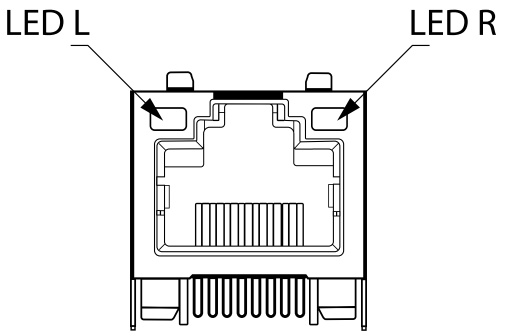
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5. Definition of RJ45

The PXM1K-ETHMULTI uses two standard RJ45 connectors to access the LAN. The mechanical and electrical characteristics of the connector are consistent with the requirements of IEC 603-7.



Top View

Pin number	Name	Description
1	TX+	Tranceive Data+
2	TX-	Tranceive Data-
3	RX+	Receive Data+
4	n/c	Not connected
5	n/c	Not connected
6	RX-	Receive Data-
7	n/c	Not connected
8	n/c	Not connected

LED_L (Yellow): Displays the speed status. When the LED is on it indicates 100Mpbs, whiles an off LED represents a speed of 10Mbps.

LED_R (Green): Displays the link and activity status. When the LED is on it indicates the link status. When the LED is flashing it indicates that there is activity.

6. Initializing the Ethernet Module

The default settings in the PXM1000 series meter are as followed:

Ethernet 1 (Static IP address)

IP Address (192.168.1.254)

Subnet Mask (255.255.255.0)

Gateway (192.168.1.1)

DNS Server 1 (8.8.8.8)

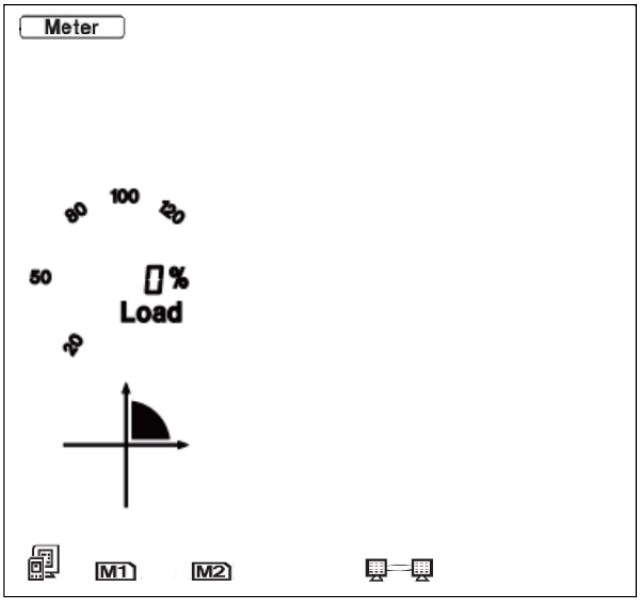
DNS Server 2 (8.8.4.4)

Modbus Port 502

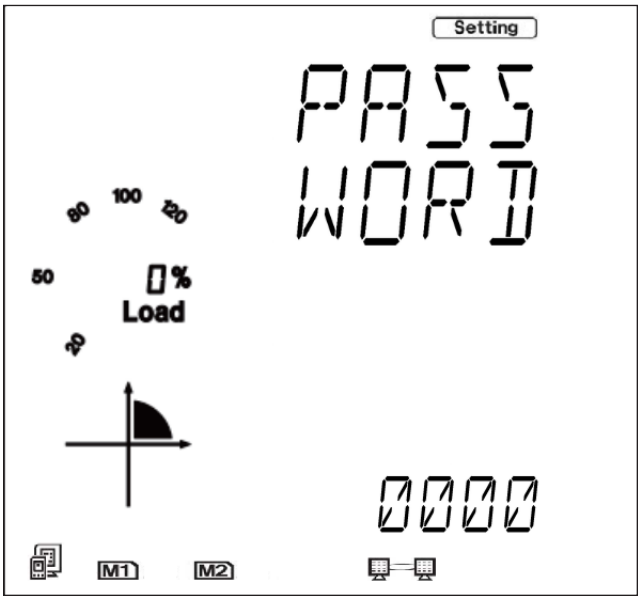
Ethernet 2 (Dynamic IP address)

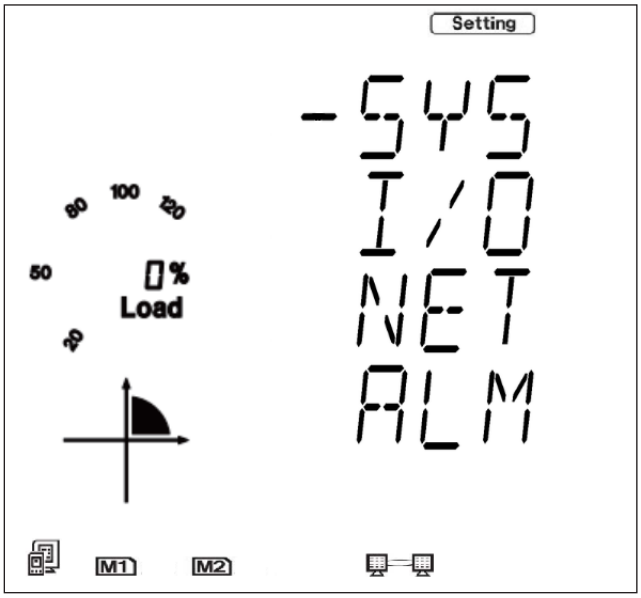
By default Ethernet 2 is configured for DHCP, meaning the network will dynamically assign its network properties. In order to view the Ethernet 2 IP address from the meters display the PXM1000 meter must be configured to the WEB2 protocol. The following explains how to change the meters protocol settings to work with the PXM1K-ETHMULTI module:

- Press the 'H' and 'V/A' buttons simultaneously on the PXM1000 series. Release the buttons and the meter will enter the meter selecting mode, as indicated by the flashing 'Meter' cursor.



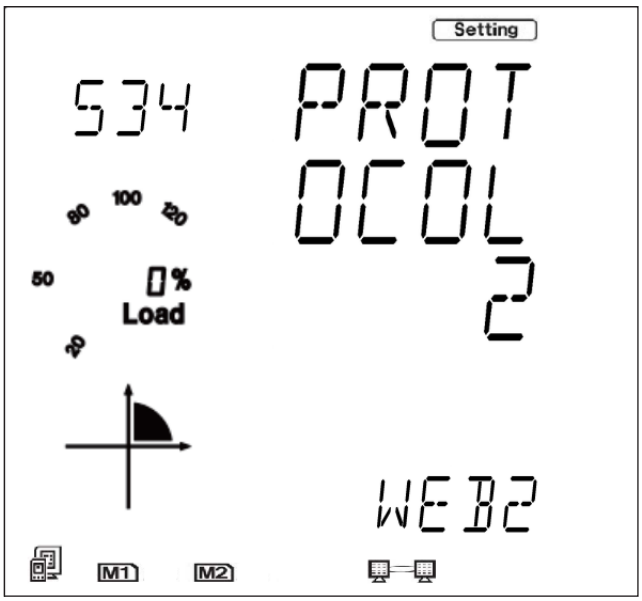
- Press the 'P' or 'E' button to move the cursor to 'Setting'. Press 'V/A' button to enter the parameter setting mode. The device address page is the first page of the 'Setting' mode. It will show the Modbus address of the meter for a second before prompting for the password of the device.
- You will be required to type in a password in the 'PASSWORD' screen. Leave the password as default '0000' and press 'V/A' to enter the parameter selection Mode.





The cursor will be on ‘SYS’. Press ‘V/A’ on this screen to get to the system settings. This will show screen ‘S01 ADDR’.

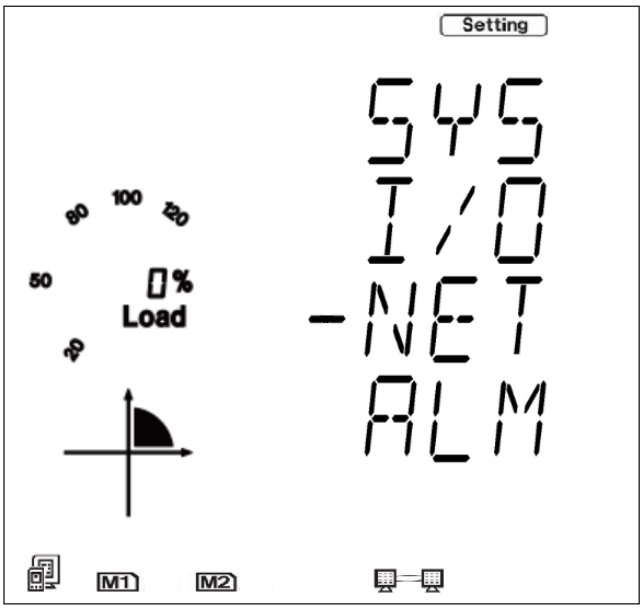
- Press the 'E' button until you get to 'S34 PROTOCOL 2'. Select the 'WEB2' protocol.
  - Press ‘V/A’ to modify the setting; the cursor should now flash.
  - Press ‘P’ or ‘E’ to select 'WEB2'.
  - Press ‘V/A’ to confirm the change.



Once the protocol 2 setting of the PXM1000 meter is configured for WEB2, users can now view all necessary settings in the meters NET settings.

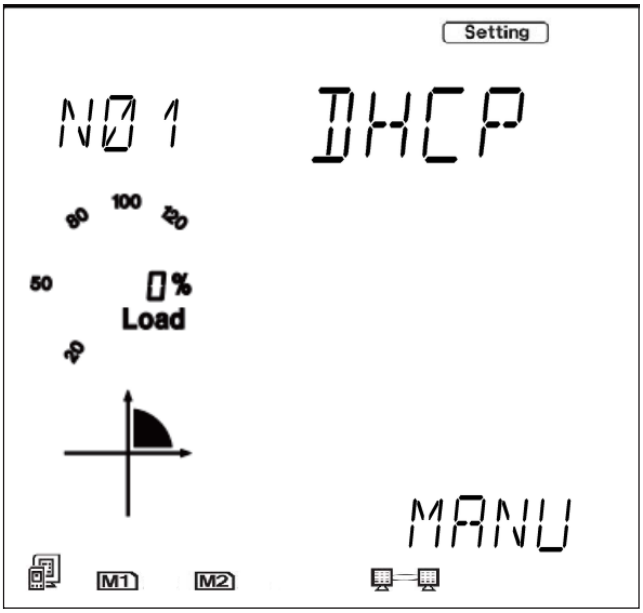
Press the 'H' button to exit the system setting, you will be directed to the parameter selection screen.

Press the 'P' or 'E' button to move the cursor to 'NET' and press the 'V/A' button to enter the Ethernet module settings.

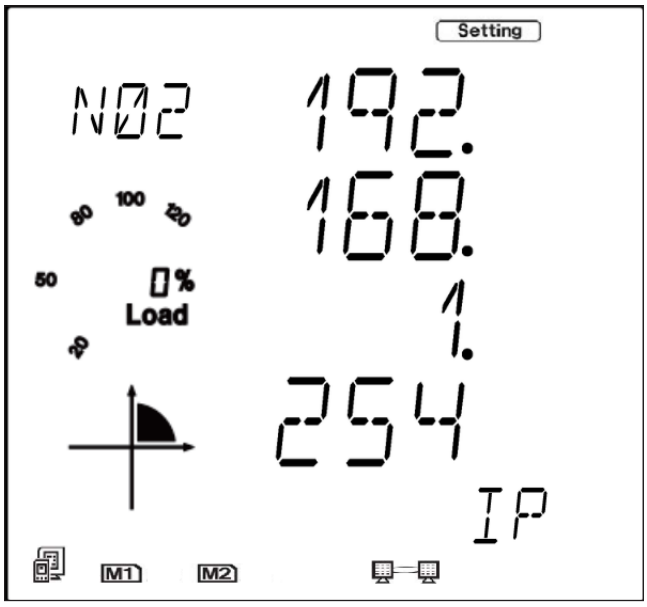


- The first page of the NET Settings will be the N01 DHCP setting. By default this is configured to Manual. Setting this configuration to Auto will allow the router to assign the meter with an IP address, whiles Manual will allow the user to configure the IP address. Press the 'V/A' button to enter edit mode. Press 'P' or 'E' to change the setting and press 'V/A' to confirm.

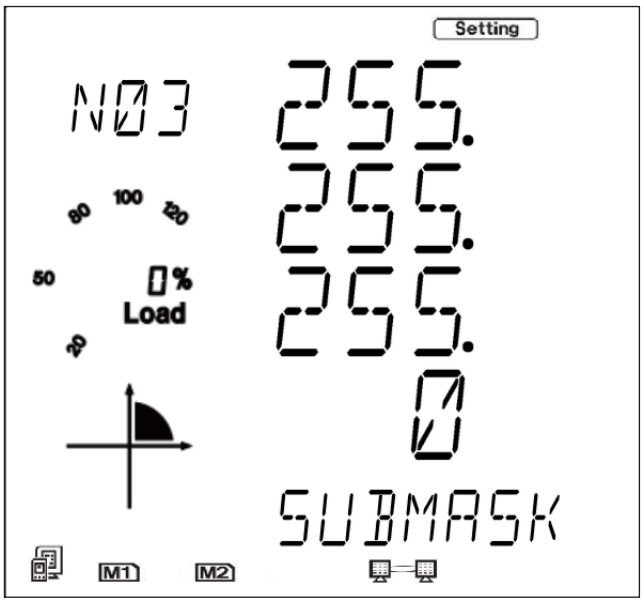
Note: If the DHCP is selected as Auto, the Ethernet module needs to be rebooted before it can be assigned with the new IP address.



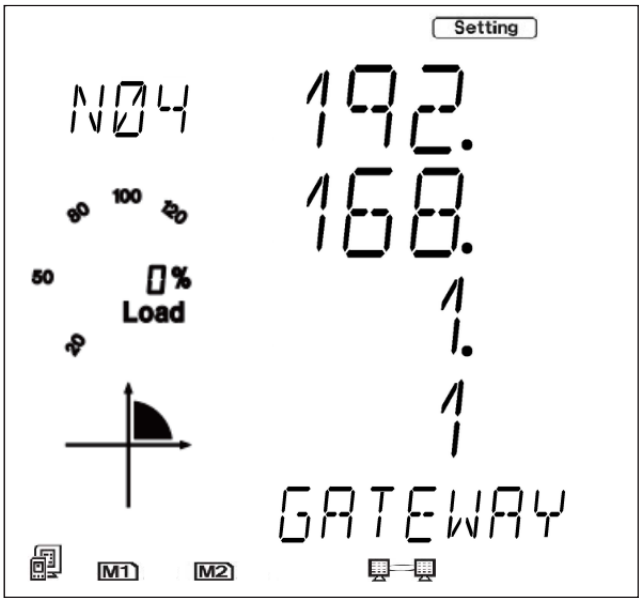
- Press 'P' to get to "N02 IP address" This is the IP address for Ethernet 1 and can be used to access the web interface of the module. Users can configured the IP address if the DHCP is configured to Manual. Press 'V/A' to configure the IP address. The cursor of the first digit will begin to flash. Press the 'H' button to scroll through the digits, press the 'P' or 'E' to change the value of the flashing cursor and press 'V/A' to confirm.



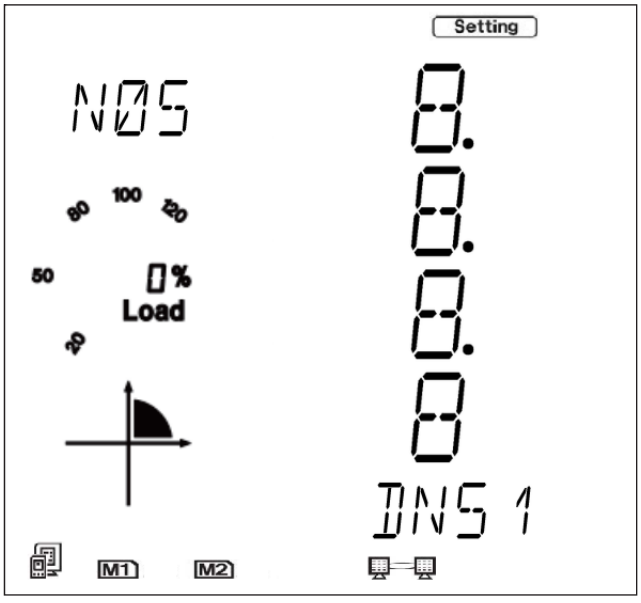
Press 'P' to get to "N03 Subnet Mask". Press 'V/A' to configure the subnet address. The cursor of the first digit will begin to flash. Press the 'H' button to scroll through the digits, press the 'P' or 'E' to change the value of the flashing cursor and press 'V/A' to confirm.



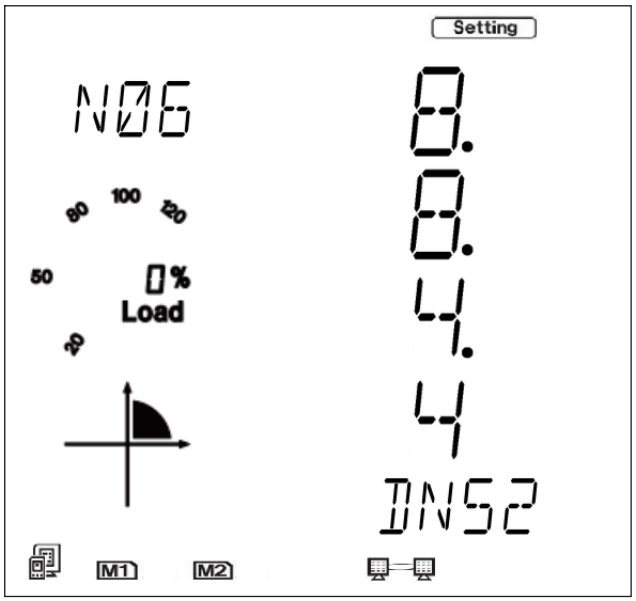
- Press 'P' to get to "N04 Gateway". Press 'V/A' to configure the gateway IP address. The cursor of the first digit will begin to flash. Press the 'H' button to scroll through the digits, press the 'P' or 'E' to change the value of the flashing cursor and press 'V/A' to confirm.



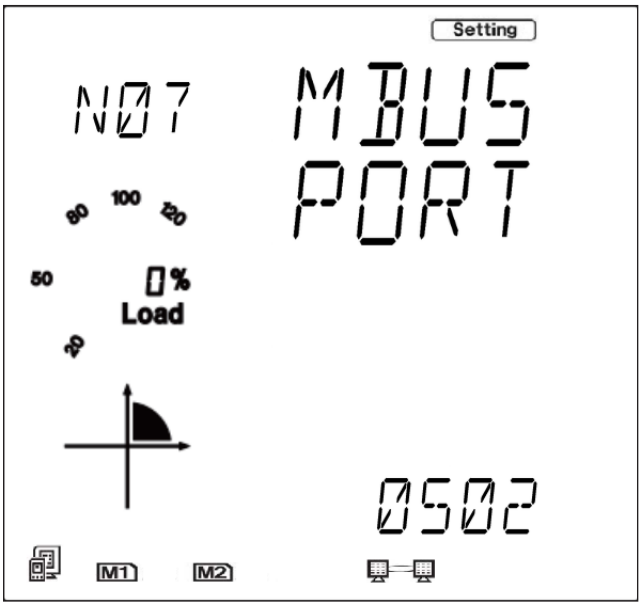
- Press 'P' to get to "N05 DNS Primary Server". Press 'V/A' to configure the DNS address. The cursor of the first digit will begin to flash. Press the 'H' button to scroll through the digits, press the 'P' or 'E' to change the value of the flashing cursor and press 'V/A' to confirm. The DNS parameters must be set correctly to use the SMTP, FTP/HTTP Post and PX-EVA functions.



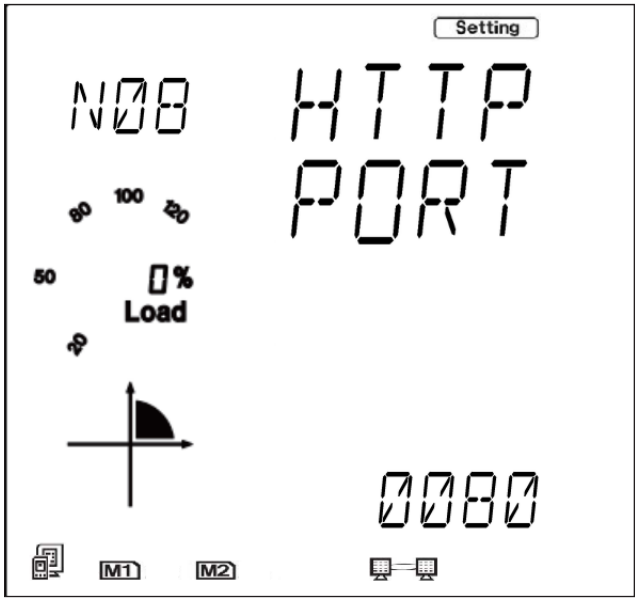
- Press 'P' to get to "N06 DNS Secondary Server". Press 'V/A' to configure the DNS address. The cursor of the first digit will begin to flash. Press the 'H' button to scroll through the digits, press the 'P' or 'E' to change the value of the flashing cursor and press 'V/A' to confirm.



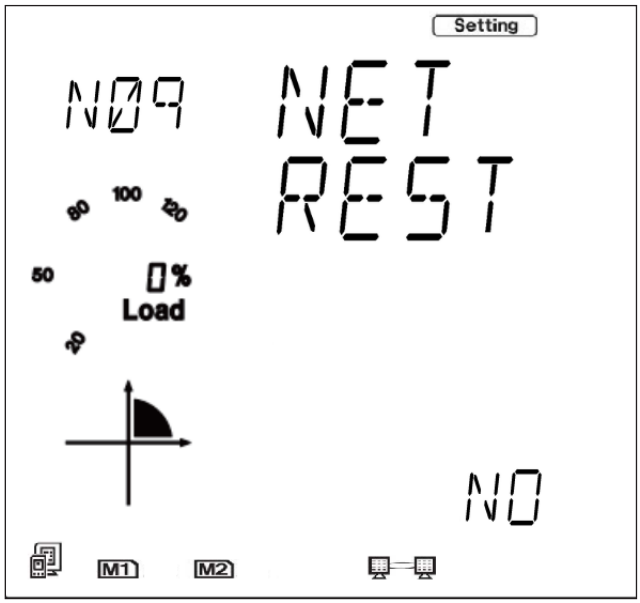
- Press 'P' to get to "N07 Modbus Port". Press 'V/A' to configure the Modbus Port. The cursor of the first digit will begin to flash. Press the 'H' button to scroll through the digits, press the 'P' or 'E' to change the value of the flashing cursor and press 'V/A' to confirm.



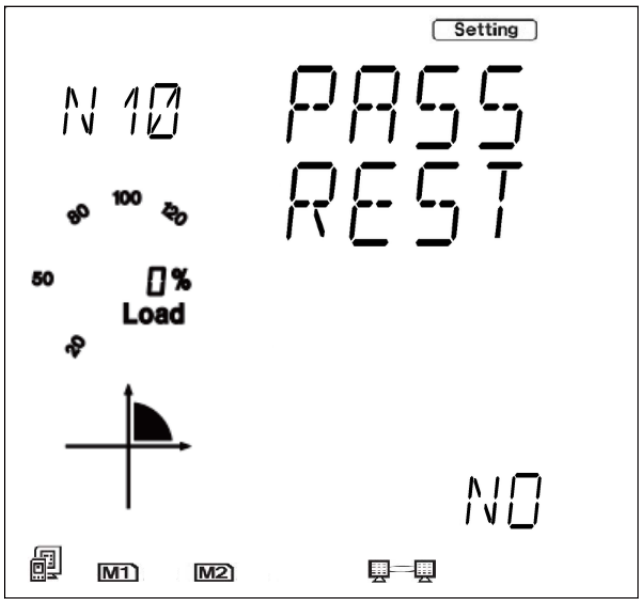
- Press 'P' to get to "N08 HTTP Port". Press 'V/A' to configure the HTTP Port. The cursor of the first digit will begin to flash. Press the 'H' button to scroll through the digits, press the 'P' or 'E' to change the value of the flashing cursor and press 'V/A' to confirm.



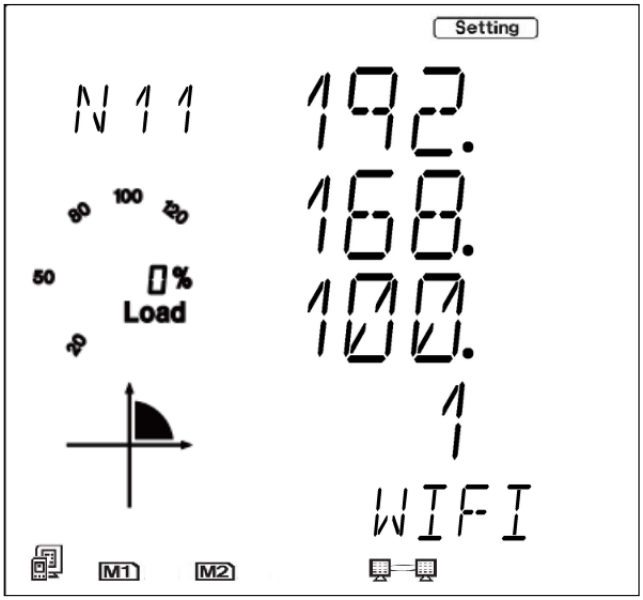
- Press 'P' to get to "N09 NET REST". After making any changes to the NET settings, users must reboot the Ethernet module from this page for the settings to take effect. Press 'V/A' to reboot the module, the cursor will begin to flash. Press the 'P' or 'E' button to change the setting to 'Reset' and press 'V/A' to confirm. The cursor will return to 'No' once the module has successfully reset.



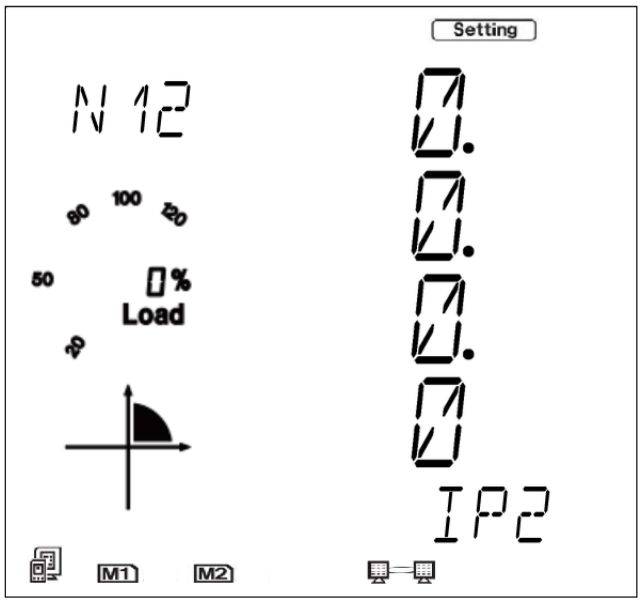
- Press 'P' to get to "N10 PASSREST". Press 'V/A' to configure the password reset. The cursor of the first digit will begin to flash. Press the 'P' or 'E' button to change the setting to 'Reset' and press 'V/A' to confirm. The cursor will return to 'No' once successful.



- Press 'P' to get to "N11 WiFi" This is the IP address of WiFi and will be the IP address to access the web interface of the module by using WiFi connection. This IP address cannot be modified from the meters display, it can only be configured on the meters web server.



- Press 'P' to get to 'N12 IP2'. This is the IP address for Ethernet port 2, it is preset as dynamic DHCP. A new IP address will be assigned to it when it is connected to the LAN via Ethernet port 2.



7. Cable

An RJ45 cable is needed to connect the meter to the network.

A shielded twisted pair cable(standard 568A or standard 568B) is recommended as reference to the EIA/TIA standard.

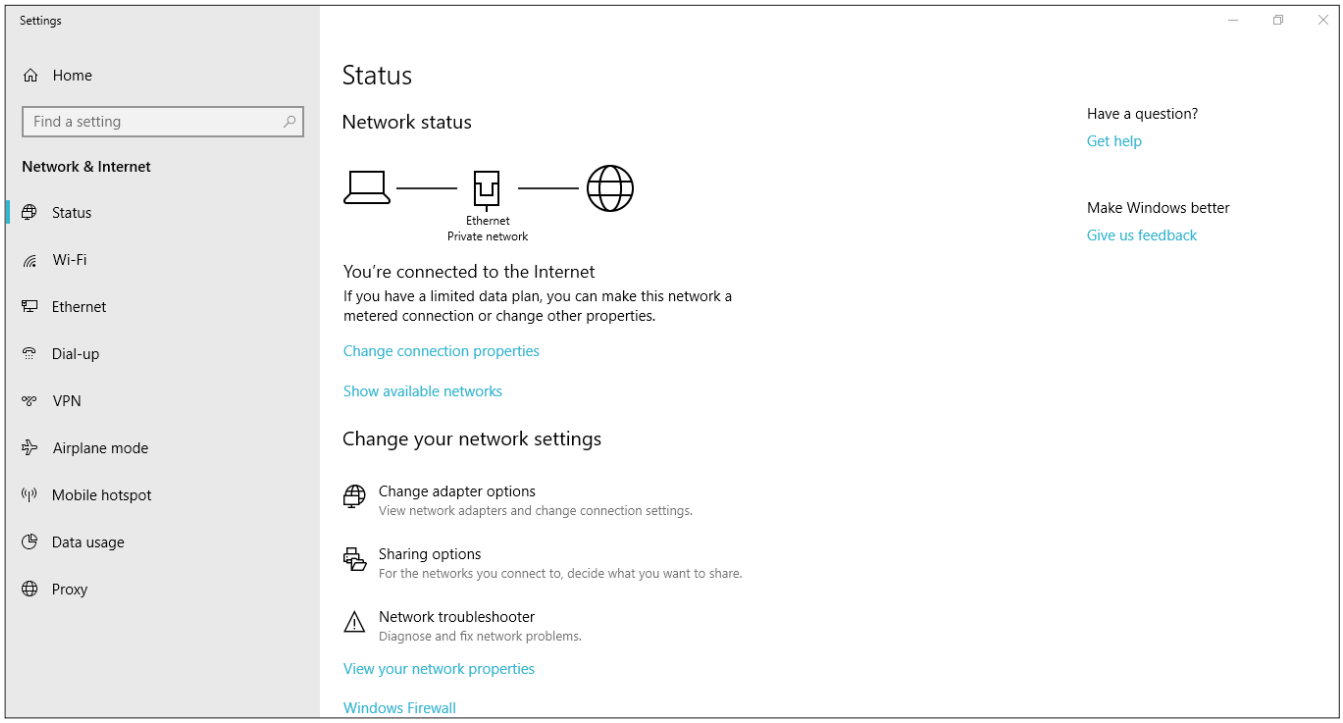
8. Connection Method

8.1 Direct Connect to a Computer

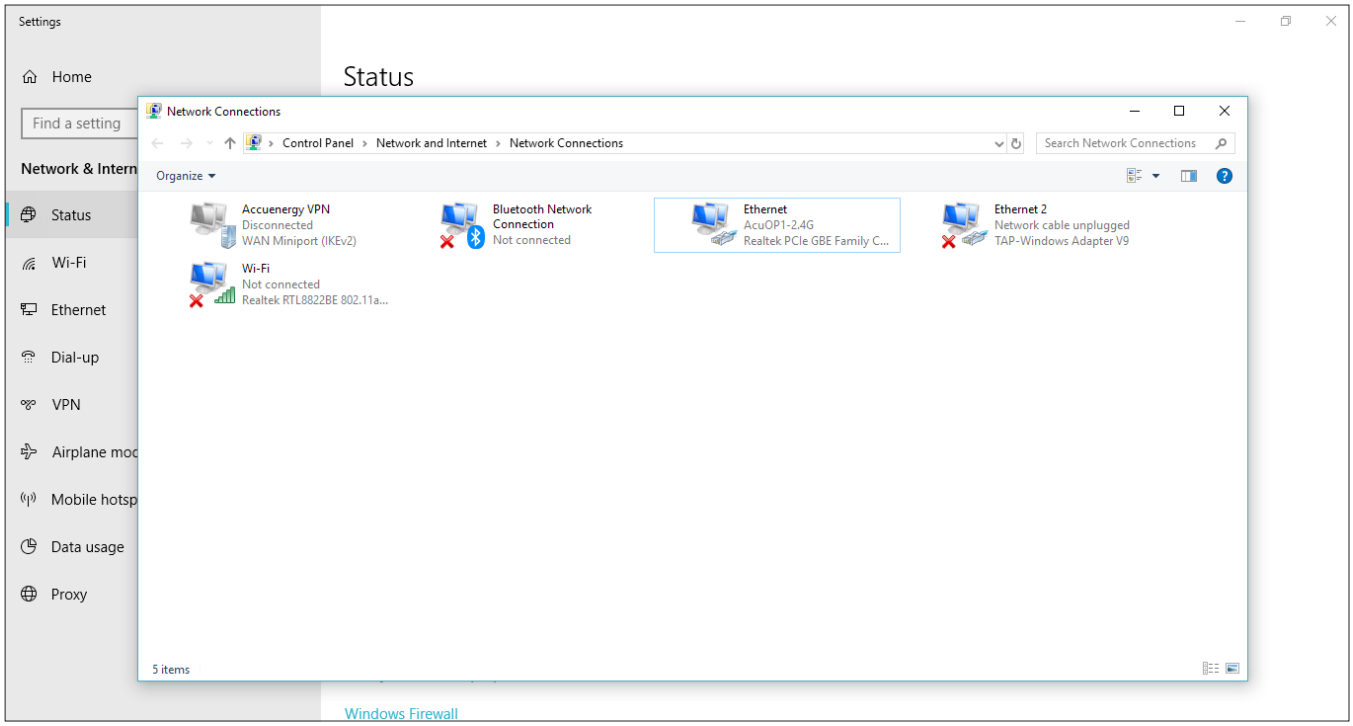
The PXM1K-ETHMULTI can be connected to a computer using a crossover cable(standard 568A). The PXM1K-ETHMULTI module supports Modbus-TCP and HTTPS Functions for this method of connection.

To connect meter directly to the computer, the computers IP must be within the same subnet as the meters IP address. The following steps outline how to change the computers IP using a computer running the Windows OS::

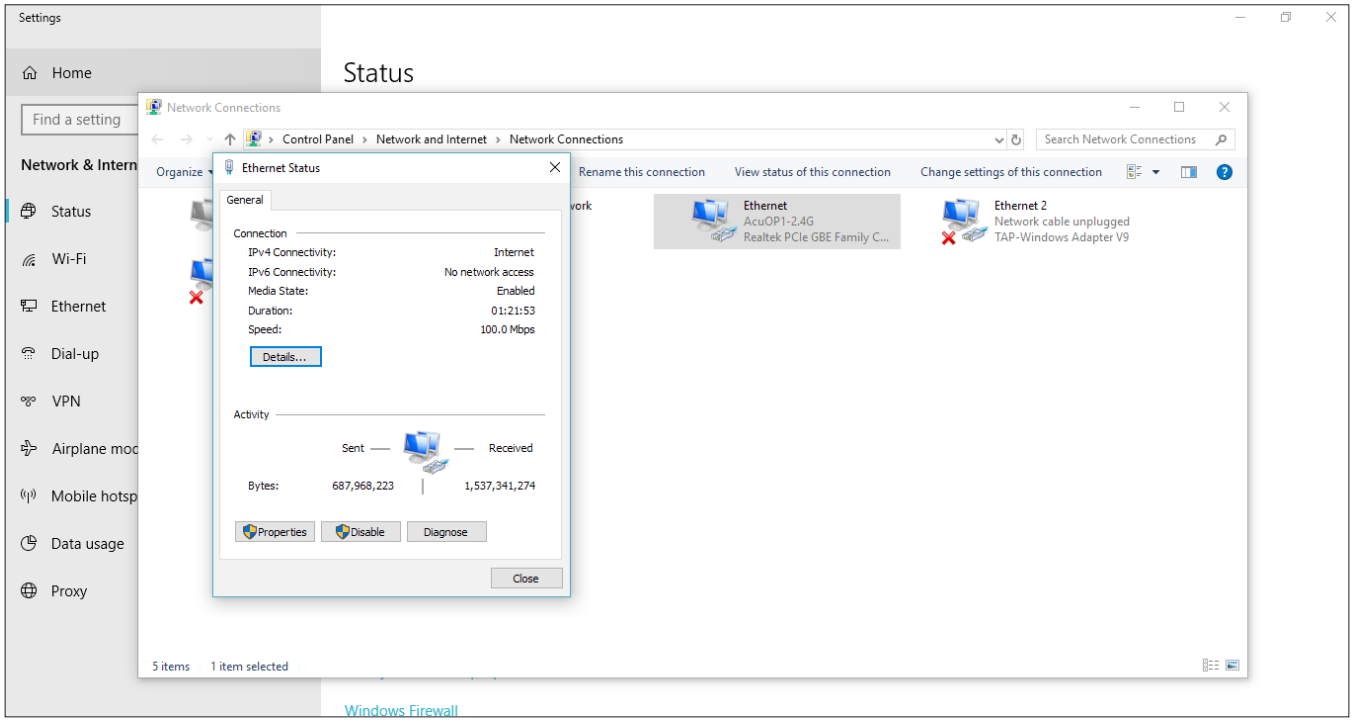
- Manually connect the meter via Ethernet cable to the computer
- Right click on the connection icon
- Select "Open Network Sharing Center"



- Click on Change adapter options

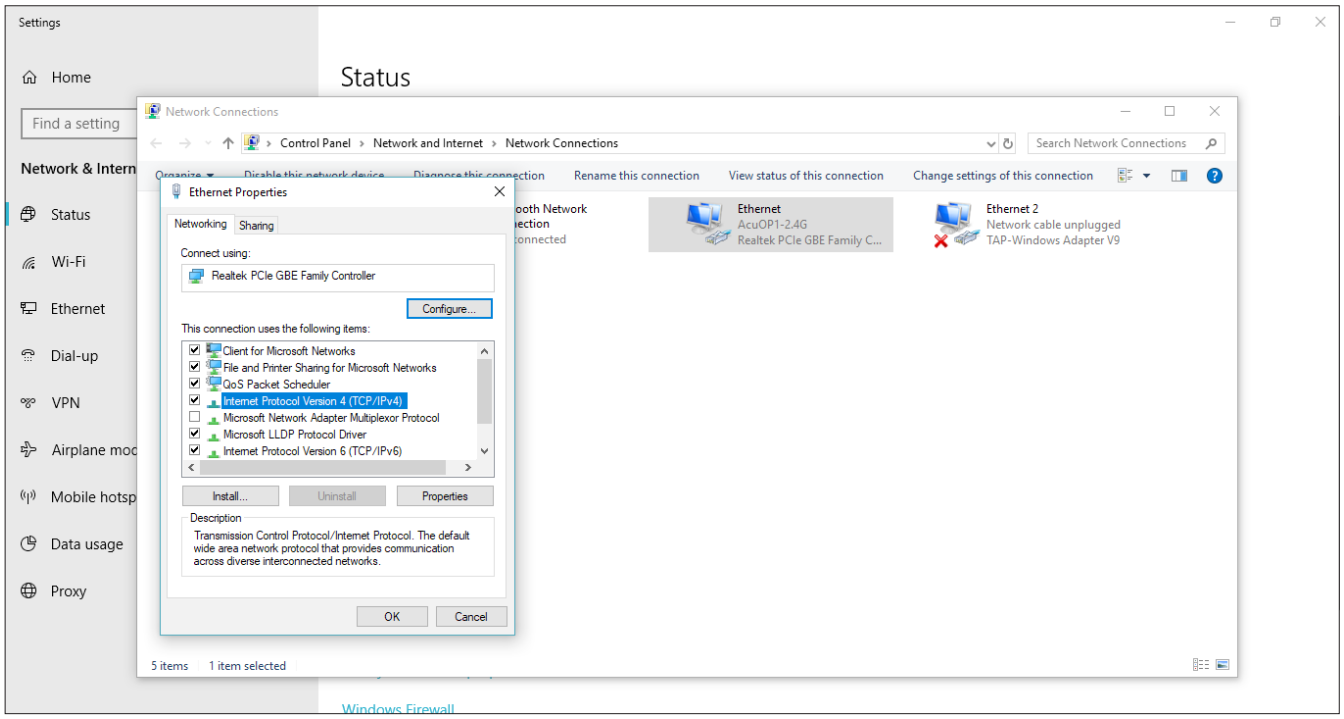


- Once there, right click on the local area connection icon and select properties.

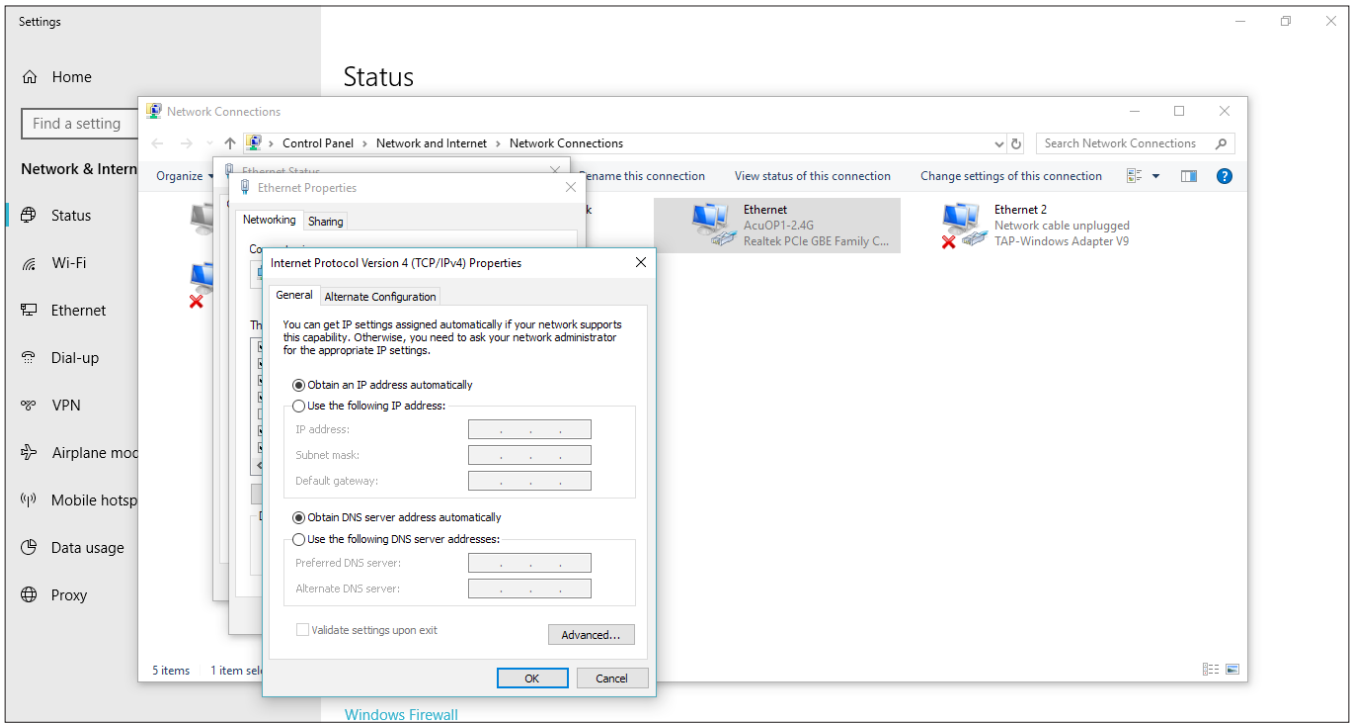


8. Connection Method

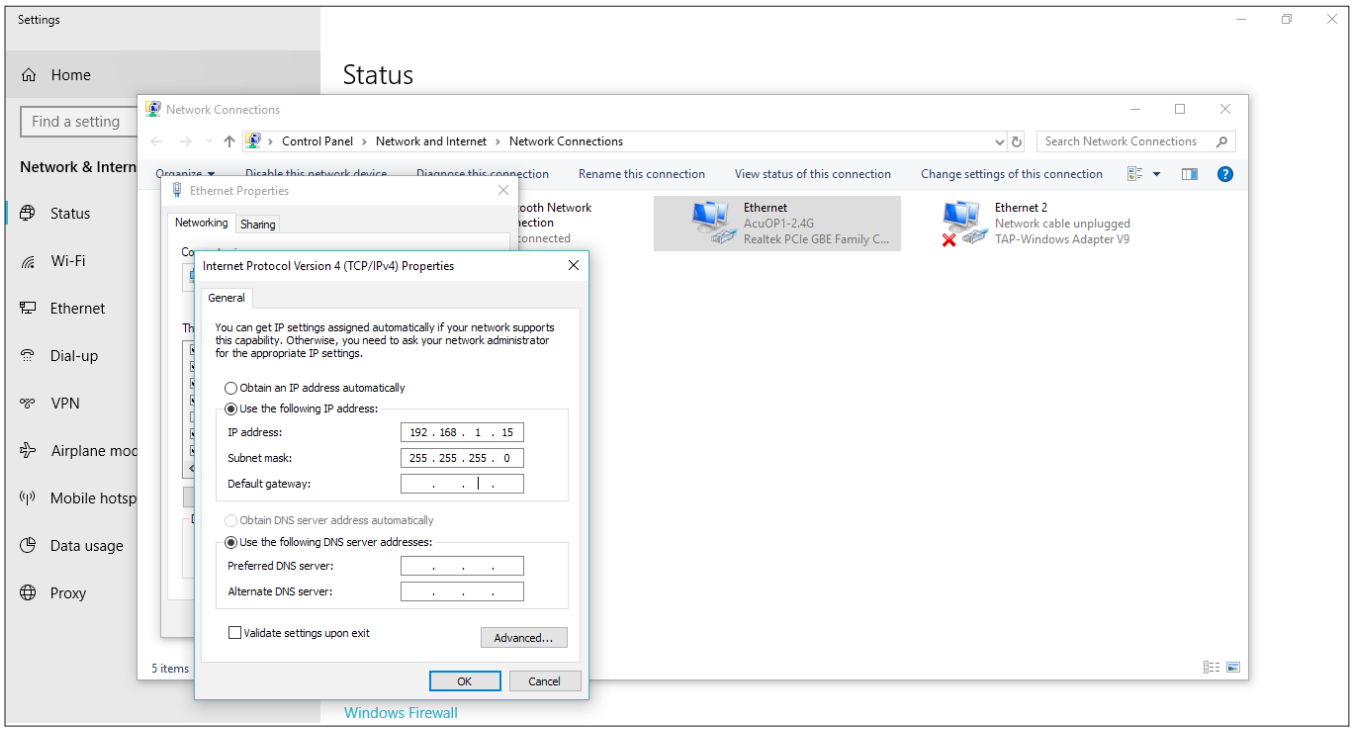
- Select the icon that says Internet Protocol Version 4 TCP/IP



- The Internet Protocol Version 4(TCP/IP) Properties box will pop up



- Click on ''Use the following IP address'' and enter in an IP number so that meter and computer are in the same local network range. For example, if the meter has IP address of 192.168.1.254, then the computer must be assigned with an IP 192.168.1.xxx, where xxx can be any number but cannot be the same as the value the meter has.



- Once you have entered in the IP address, press the Tab key on your keyboard until you hit the bottom and click OK
- Before selecting the OK button make note of the IP address you have assigned to the meter and then press OK.

Note: The meter and computer cannot have the same IP address, they must be different.

8.2 Direct Connect to a Router/Switch

The PXM1K-ETHMULTI can be connected to a router or switch using a patch cable. The DHCP can be configured to Auto to have the router assign the meter with an IP address or the DHCP can be configured to Manual to set an IP address and network settings manually.

PXM1K-ETHMULTI has two Ethernet ports, Ethernet 1 is set to have the static DHCP, and Ethernet 2 is set to have the dynamic DHCP. Both of the Ethernet ports have the same functionalities, you can use either of them according to the requirement.

8.3 Connect through WiFi

The PXM1K-ETHMULTI can be connected through WiFi network.

By default the PXM1K-ETHMULTI will be in Access Point mode with default IP address of 192.168.100.1. Ensure the device connecting to the PXM1K-ETHMULTI has DHCP enabled or it should be in the same subnet as the PXM1K-ETHMULTI. The module will appear in the WiFi network as PXM1K-ETHMULTI-WIFI-(serial number of module) as the SSID or name of the wireless network. By default, the network key or password will be ''pxm1k-ethmulti''.

- Once connected to the network, open an internet browser and type in the IP address of the WIFI module: 192.168.100.1
- Log in at Admin access level, using the default password of 'admin'.

9. Description of Modbus-TCP Protocol

The Modbus-TCP protocol is used as one of the communication protocols in the PXM1K-ETHMULTI. The protocol establishes a client and server connection in Ethernet. The client device first sets up a TCP-IP link with server device. The client device then sends a request to the server device and the server device in return sends a response to the client device. Figure below shows how the Modbus-TCP protocol works.



9.1 Protocol

9.1.1 Data Frame Format

MBAP Header	Function	Data
7x8 bits	8-bits	Nx8 bits

9.1.2 Modbus Application Header (MBA Header) Field

Modbus application header field is the start of the data frame, and consists of seven bytes.

Field	Length	Description
Transaction Identifier	2 Bytes	Identification of a Modbus Request/Response transaction
Protocol Identifier	2 Bytes	Modbus Protocol = 0
Length	2 Bytes	Number of following bytes
Unit Identifier	1 Byte	Server address, in the range of 0-247 decimal

9.1.3 Function Field

The function code field of a message frame contains eight bits. Valid codes are in the range of 1-255. When a message is sent from a client to a server device, the function code field tells the server what kinds of action to perform.

Code	Meaning	Data
01	Read Relay Output Status	Obtain current status of Relay Output
02	Read Digital Input (DI) Status	Obtain current status of Digital Input
03	Read Data	Obtain current binary value in one or more registers
05	Control Single Relay Output	Force Relay to a state of ON or OFF
16	Write Multiple Registers	Place specific value into a series of consecutive multiple registers

9.1.4 Data Field

The data field is constructed using sets of two hexadecimal digits, in the range of 00 to FF. The data field of messages sent from a client to server contains additional information which the server must use to take the action defined by the function code. This can include information such as the register addresses, the quantity of registers to query and the count of the actual number of data bytes. For example, if the client requests a server to read a group of holding registers(function code 03), the data field specifies the starting register and how many registers are to be read.

If the client needs to write data(function code 10 hexadecimal) to a group of registers in the server, the data field specifies the starting register, how many registers to write, the count of data bytes to follow in the data field and the data to be written into the registers.

9.2 Format of Communication

9.2.1 Explanation of Frame

Transaction identifier hi	Transaction identifier lo	Protocol identifier hi	Protocol identifier lo	Length hi	Length lo	Unit identifier
0H	00H	00H	00H	00H	06H	01H

Function Code	Data start register hi	Data start register lo	Data # of registers hi	Data # of registers lo
03H	40H	00H	00H	48H

The meaning of each abbreviated field above is:

Transaction identifier hi: High byte of transaction identifier

Transaction identifier lo: Low byte of transaction identifier

Protocol identifier hi: High byte of protocol identifier

Protocol identifier low: Low byte of protocol identifier

Length hi: High byte of length

Length lo: Low byte of length

Unit identifier: Server address

Fun: Function code

Data start register hi: High byte of starting register address

Data start register lo: Low byte of starting register address

Data #of registers hi: High byte of number of registers

Data #of registers lo: Low byte of number of registers

1) Read Status of Relay (Function code 01)

Function Code 1

This function code is used to read the relay output status in the PXM1000 series meter.

1=On 0=Off

There are 8 relay outputs in the PXM1000 series meter and they start at address 0000H.

The following query is to read 2 relay output status of the PXM1000 series address 1.

Query

Transaction identifier hi	Transaction identifier lo	Protocol identifier hi	Protocol identifier lo	Length hi	Length lo	Unit identifier
00H	00H	00H	00H	00H	06H	01H

Fun	Data start register hi	Data start register lo	Data # of registers hi	Data # of registers lo
01H	00H	00H	00H	02H

Response

The PXM1000 series meter responds back with the MBAP header, function code, quantity of data bytes and the data.

An example of response to read the status of the first 2 relay outputs starting at 0000H is shown below. The status of relay output 1 and 2 corresponds to the last 2 bits of data.

Relay 1: bit0 Relay 2: bit1

Transaction identifier hi	Transaction identifier lo	Protocol identifier hi	Protocol identifier lo	Length hi	Length lo	Unit identifier
00H	00H	00H	00H	00H	04H	01H

Fun	Byte count	Data
01H	01H	02H

The content of the data is,

7	6	5	4	3	2	1	0
0	0	0	0	0	0	1	0

MSB LSB

(Relay 1 = OFF, Relay 2 = ON)

2) Read Status of DI (Function Code 02)

Function Code 2

1=On 0=Off

There are 28 DIs in the PXM1000 series meter starting at address 0000H.

The following query is to read 4 DI statuses of PXM1K-1 module with logic address of 1 in the PXM1000 series meter.

Query

Transaction identifier hi	Transaction identifier lo	Protocol identifier hi	Protocol identifier lo	Length hi	Length lo	Unit identifier
00H	00H	00H	00H	00H	06H	01H

Fun	Data start register hi	Data start register lo	Data # of registers hi	Data # of registers lo
02H	00H	00H	00H	04H

Response

The response includes the MBAP header, function code, quantity of data characters and the data.

An example response from the meter to read the status of 4 DIs(DI1-On, DI2=On, DI3=On, DI4=On) is shown below. The status of each corresponds to the last 4 bits of the data.

DI1: bit0 DI2: bit1 DI3: bit2 DI4: bit3

Transaction identifier hi	Transaction identifier lo	Protocol identifier hi	Protocol identifier lo	Length hi	Length lo	Unit identifier
00H	00H	00H	00H	00H	04H	01H

Fun	Byte count	Data
02H	01H	0FH

The content of the data is,

7	6	5	4	3	2	1	0
0	0	0	0	1	1	1	1

MSB LSB

3) Read Data (Function Code 03)

Function Code 3

Query

This function allows the user to obtain the measurement results of the PXM1000 series meter.

Below is an example to read 6 registers corresponding to the device clock of the meter, starting at 1040H.

Transaction identifier hi	Transaction identifier lo	Protocol identifier hi	Protocol identifier lo	Length hi	Length lo	Unit identifier
00H	00H	00H	00H	00H	06H	01H

Fun	Data start register hi	Data start register lo	Data # of registers hi	Data # of registers lo
03H	10H	40H	00H	06H

An example response is provided to read the time (2006-12-18 14:15:20).

Transaction identifier hi	Transaction identifier lo	Protocol identifier hi	Protocol identifier lo	Length hi	Length lo	Unit identifier
00H	00H	00H	00H	00H	0FH	01H

Fun	Byte count	Data1 hi	Data1 lo	Data2 hi	Data2 lo	Data3 hi	Data3 lo	Data4 hi	Data4 lo	Data5 hi	Data5 lo	Data6 hi	Data6 lo
03H	0CH	07H	D6H	00H	0CH	00H	12H	00H	0EH	00H	0FH	00H	14H

4) Control Relay (Function Code05)

Function Code 5

Query

This function code enables the control of a single relay output in the PXM1000 series meter. Any relay output in the PXM1000 series meter can be controlled on or off starting at 0000H.

Sending the data 'FF00H' will set they relay output on and sending '0000H' will turn it off; all other values are illegal and will not affect they relay output status.

The example below is a request to a PXM1000 series meter to turn on relay output 1.

Transaction identifier hi	Transaction identifier lo	Protocol identifier hi	Protocol identifier lo	Length hi	Length lo	Unit identifier
00H	00H	00H	00H	00H	06H	01H

Fun	Data start register hi	Data start register lo	Data # of registers hi	Data # of registers lo
05H	00H	00H	FFH	00H

Response

The normal response to the command request is to retransmit the message as received after the relay output status has been altered.

Transaction identifier hi	Transaction identifier lo	Protocol identifier hi	Protocol identifier lo	Length hi	Length lo	Unit identifier
00H	00H	00H	00H	00H	06H	01H

Fun	Data start register hi	Data start register lo	Value hi	Value lo
05H	00H	00H	FFH	00H

5) Preset/Reset Multi-Register (Function Code 16)

Function Code 16

Query

This function code allows the user to modify the contents of a register. The example below is a request to an PXM1000 series meter with device address 1 to preset the CT1(500) and CT2(5) registers. The CT1 data address is 1008H and CT2 is at 1009H.

Transaction identifier hi	Transaction identifier lo	Protocol identifier hi	Protocol identifier lo	Length hi	Length lo	Unit identifier
00H	00H	00H	00H	00H	0BH	01H

Fun	Data start register hi	Data start register lo	Data # of registers hi	Data # of registers lo	Byte count	Value1 hi	Value1 lo	Value2 hi	Value2 lo
10H	10H	08H	00H	02H	04H	01H	F4H	00H	05H

Response

The normal response to a preset Multi-Register request including the MBAP Header, function code, data start register and the number of registers is shown below.

Transaction identifier hi	Transaction identifier lo	Protocol identifier hi	Protocol identifier lo	Length hi	Length lo	Unit identifier
00H	00H	00H	00H	00H	06H	01H

Fun	Data start register hi	Data start register lo	Data # of registers hi	Data # of registers lo
10H	10H	08H	00H	02H

10. Web Interface Readings and Parameter Settings

The PXM1K-ETHMULTI module supports the HTTPS protocol to allow for the use of a web interface. The user will need to access the PXM1K-ETHMULTI web interface to configure the module and use its functions.

The PXM1K-ETHMULTI web interface allows for different user access levels.

To access the web interface the IP address for the PXM1K-ETHMULTI either Ethernet 1, Ethernet 2 or a WiFi IP address must be known..

10.1 User Access Login

Enter the correct IP address of the module in the search bar of the internet browser to access the web interface of the PXM1K-ETHMULTI

The user will be redirected to a web page prompting to select the Access Level and enter appropriate password for that level. Note that the default passwords should be changed when the meter is commissioned to a non-trival password.

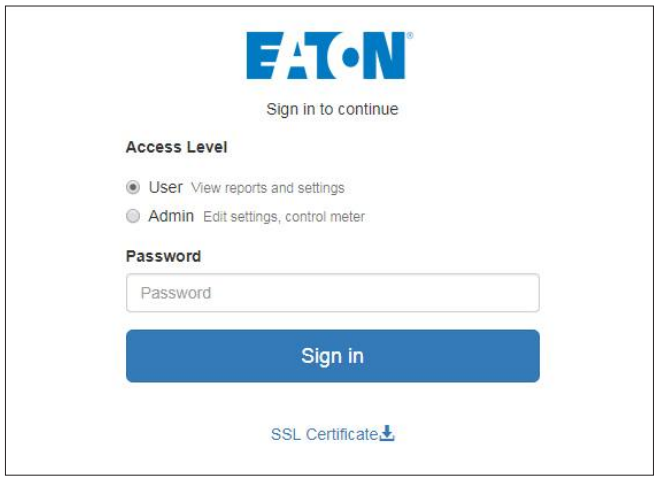
The User level is ideal for users who need only to take readings and view status from the meter.

The default password for the User level is **view**.

It is recommended that no more than 5 users are logged in at the same time for this level to ensure optimal performance of web interface.

The Admin level is ideal for users who need access to configurations on the meter or the web interface and to view readings.

The default password for the Admin level is **admin**.



The two different access levels are summarized below:

Access Level	Default Password	Read Parameter/Status	Configure Settings
User	view	Yes	No
Admin	admin	Yes	Yes

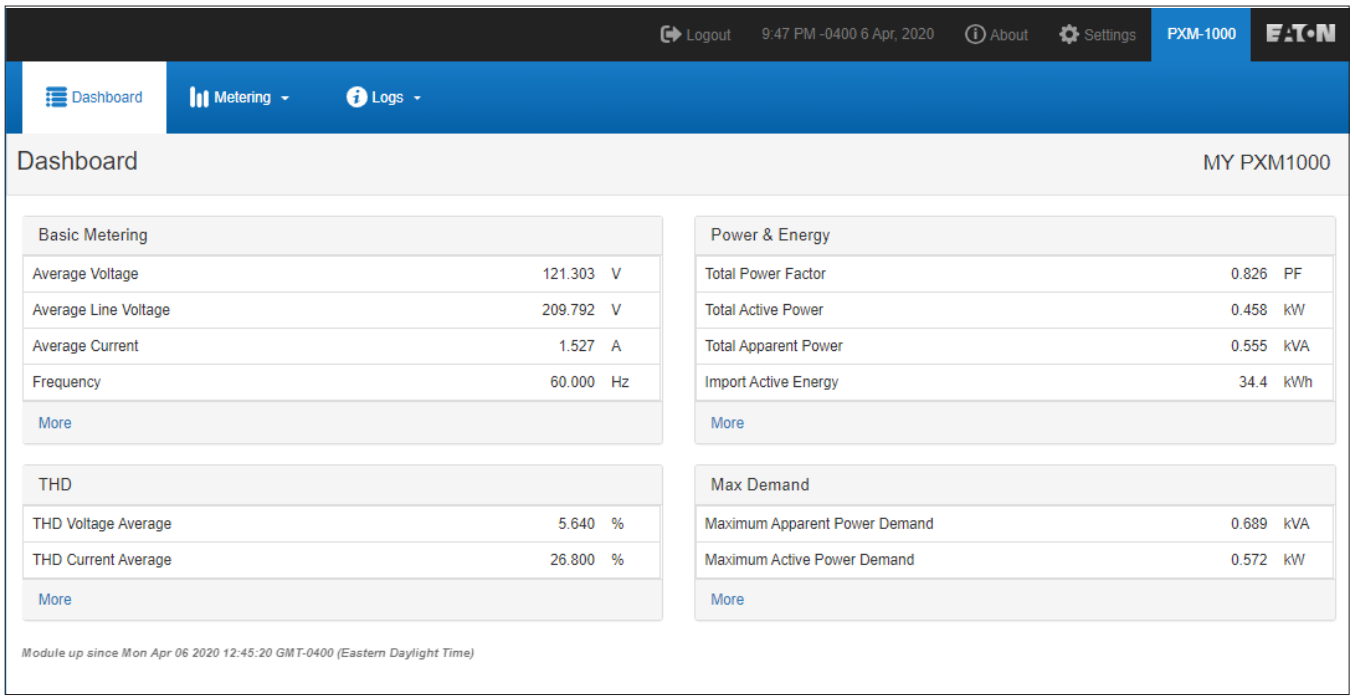
10.2 Dashboard

In the dashboard, the user will find the tabs to access different pages in the web interface such as '**Metering**', '**Logs**', and '**Settings**'. The dashboard is the first page the user will see once they have entered the correct password for the appropriate access level and is the same for both access levels.

The dashboard displays selected parameters from the different groups of metering parameters such as '**Basic Metering**', '**Power & Energy**', '**THD**' and '**Max Demand**'. By clicking on "Full report" under any one of these four metering parameter groups the user will be redirected to the web page which contains all the parameters supported by that metering parameter group.

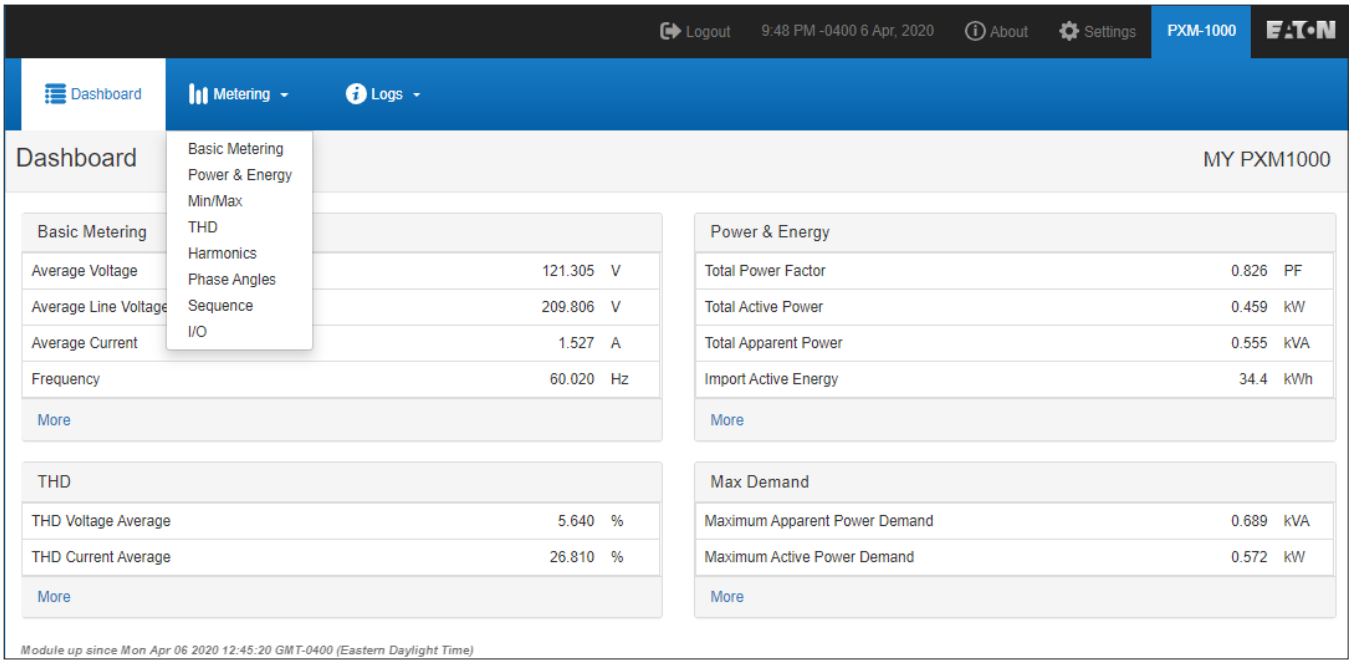
The dashboard also displays how long the PXM1K-ETHMULTI module has been connected to the network since the last reboot of the module in the bottom let corner of the page.

The parameters on this page are updated every 5 sec.



10.3 Metering Web Page

Click on the 'Metering' tab to visit the metering data web pages. There are eight kinds of metering parameter web pages. They are "Basic Metering", "Power & Energy", "Min/Max", 'THD', 'Harmonics', "Phase Angles", ''Sequence'' and ''I/O''. Each web page shows data from the PXM1000 series meter.

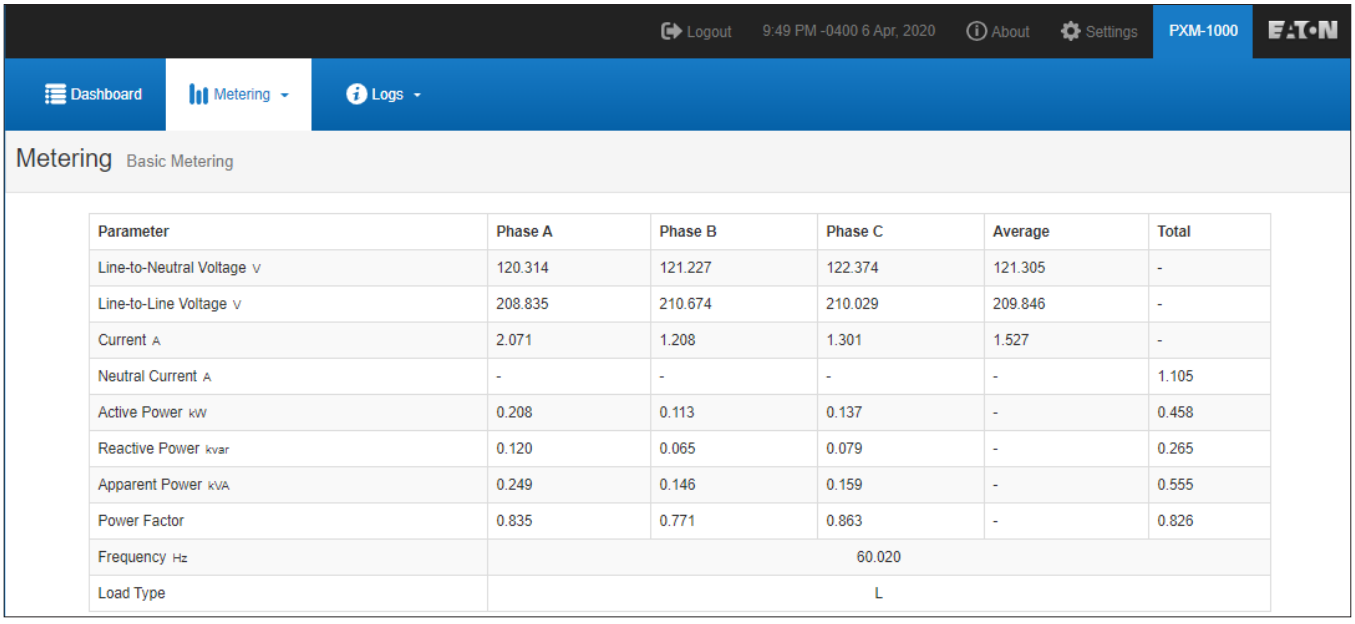


10.3.1 Basic Metering

The Basic Metering webpage includes the data of real-time parameters for the PXM1000 series meter. This includes the Line Voltages, Phase Voltages, Current, Neutral Current, Active, Reactive and Apparent Power, Power Factor, Frequency and Load type.

The parameters on this page are updated every 1 sec.

The values displayed in this webpage will depend on the wiring configuration mode of the meter. For example, if the meter is configured as '2LL' or '3LL' then the metering webpage will not display the phase readings, only the total values will be shown.



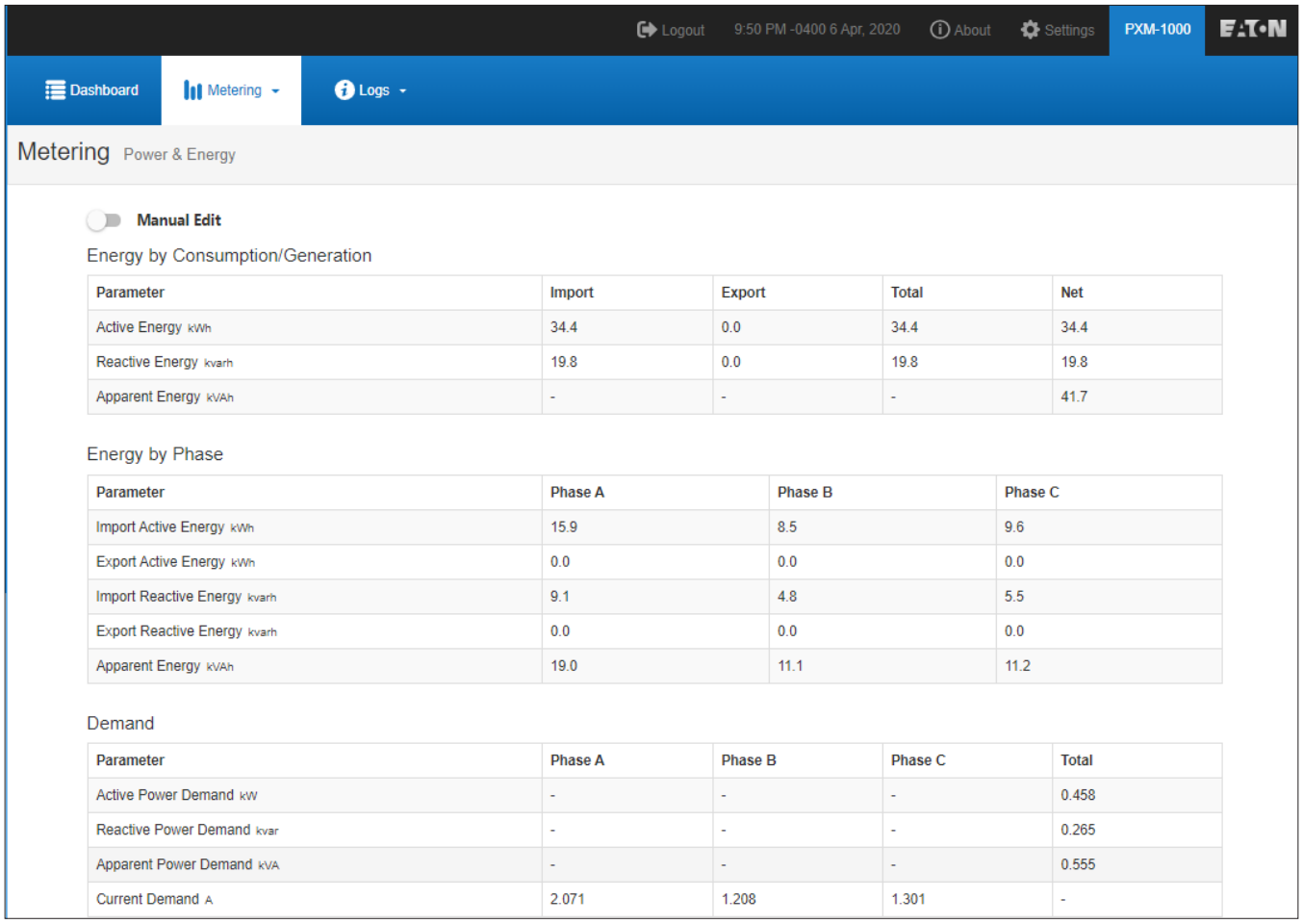
Parameter	Phase A	Phase B	Phase C	Average	Total
Line-to-Neutral Voltage V	120.314	121.227	122.374	121.305	-
Line-to-Line Voltage V	208.835	210.674	210.029	209.846	-
Current A	2.071	1.208	1.301	1.527	-
Neutral Current A	-	-	-	-	1.105
Active Power kW	0.208	0.113	0.137	-	0.458
Reactive Power kvar	0.120	0.065	0.079	-	0.265
Apparent Power kVA	0.249	0.146	0.159	-	0.555
Power Factor	0.835	0.771	0.863	-	0.826
Frequency Hz	60.020				
Load Type	L				

10.3.2 Power & Energy

The Power & Energy webpage shows the energy data for the PXM1000 series meter such as the Active and Reactive energy that is consumed and delivered as well as the Apparent energy per phase and total.

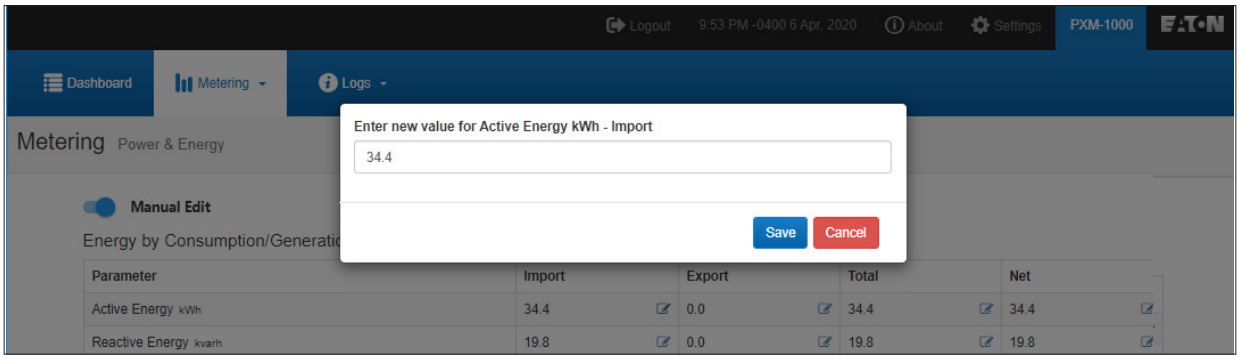
This webpage also shows the Demand parameters for the Active, Reactive and Apparent Power as well as the three phase Current demands.

The parameters in this webpage are updated every 5 sec.



From the Power & Energy page user have the option to edit and write new Energy values. Simply click on the 'Manual Edit' button to edit the energy values.

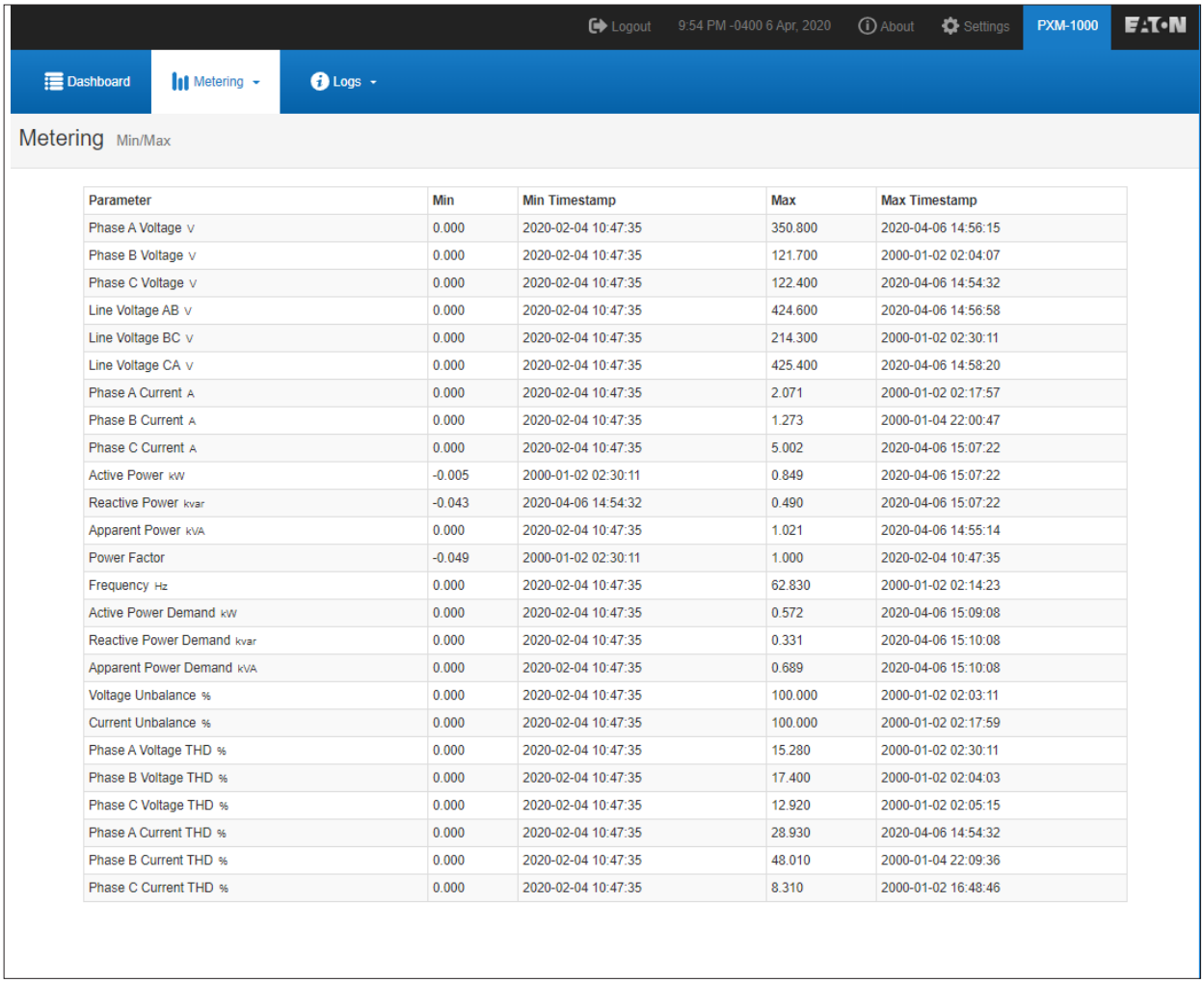
This will be useful for users who want to start and monitor energy accumulation at a certain kilowatt hour reading. All energy parameters including individual phase, total, import, export, and net for the real, reactive and apparent energies support this feature.



10.3.3 Min/Max

The Min/Max page shows the maximum and minimum statistics that the meter has records since the life time of the meter or from the last reset of the min/max statistics as well as the timestamps they were recorded at.

The parameters in this web page are updated every 10 sec.

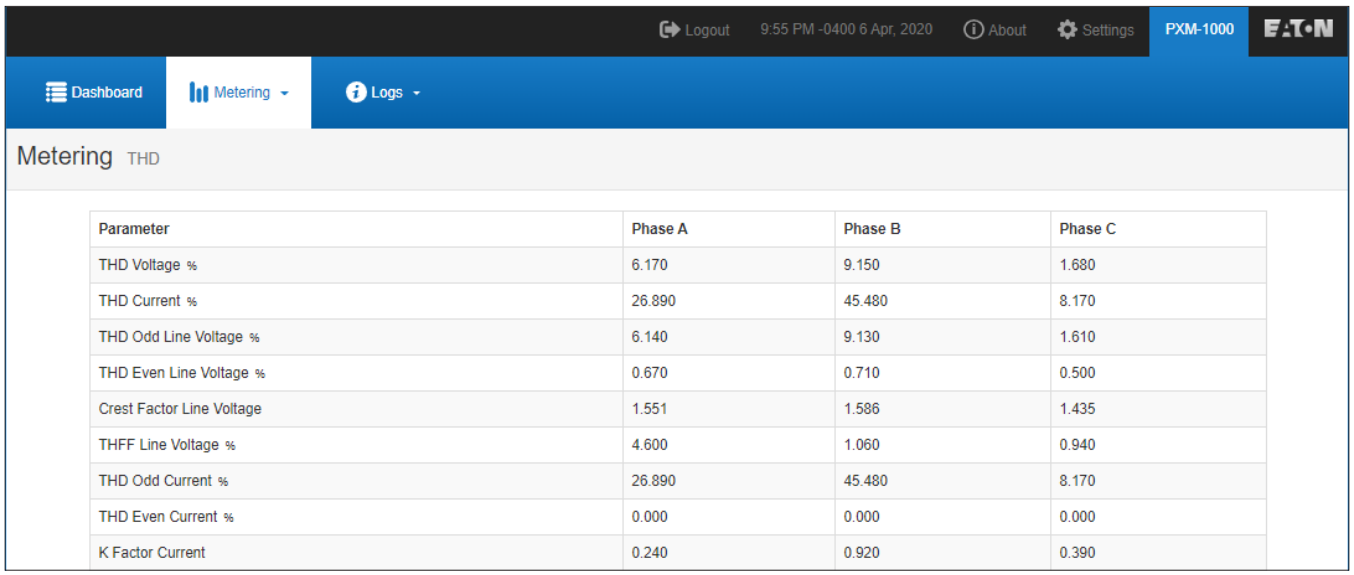


Parameter	Min	Min Timestamp	Max	Max Timestamp
Phase A Voltage V	0.000	2020-02-04 10:47:35	350.800	2020-04-06 14:56:15
Phase B Voltage V	0.000	2020-02-04 10:47:35	121.700	2000-01-02 02:04:07
Phase C Voltage V	0.000	2020-02-04 10:47:35	122.400	2020-04-06 14:54:32
Line Voltage AB V	0.000	2020-02-04 10:47:35	424.600	2020-04-06 14:56:58
Line Voltage BC V	0.000	2020-02-04 10:47:35	214.300	2000-01-02 02:30:11
Line Voltage CA V	0.000	2020-02-04 10:47:35	425.400	2020-04-06 14:58:20
Phase A Current A	0.000	2020-02-04 10:47:35	2.071	2000-01-02 02:17:57
Phase B Current A	0.000	2020-02-04 10:47:35	1.273	2000-01-04 22:00:47
Phase C Current A	0.000	2020-02-04 10:47:35	5.002	2020-04-06 15:07:22
Active Power kW	-0.005	2000-01-02 02:30:11	0.849	2020-04-06 15:07:22
Reactive Power kvar	-0.043	2020-04-06 14:54:32	0.490	2020-04-06 15:07:22
Apparent Power kVA	0.000	2020-02-04 10:47:35	1.021	2020-04-06 14:55:14
Power Factor	-0.049	2000-01-02 02:30:11	1.000	2020-02-04 10:47:35
Frequency Hz	0.000	2020-02-04 10:47:35	62.830	2000-01-02 02:14:23
Active Power Demand kW	0.000	2020-02-04 10:47:35	0.572	2020-04-06 15:09:08
Reactive Power Demand kvar	0.000	2020-02-04 10:47:35	0.331	2020-04-06 15:10:08
Apparent Power Demand kVA	0.000	2020-02-04 10:47:35	0.689	2020-04-06 15:10:08
Voltage Unbalance %	0.000	2020-02-04 10:47:35	100.000	2000-01-02 02:03:11
Current Unbalance %	0.000	2020-02-04 10:47:35	100.000	2000-01-02 02:17:59
Phase A Voltage THD %	0.000	2020-02-04 10:47:35	15.280	2000-01-02 02:30:11
Phase B Voltage THD %	0.000	2020-02-04 10:47:35	17.400	2000-01-02 02:04:03
Phase C Voltage THD %	0.000	2020-02-04 10:47:35	12.920	2000-01-02 02:05:15
Phase A Current THD %	0.000	2020-02-04 10:47:35	28.930	2020-04-06 14:54:32
Phase B Current THD %	0.000	2020-02-04 10:47:35	48.010	2000-01-04 22:09:36
Phase C Current THD %	0.000	2020-02-04 10:47:35	8.310	2000-01-02 16:48:46

10.3.4 THD

The THD web page shows the power quality data such as the THD, THFF, Crest and K Factor for both the voltage and current.

The parameters in this web page are updated every 15 sec.

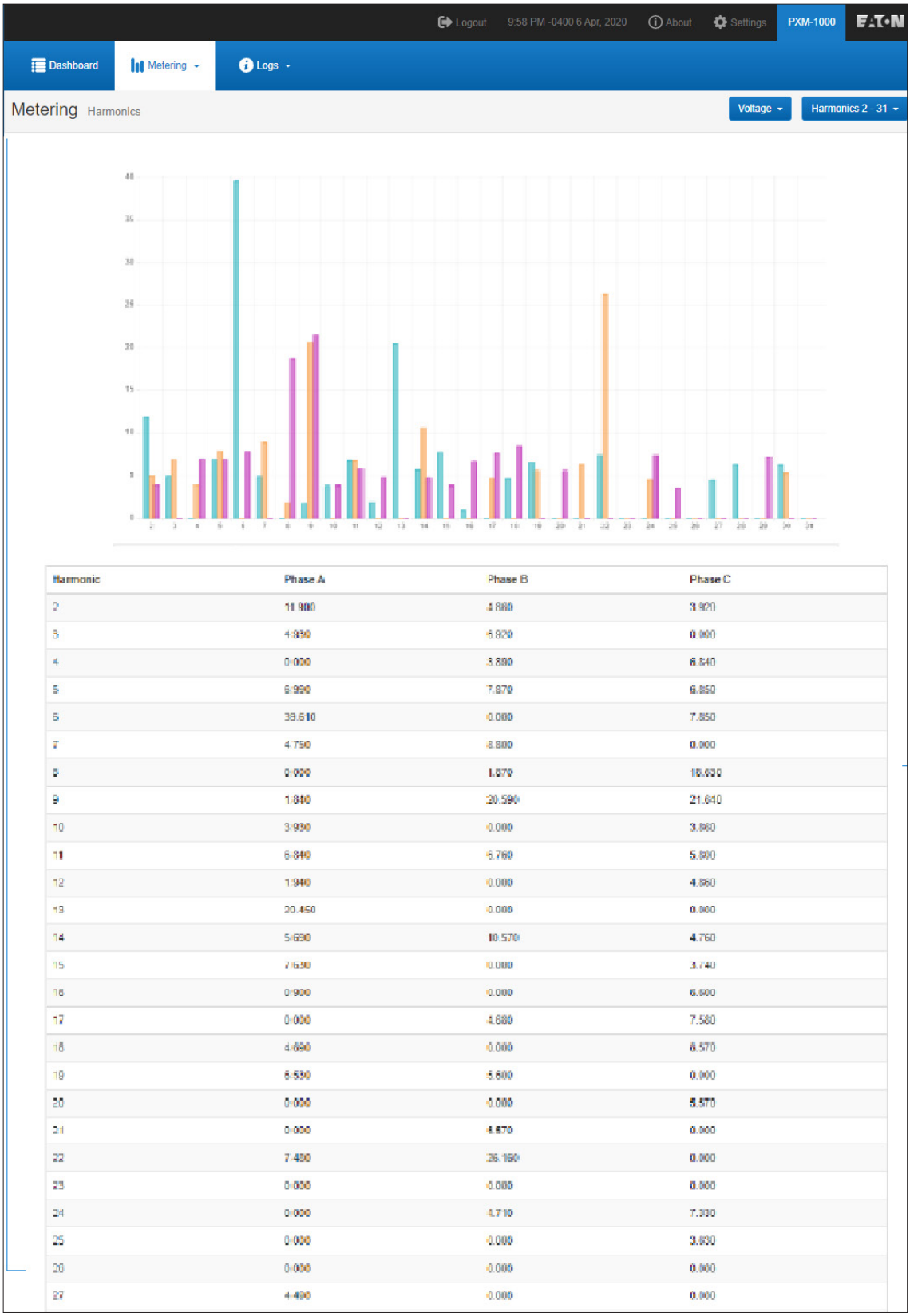


Parameter	Phase A	Phase B	Phase C
THD Voltage %	6.170	9.150	1.680
THD Current %	26.890	45.480	8.170
THD Odd Line Voltage %	6.140	9.130	1.610
THD Even Line Voltage %	0.670	0.710	0.500
Crest Factor Line Voltage	1.551	1.586	1.435
THFF Line Voltage %	4.600	1.060	0.940
THD Odd Current %	26.890	45.480	8.170
THD Even Current %	0.000	0.000	0.000
K Factor Current	0.240	0.920	0.390

10.3.5 Harmonics

The Harmonics web page will show the harmonics of the voltage and the current waveform being measured. It will display the harmonics of each phase in graphical and tabular format. Select between voltage and current to view their respective harmonics as well as between 2nd - 31st harmonics or 32nd - 63rd from the drop down list.

The parameters in this web page are updated every 15 sec.

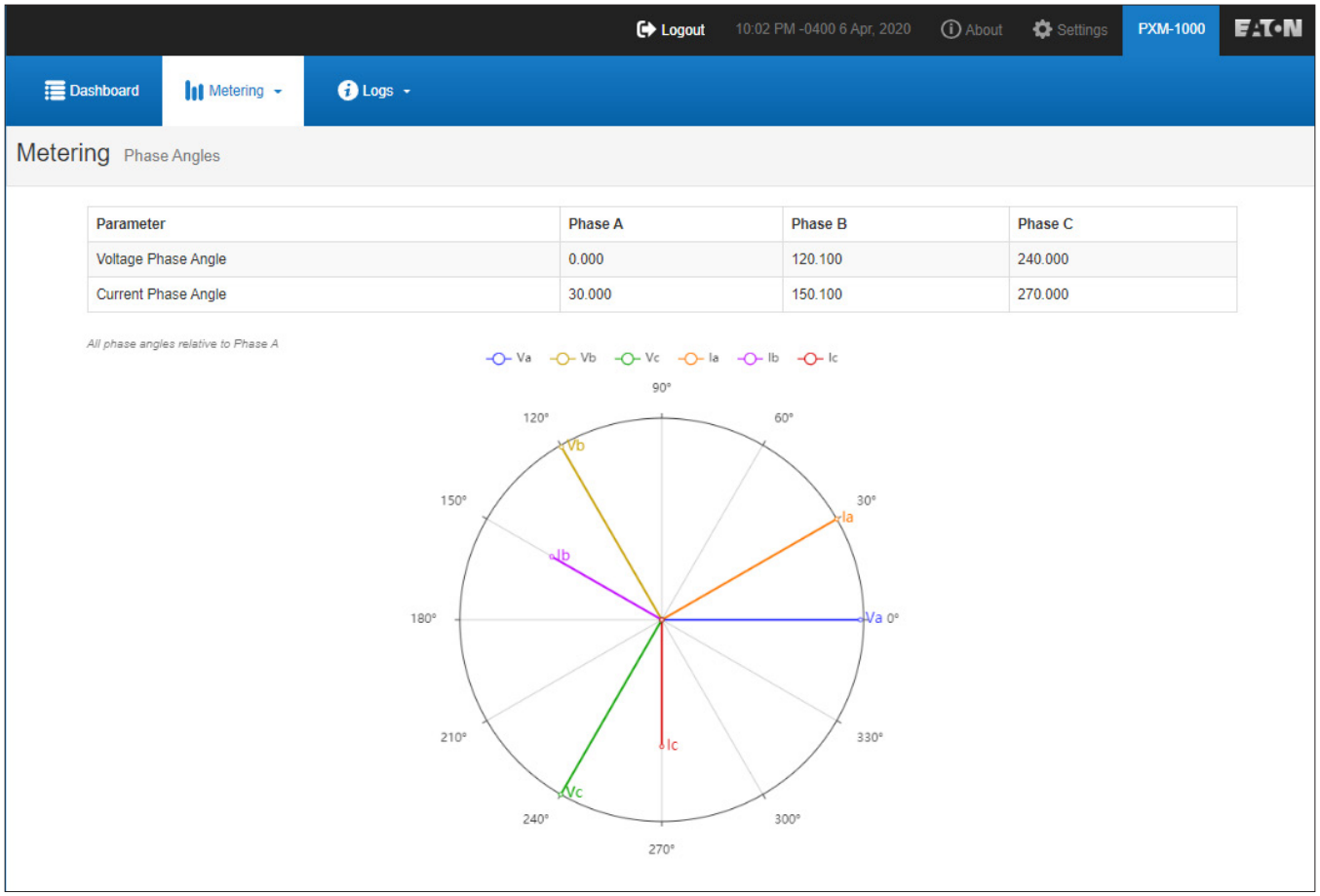


Harmonic	Phase A	Phase B	Phase C
2	11.900	4.860	3.920
3	4.830	6.820	0.000
4	0.000	3.800	6.840
5	9.990	7.870	6.850
6	35.610	0.000	7.850
7	4.750	8.800	0.000
8	0.000	1.870	10.000
9	1.840	20.590	21.640
10	3.930	0.000	3.860
11	6.840	6.760	5.800
12	1.940	0.000	4.860
13	20.450	0.000	0.000
14	5.690	10.570	4.760
15	7.630	0.000	3.740
16	0.900	0.000	6.600
17	0.000	4.680	7.580
18	4.690	0.000	6.570
19	6.530	5.800	0.000
20	0.000	0.000	5.570
21	0.000	4.570	0.000
22	7.480	26.160	0.000
23	0.000	0.000	0.000
24	0.000	4.710	7.330
25	0.000	0.000	3.630
26	0.000	0.000	0.000
27	4.490	0.000	0.000

10.3.6 Phase Angles

The Phase Angles web page will show the phase angles of the voltage and current waveform being measured. This page provides a visual diagram of the phase angles with respect to the voltage connected to the Phase A voltage input.

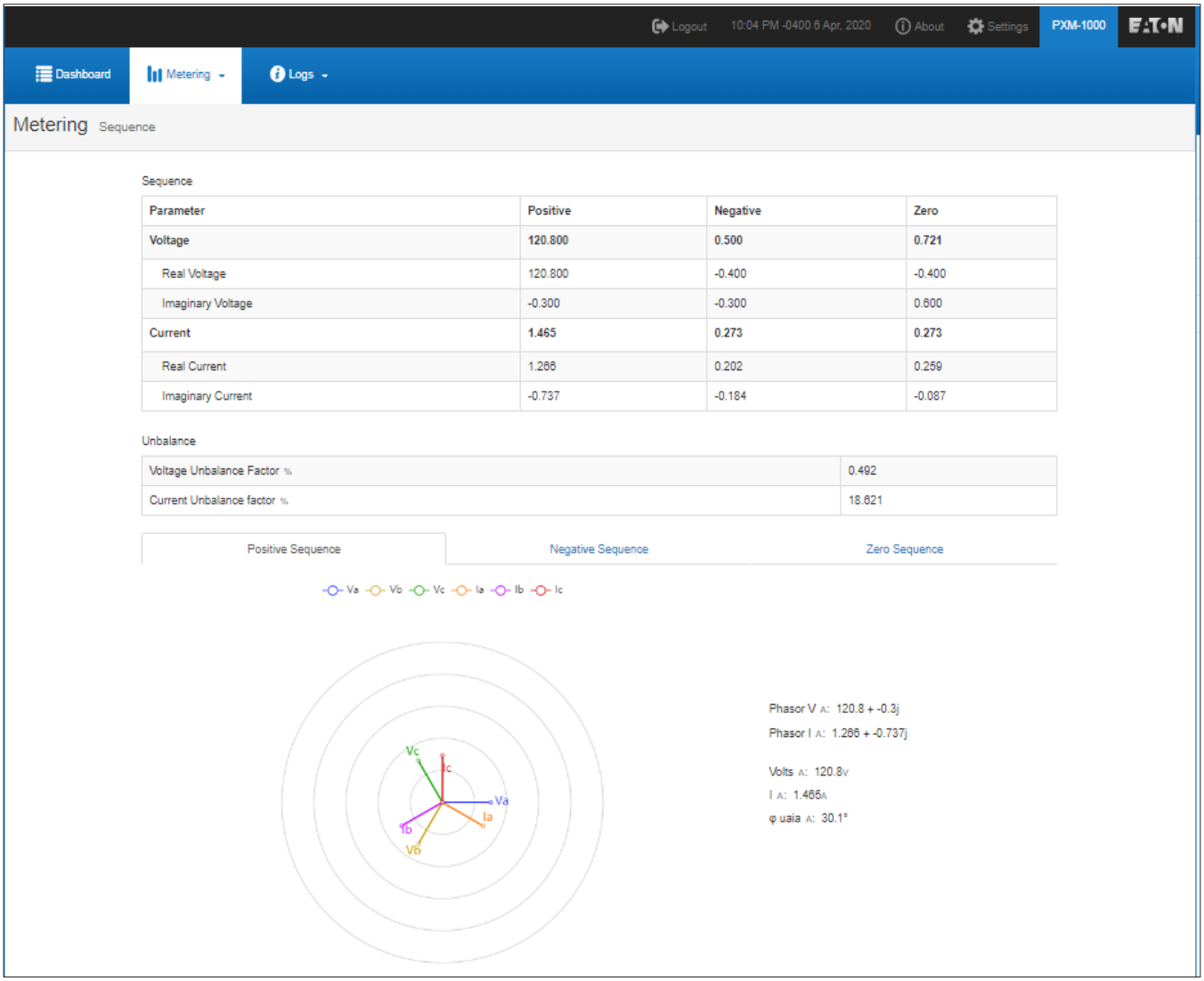
The parameters in this web page are updated every 10 sec.



10.3.7 Sequence

The Sequence web page will show the positive, negative and zero components of the voltage and current waveform being measured.

The parameters in this web page are updated every 10 sec.



10.3.8 I/O

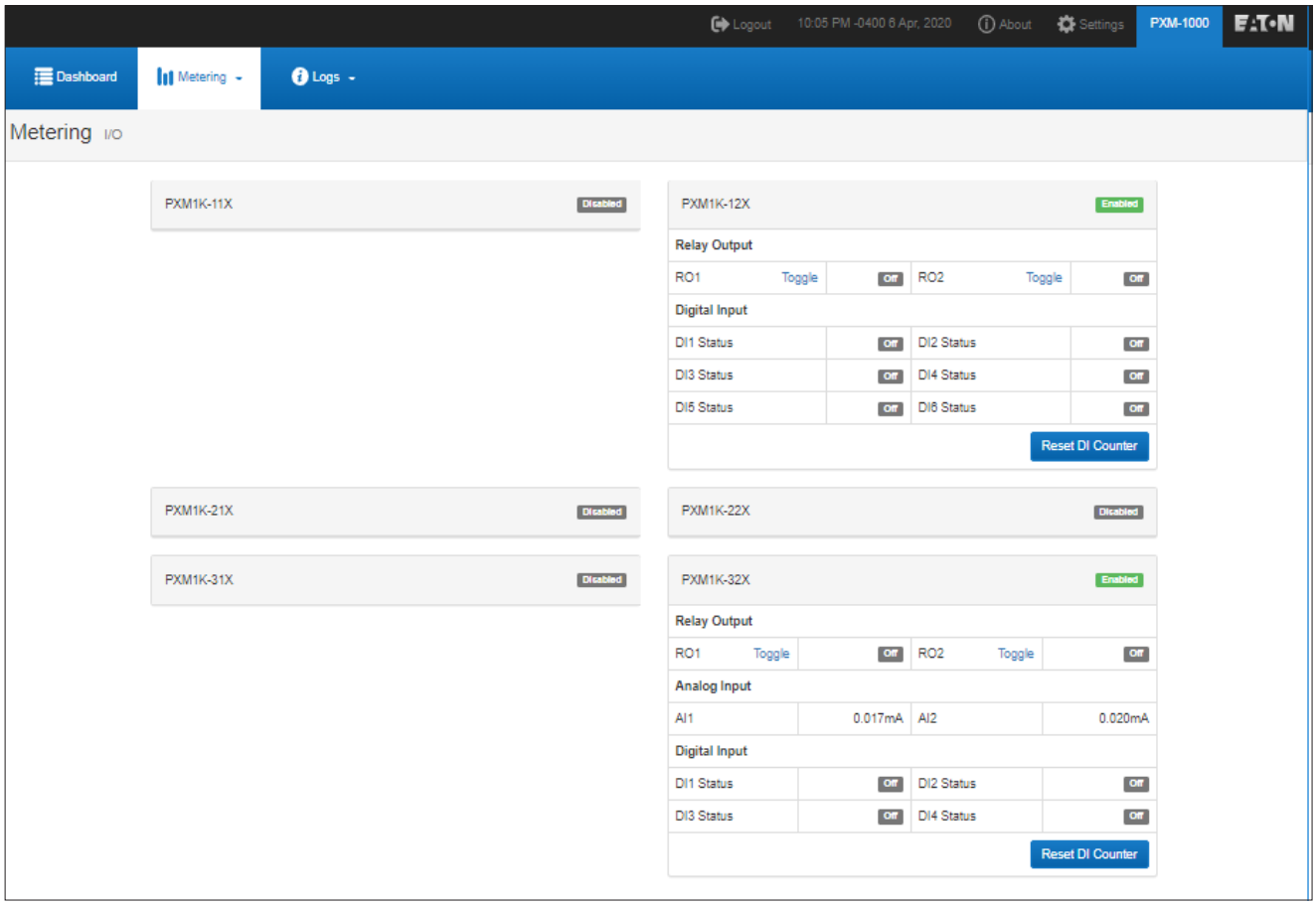
The I/O web page displays the status of the I/O modules that are connected and their values depending on the model of the module that is connected to the meter. I.E. The PXM1K-1 module will display the Relay Output status(on/off), DI status/counter. The I/O module can be configured in the general settings section of the web interface which is discussed later in the manual.

The parameters in this web page are updated every 5 sec.

10. Web Interface Readings and Parameter Settings

From the IO readings page, users can perform the following functions depending on the IO module being used:

IO Module	Functions
PXM1K-1x	• Read Digital Input Status or Digital Input Count value depending on the mode configured (Channels 1-6) • Read the Relay Output Channel 1 and 2 Status ON/OFF • Toggle Relay Output Channels 1 and 2 ON/OFF **Note:** Relay must be configured for Control Mode for toggle function to work • Reset DI Counter back to 0 (Counters 1-6)
PXM1K-2x	• Read Digital Input Status or Digital Input Count value depending on the mode configured (Channels 1-4) • Read value of the Analog Output Channels 1 and 2, either 4-20mA, 0-20mA, 0-5V, 1-5V signals • Reset DI Counter back to 0 (counters (1-4)
PXM1K-3x	• Read Digital Input Status or Digital Input Count value depending on the mode configured (Channels 1-4) • Read the Relay Output Channel 1 and 2 Status ON/OFF • Toggle Relay Output Channels 1 and 2 ON/OFF **Note:** Relay must be configured for Control Mode for toggle function to work • Read Analog Input Channels 1 and 2, either 4-20mA, 0-20mA, 0-5V, 1-5V signals • Reset DI Counter back to 0 (Counters 1-4)

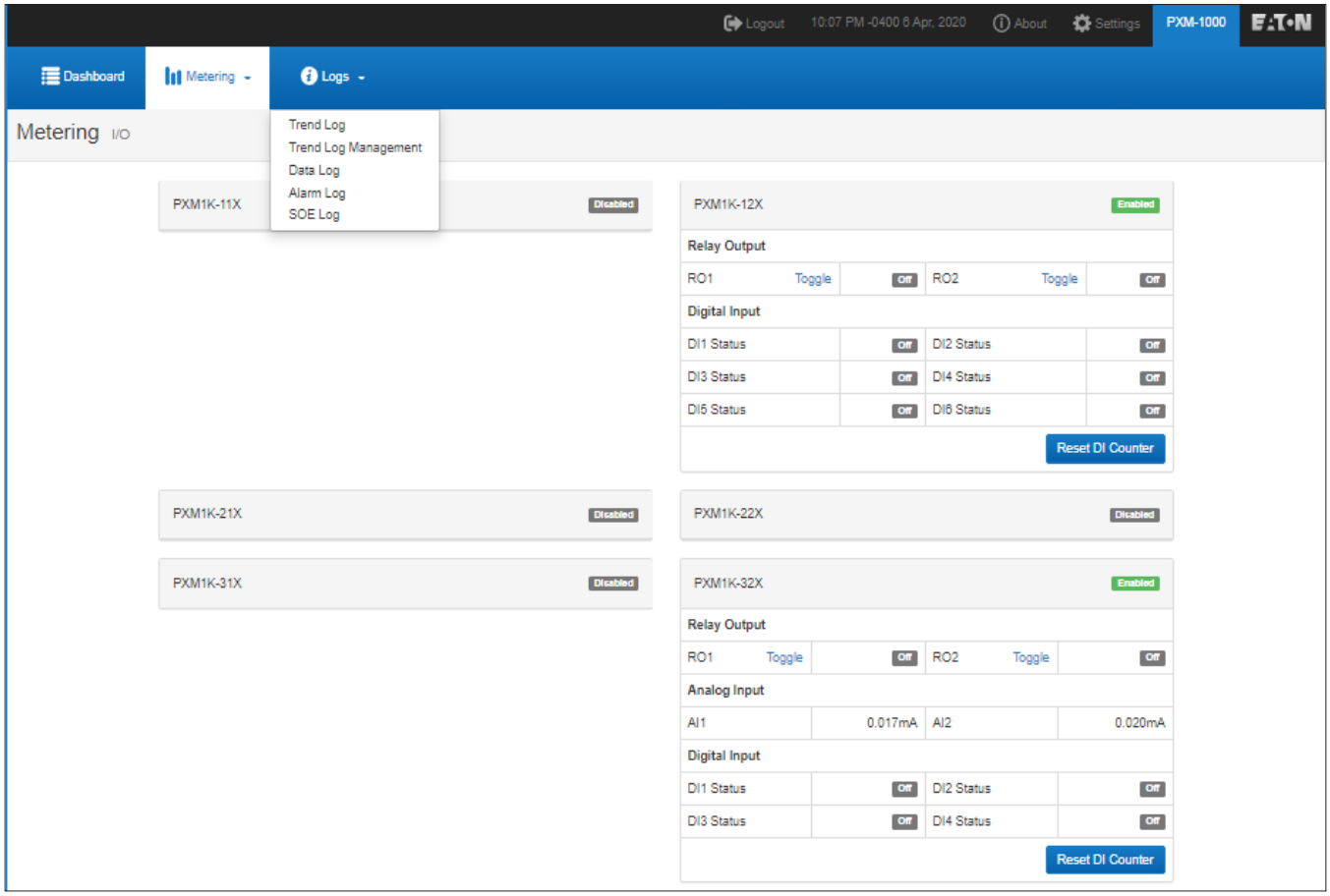


10.4 Logs

Click on the 'Logs' tab to visit the metering logs web pages.

There are five kinds of logs that can be viewed, they are "Trend Log", "Data Log", "Alarm Log", "SOE Log" and "Waveform Log"(Only available in PXM1300 model).

Each web page shows data from the PXM1000 series meter.



10.4.1 Trendlog

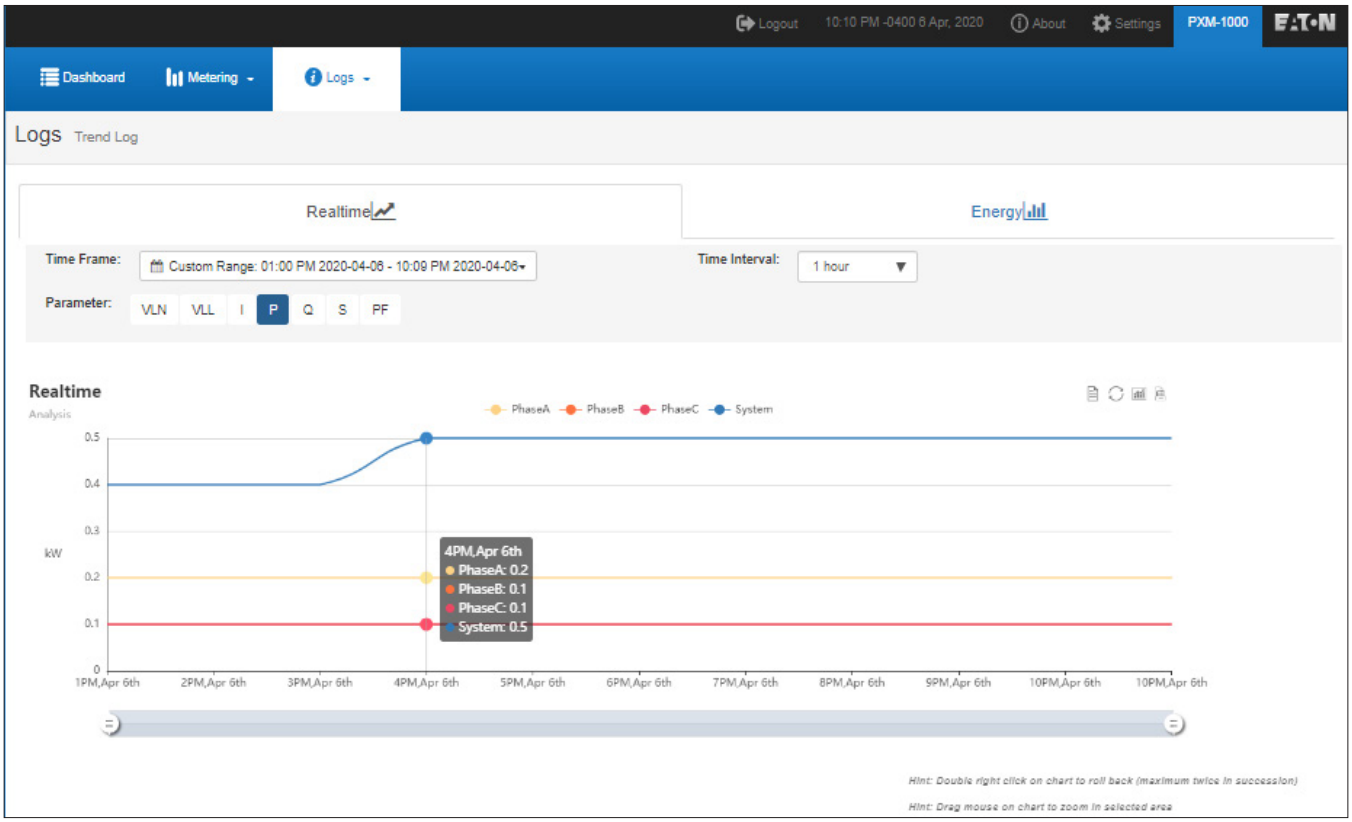
The Trend Log web page includes the real-time and energy trend diagram. The real-time trend log diagram can be selected to show the following parameters phase voltage, line voltage,current, active power, reactive power, apparent power and power factor for each phase as well as the totals. The energy trend log shows the imported and exported active energy, reactive energy, total energy, net energy and apparent energy.

The data of the trend log can be previewed and downloaded as a .csv file by clicking the **'Data Review'** and **'Data'** icons on the right top side of the diagram. The trend log diagram can also be saved as an image by clicking the **'Image'** icon.

Realtime

The real time parameters can be trended at different time intervals depending on the Time Frame selected. Listed below are the time intervals for each possible time frame setting:

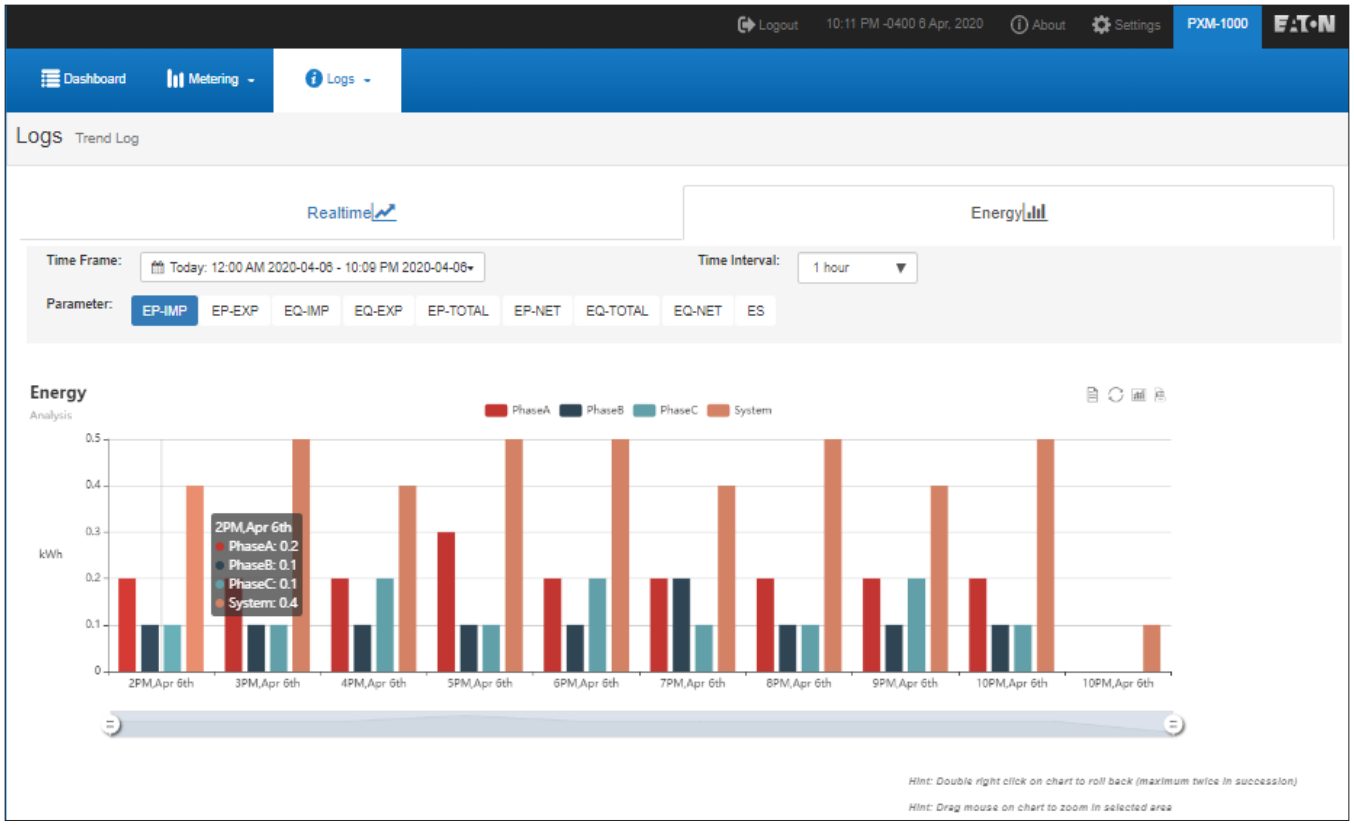
Time Frame	Time Intervals
Last 10 minutes	1 second 15 seconds 1 minute
Last 1 Hour	1 minute
Today	15 minutes 1 hour
Yesterday	15 minutes 1 hour
Last 7 days	15 minutes 1 hour 1 day
Last 30 days	1 hour 1 day
Last Month	1 hour 1 day
Custom Range	Dependent on range specified



Energy

Similarly the energy parameters can be trended at different time intervals depending on the Time Frame selected. The table below displays the time intervals:

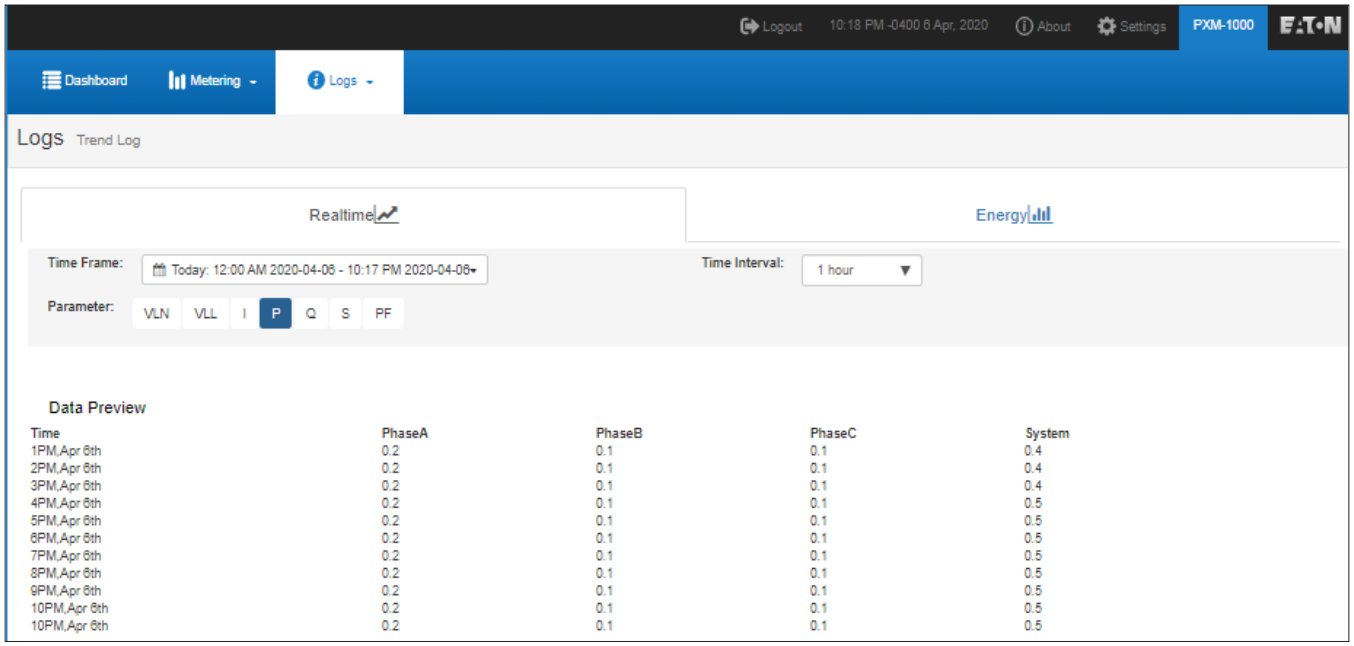
Time Frame	Time Intervals
Last 10 minutes	15 seconds 1 minute
Last 1 Hour	1 minute
Today	15 minutes 1 hour
Yesterday	15 minutes 1 hour
Last 7 days	15 minutes 1 hour 1 day
Last 30 days	1 hour 1 day
This Month	1 hour 1 day
Last Month	1 hour 1 day
Last Year	1 day 1 month
Custom Range	Dependent on range specified



Data Preview

The data preview allows the user to view the graphical data in tabular form.

User can also download this data into a csv file for further examination.



Time	PhaseA	PhaseB	PhaseC	System
1PM,Apr 6th	0.2	0.1	0.1	0.4
2PM,Apr 6th	0.2	0.1	0.1	0.4
3PM,Apr 6th	0.2	0.1	0.1	0.4
4PM,Apr 6th	0.2	0.1	0.1	0.5
5PM,Apr 6th	0.2	0.1	0.1	0.5
6PM,Apr 6th	0.2	0.1	0.1	0.5
7PM,Apr 6th	0.2	0.1	0.1	0.5
8PM,Apr 6th	0.2	0.1	0.1	0.5
9PM,Apr 6th	0.2	0.1	0.1	0.5
10PM,Apr 6th	0.2	0.1	0.1	0.5
10PM,Apr 6th	0.2	0.1	0.1	0.5

10.4.2 Trendlog Management

The trendlog management page allows the user to download data from the meters data base. The trendlog management page acts as a back up to the data logs for users.

Log Param Type: Users can select which data they want to download from the meter. In the drop down menu there is a timestamp range to show the available data.

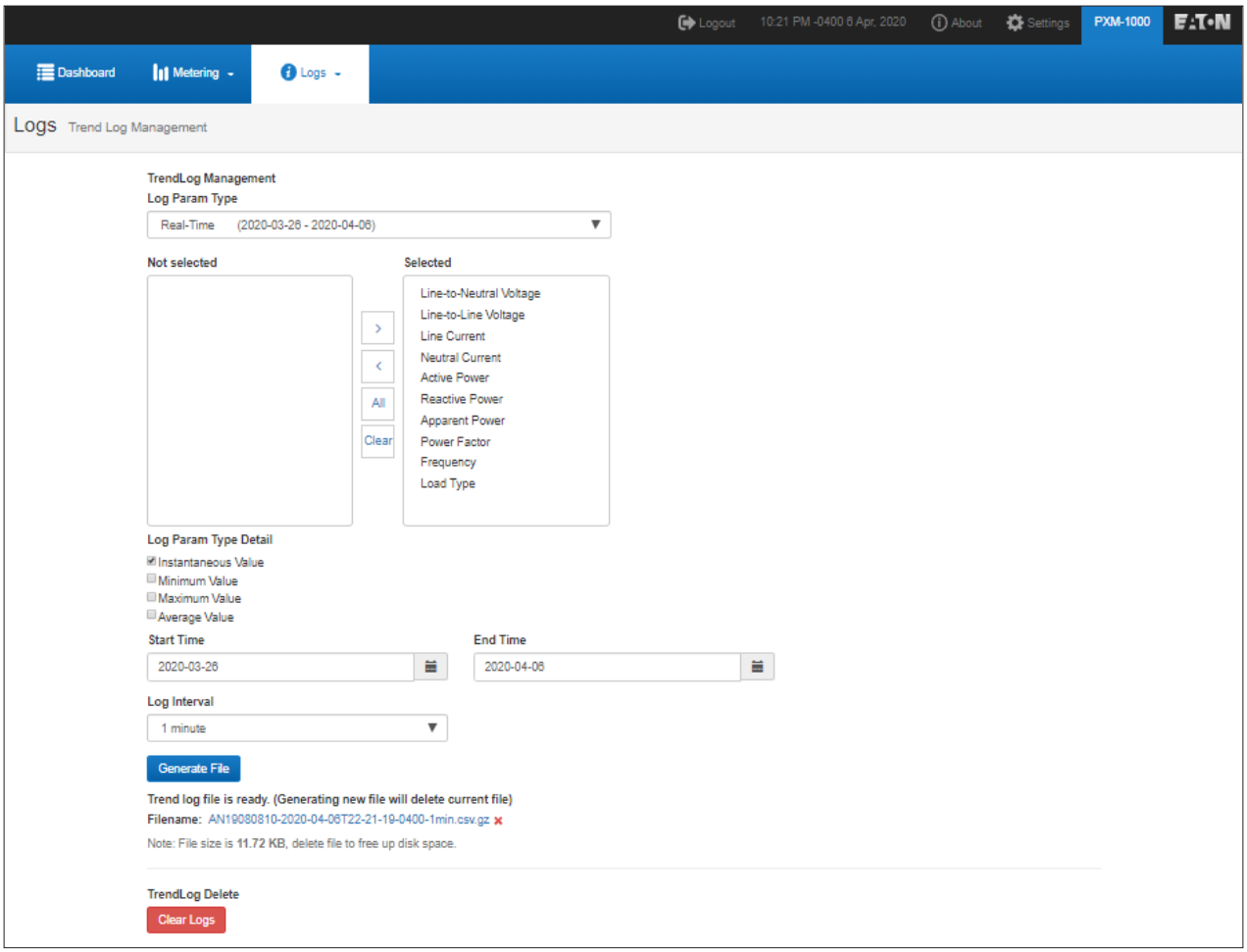
Note: Energy data will remain in meter data base for up to 3 years, whereas all other metering data will remain in the meter data base for up to 1 month before overriding.

Log Param Type Detail: This setting allows users to modify what values they see in the data log. Users can select the following parameter details:

- Instantaneous
- Minimum
- Maximum
- Average

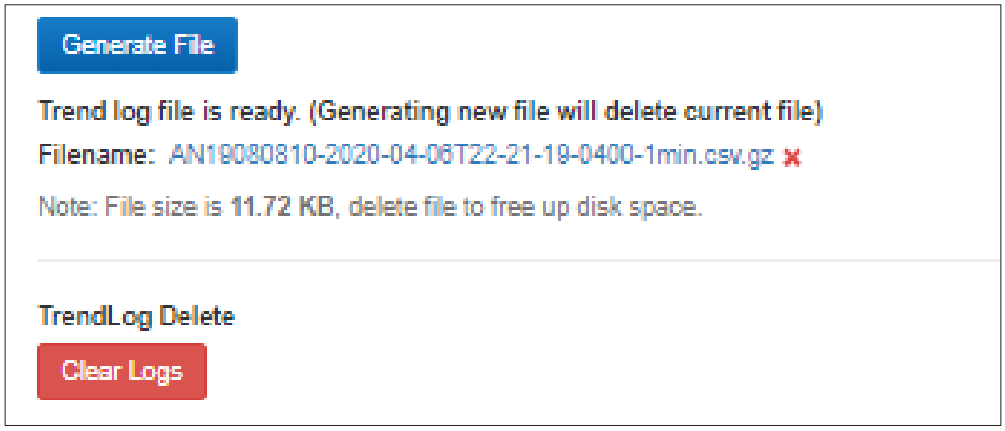
Only the 'Real-time' and 'Demand' parameters support the minimum, maximum and average parameter type details. All other parameter types such as Energy, Power Quality, and IO only support the instantaneous values displayed in the the data log.

The log file will be downloaded as a .gz file and will need to be unzipped in order to view the csv file.



Clear Logs

The clear logs function allows the user to clear and remove all metering data stored on the module database. This will allow users to clear all readings and historical data without resetting all features and functions. Users can clear the logs by clicking on the button at the bottom of the Trendlog page.



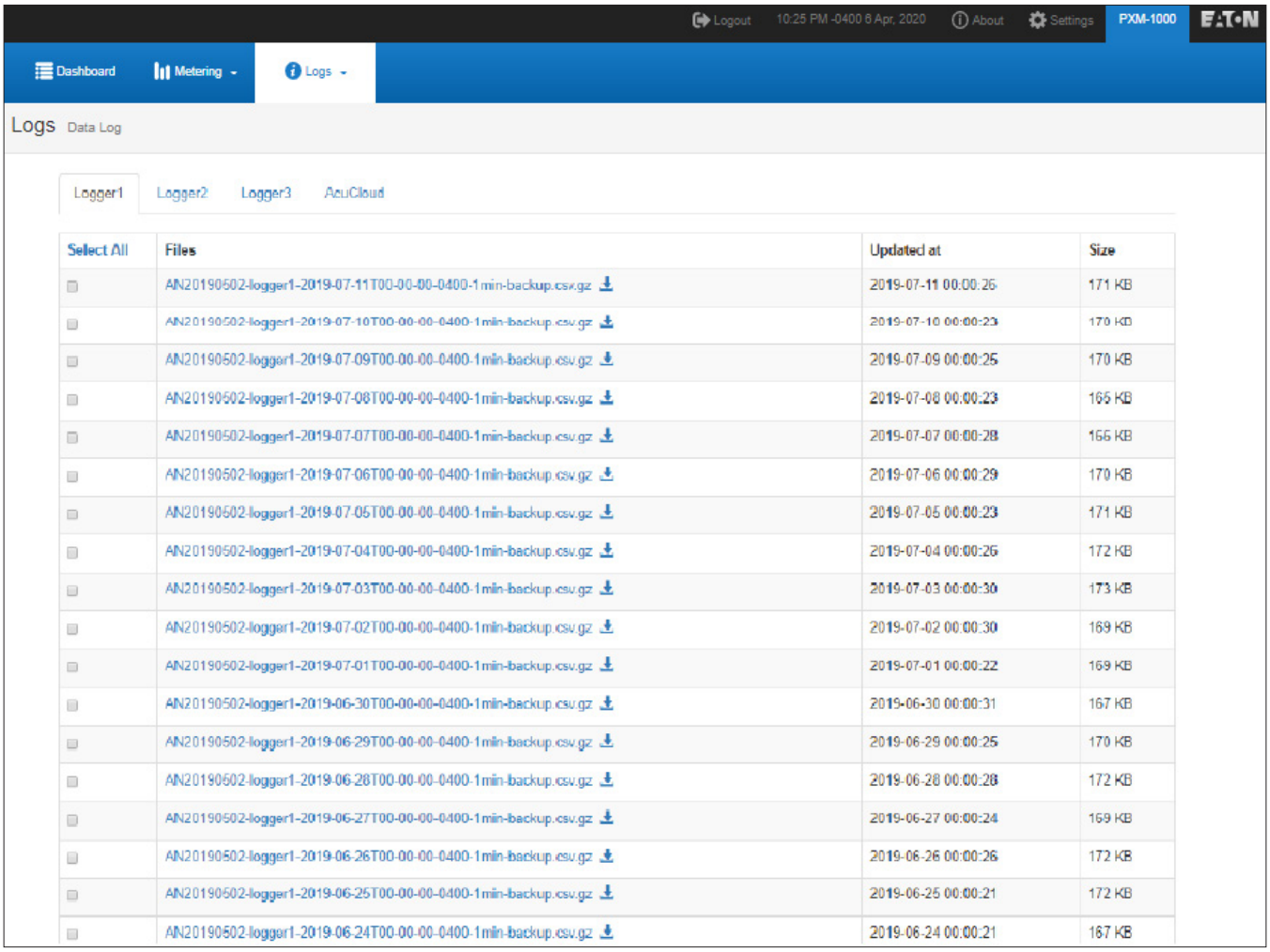
To clear the logs click on "Clear Logs", a warning message is displayed notifying users that this action is irreversible once done.

Note: This cannot be undone, once the trend log is cleared all data in meter database is cleared.

10.4.3 Data Log

The data log web page includes all the data file for three different loggers and PX-EVA.

You can select the different loggers by clicking the logger tab. After the logger is selected, the log file for this logger will show on the screen with the update time and file size. To download the file, click on the download icon to save the file in the computer. The data log will be saved as a compressed csv file.

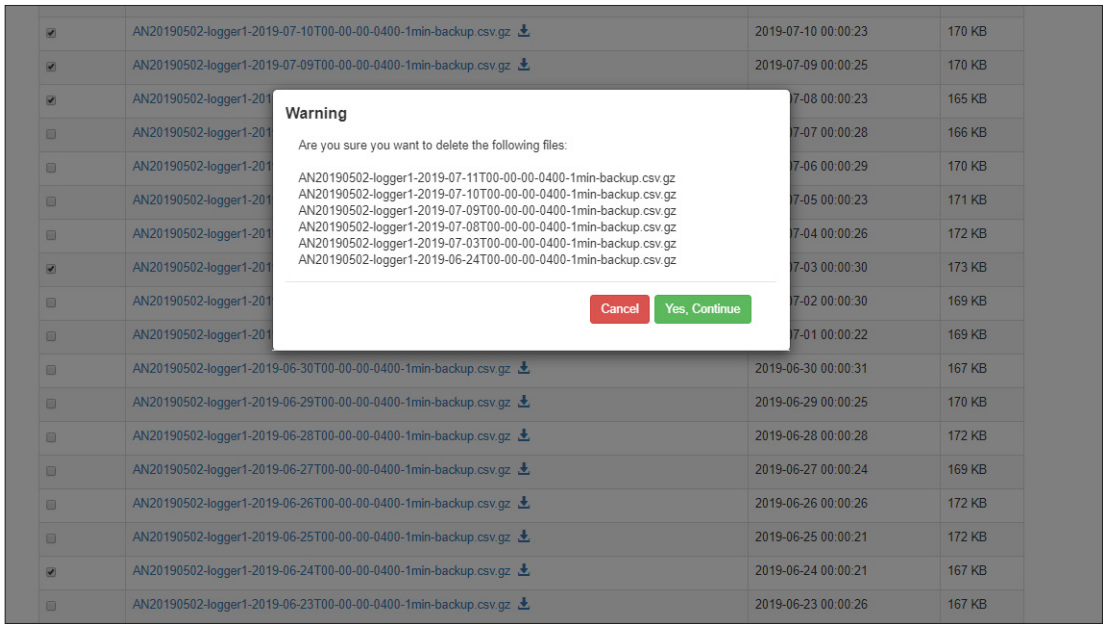


Deleting Data Logs

To delete the data logs users can check the box next to the data log file and click on the 'Delete Selected' button at the bottom of the page.

Users will be prompted by a window asking to confirm the data log delete.

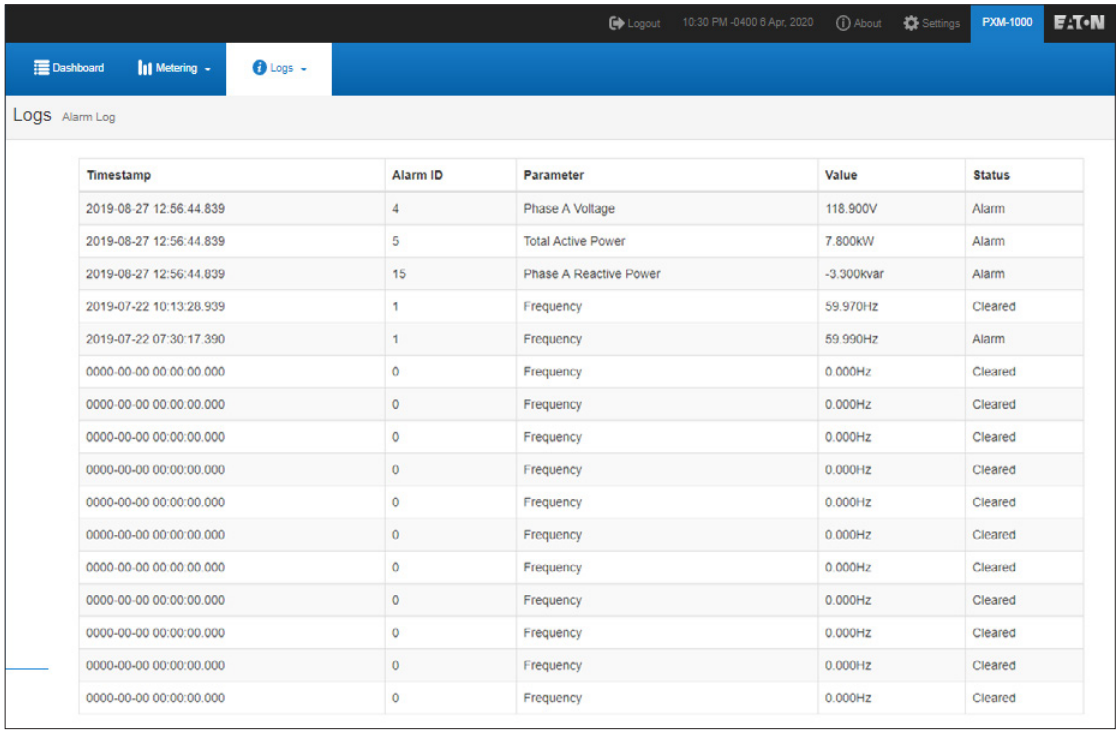
Note: Deleting the data log is permanent, this cannot be undone once deleted.



10.4.4 Alarm Log

The Alarm Log web page provides the user with a summary of the alarm events that have occurred with the meter. It will show the status of up to 16 alarm events indicating the alarm ID, status, parameter, value that exceeded or went below the threshold and the time-stamp of the alarm event.

Once all 16 alarm events are full, the newest alarm event will then wrap around to alarm 1. The parameters in the alarm status web page are updated every 10 seconds.

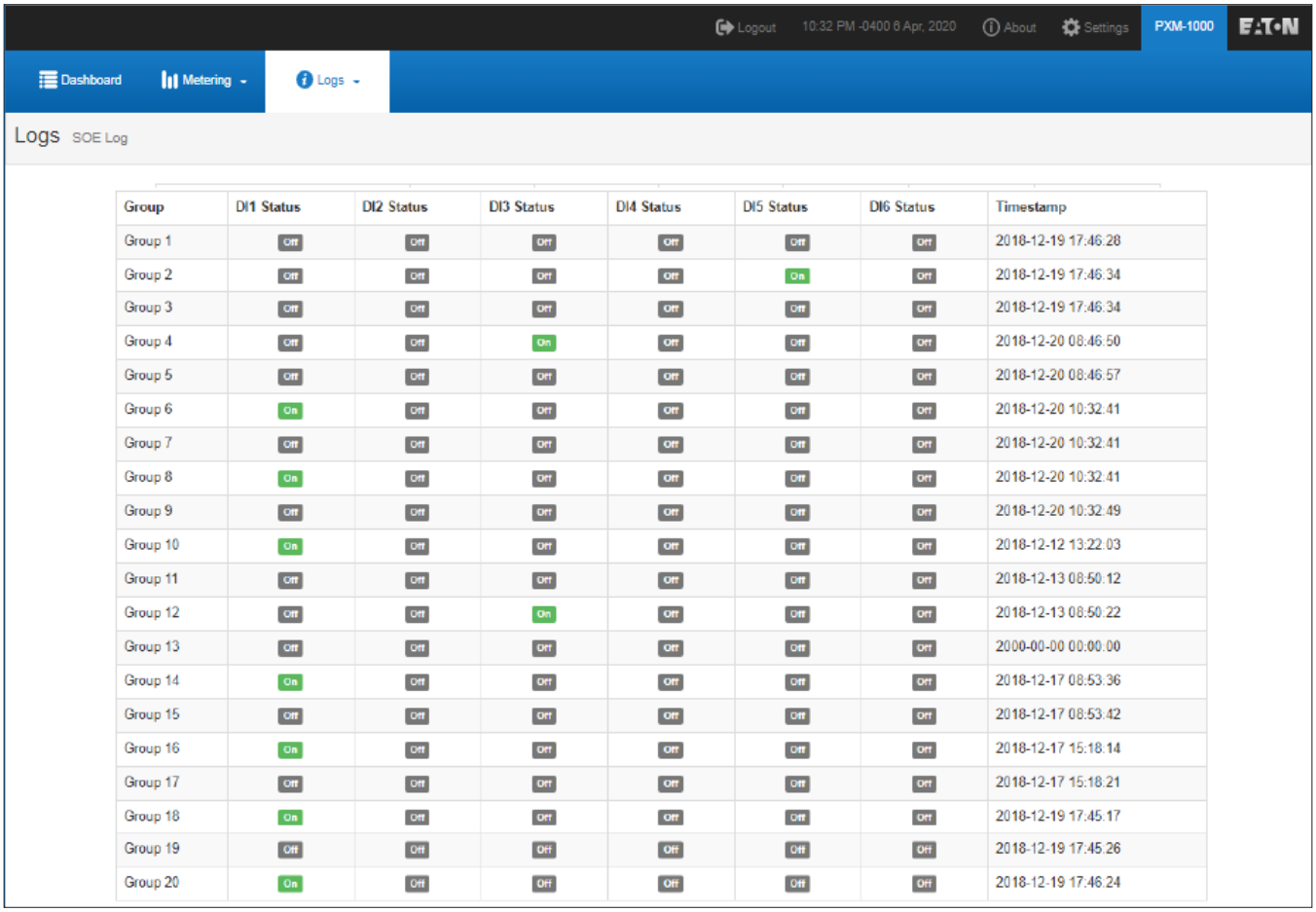


Timestamp	Alarm ID	Parameter	Value	Status
2019-08-27 12:56:44.839	4	Phase A Voltage	118.900V	Alarm
2019-08-27 12:56:44.839	5	Total Active Power	7.800kW	Alarm
2019-08-27 12:56:44.839	15	Phase A Reactive Power	-3.300kvar	Alarm
2019-07-22 10:13:28.939	1	Frequency	59.970Hz	Cleared
2019-07-22 07:30:17.390	1	Frequency	59.990Hz	Alarm
0000-00-00 00:00:00.000	0	Frequency	0.000Hz	Cleared
0000-00-00 00:00:00.000	0	Frequency	0.000Hz	Cleared
0000-00-00 00:00:00.000	0	Frequency	0.000Hz	Cleared
0000-00-00 00:00:00.000	0	Frequency	0.000Hz	Cleared
0000-00-00 00:00:00.000	0	Frequency	0.000Hz	Cleared
0000-00-00 00:00:00.000	0	Frequency	0.000Hz	Cleared
0000-00-00 00:00:00.000	0	Frequency	0.000Hz	Cleared
0000-00-00 00:00:00.000	0	Frequency	0.000Hz	Cleared
0000-00-00 00:00:00.000	0	Frequency	0.000Hz	Cleared
0000-00-00 00:00:00.000	0	Frequency	0.000Hz	Cleared
0000-00-00 00:00:00.000	0	Frequency	0.000Hz	Cleared
0000-00-00 00:00:00.000	0	Frequency	0.000Hz	Cleared

10.4.5 SOE (input log) Log

The SOE web page will display the Sequence of Event log for the enabled I/O module that is attached to the PXM1000 series meter with timestamps and will display the DI status for up to 20 SOE events. The SOE must be enabled from the PXMview software.

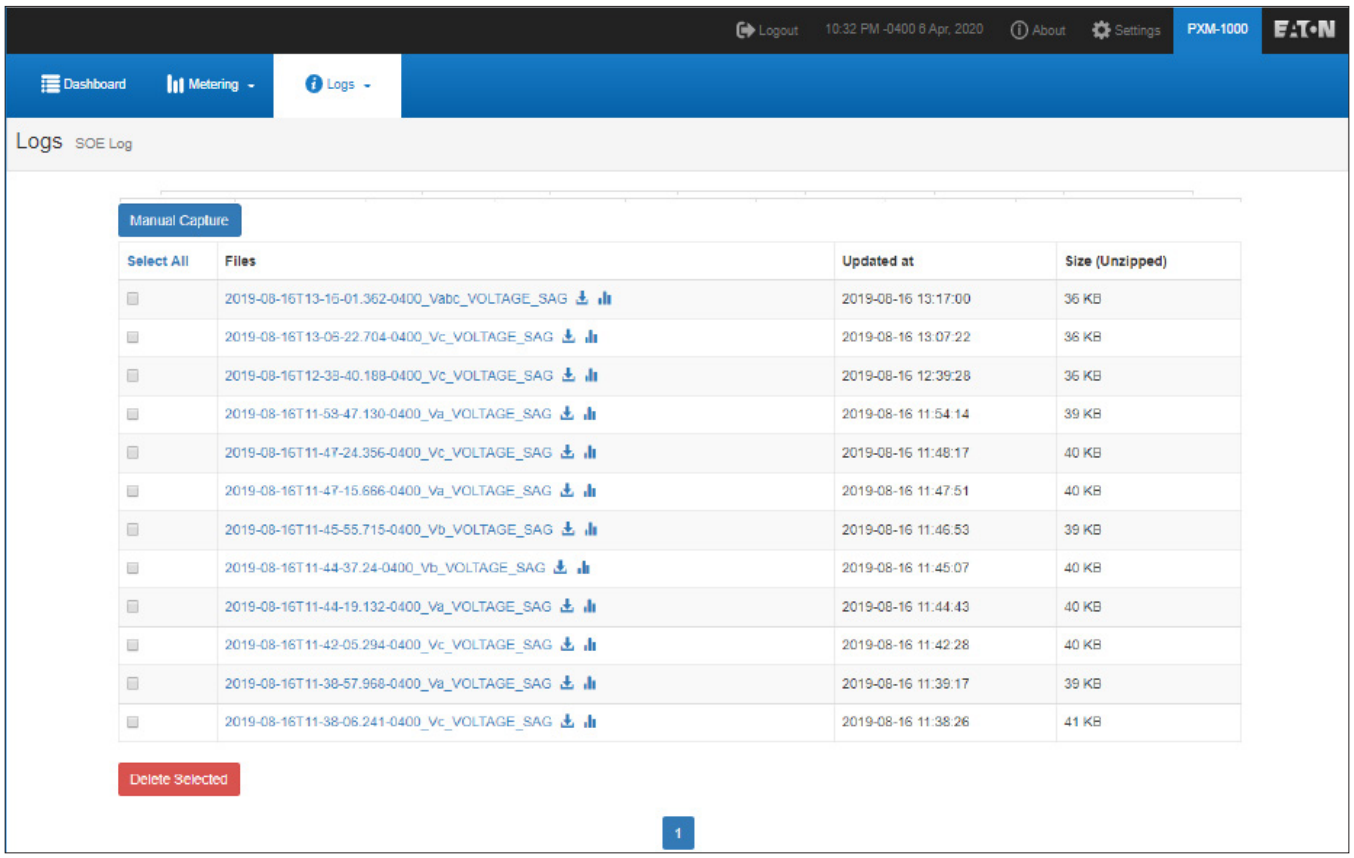
The SOE log parameters are updated every 10 sec.



Group	DI1 Status	DI2 Status	DI3 Status	DI4 Status	DI5 Status	DI6 Status	Timestamp
Group 1	Off	Off	Off	Off	Off	Off	2018-12-19 17:46:28
Group 2	Off	Off	Off	Off	On	Off	2018-12-19 17:46:34
Group 3	Off	Off	Off	Off	Off	Off	2018-12-19 17:46:34
Group 4	Off	Off	On	Off	Off	Off	2018-12-20 08:46:50
Group 5	Off	Off	Off	Off	Off	Off	2018-12-20 08:46:57
Group 6	On	Off	Off	Off	Off	Off	2018-12-20 10:32:41
Group 7	Off	Off	Off	Off	Off	Off	2018-12-20 10:32:41
Group 8	On	Off	Off	Off	Off	Off	2018-12-20 10:32:41
Group 9	Off	Off	Off	Off	Off	Off	2018-12-20 10:32:49
Group 10	On	Off	Off	Off	Off	Off	2018-12-12 13:22:03
Group 11	Off	Off	Off	Off	Off	Off	2018-12-13 08:50:12
Group 12	Off	Off	On	Off	Off	Off	2018-12-13 08:50:22
Group 13	Off	Off	Off	Off	Off	Off	2000-00-00 00:00:00
Group 14	On	Off	Off	Off	Off	Off	2018-12-17 08:53:36
Group 15	Off	Off	Off	Off	Off	Off	2018-12-17 08:53:42
Group 16	On	Off	Off	Off	Off	Off	2018-12-17 15:18:14
Group 17	Off	Off	Off	Off	Off	Off	2018-12-17 15:18:21
Group 18	On	Off	Off	Off	Off	Off	2018-12-19 17:45:17
Group 19	Off	Off	Off	Off	Off	Off	2018-12-19 17:45:26
Group 20	On	Off	Off	Off	Off	Off	2018-12-19 17:46:24

10.4.6 Waveform Log

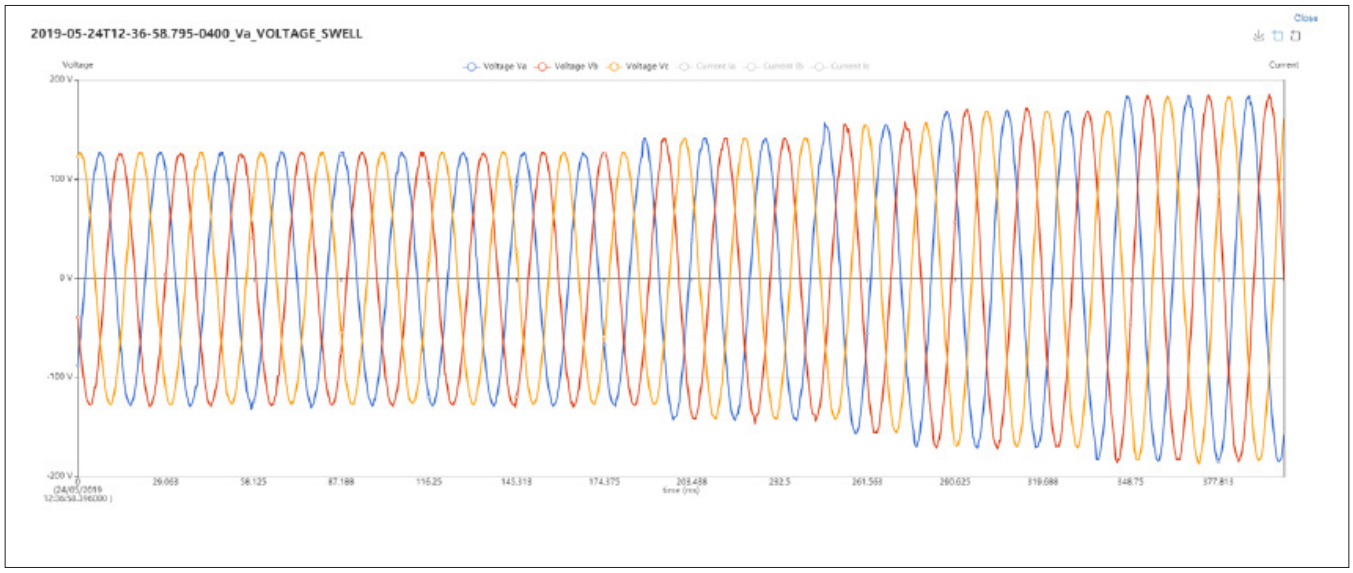
The waveform log is available only on the PXM1300 model of the PXM1000 series meter. This meter supports a waveform capture function that allows users to capture and record 10 cycles before and after the triggering point whether it be a voltage sag, swell, or over current. The waveform log on the web interface allows users to view these waveforms whenever a power quality event has occurred. The log displays is a table that includes the waveform files, the time the waveform is updated at, and the size of the file. The waveform file name includes the timestamp when the event occurred as well as the parameter name/ event name that triggered the power quality event.



To view the waveform click on the icon. Users can select which voltage or current phase they want to view on the graph during the power quality event.

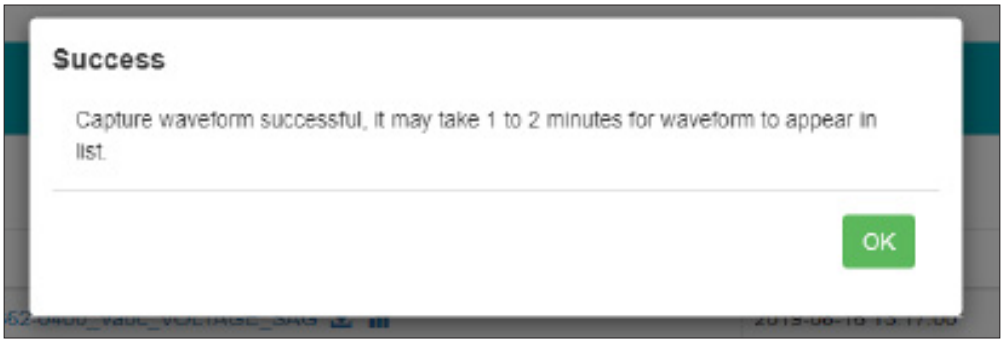
An image of the graph can be downloaded by clicking on the button.

Users can zoom into the graph for further detail by clicking on the . The graph can be reset back by double clicking the button.



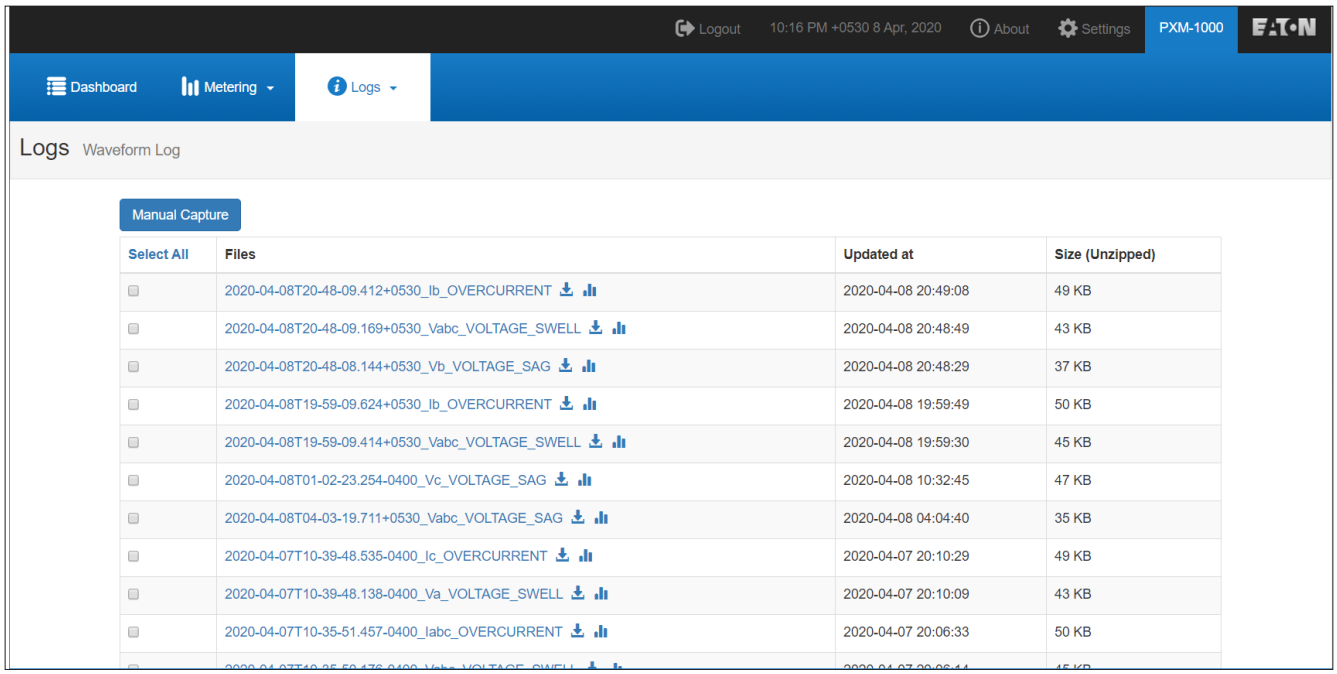
Users can select 'Close' to close the waveform and navigate back to the waveform Log.

Users can perform a manual waveform capture by clicking on 'Manual Capture'. This will manually capture voltage and current waveforms of the system being monitored. Once the waveform is capture it will take approximately 1-2 minutes to appear in the waveform log.

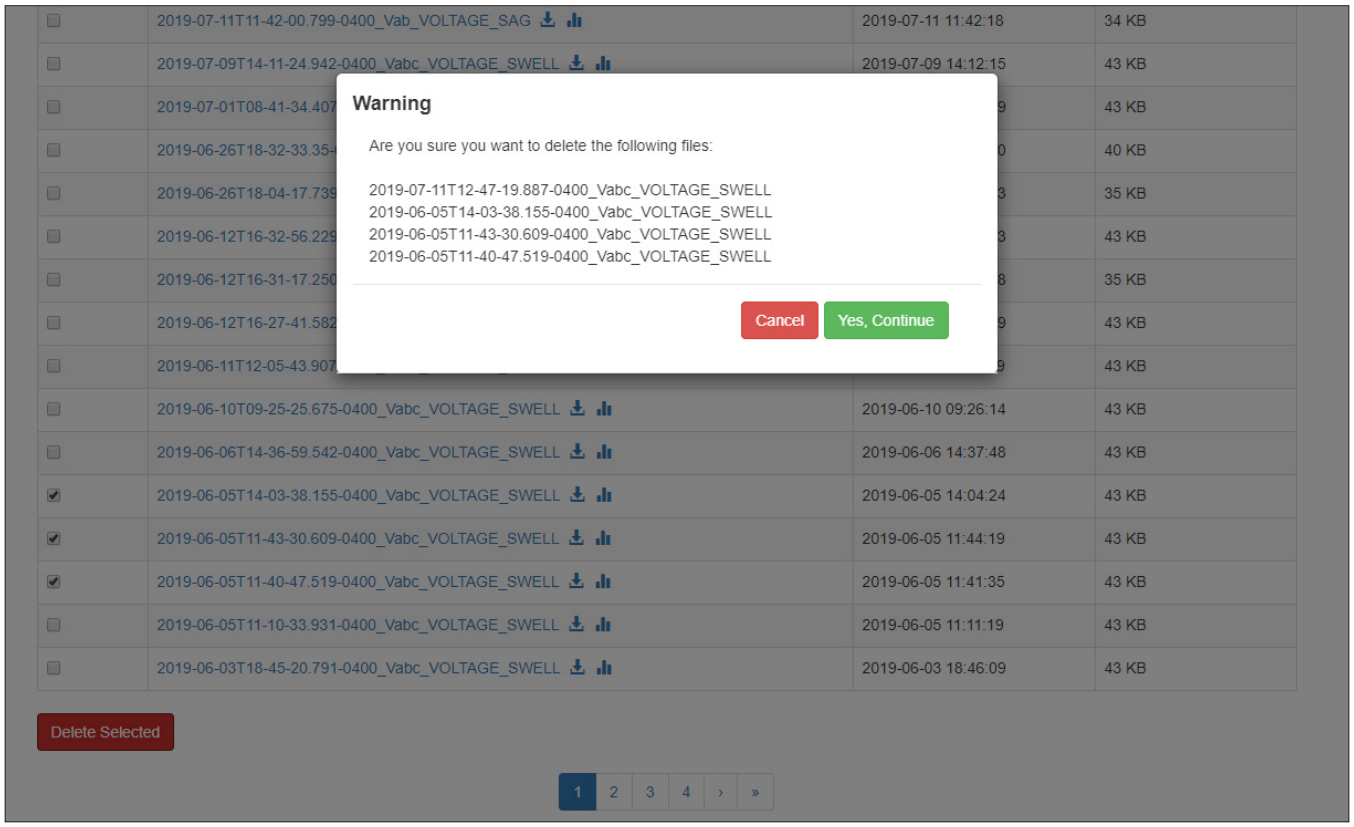


The file name for the manually captured waveform will include the timestamp the capture occurred and the reason will say 'MANUAL' at the end of the file.

10. Web Interface Readings and Parameter Settings



In the waveform log page, user also have the option to select and delete waveform files. Simply click on the check box next to the file to select it, alternatively users can click on the 'Select All' button. Once all files are selected click on delete to remove the files from the waveform log.



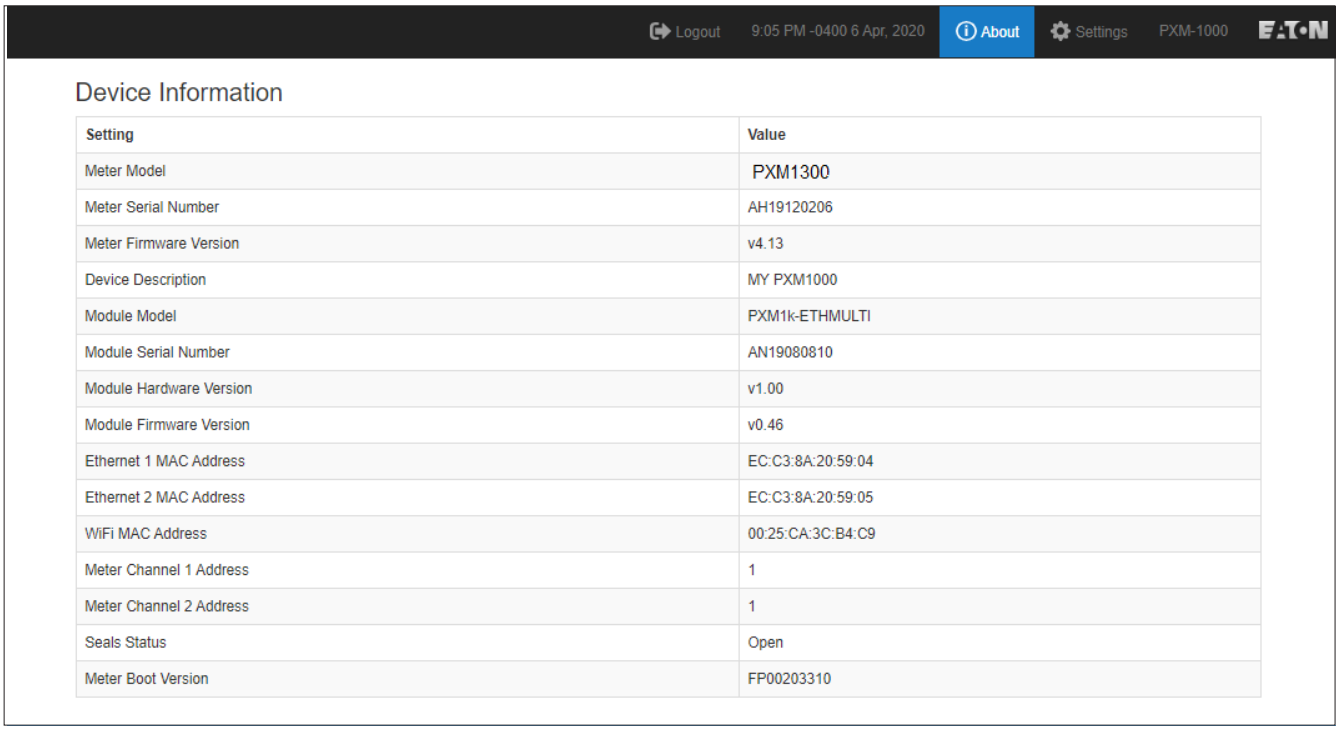
COMTRADE

The PXM1K-ETHMULTI allows users to download the waveform data as a COMTRADE file. COMTRADE is a file format for storing waveform data related to transient power system disturbances.

The file can be downloaded from the waveform log by clicking on the file name or by clicking on the button. Once downloaded the file can be read using a COMTRADE file reader where users can further analyze the waveform data provided from the meter.

10.5 About

The About tab located at the top right corner of the web interface allows users to view the Device Information page. This page provides users with information about the PXM1000 series meter and the PXM1K-ETHMULTI module. The Device Information contains the model of the PXM1000 meter, serial number, firmware version and the meter addresses. It also contains the serial number, firmware version, hardware version and the MAC addresses of the PXM1K-ETHMULTI module.



Setting	Value
Meter Model	PXM1300
Meter Serial Number	AH19120206
Meter Firmware Version	v4.13
Device Description	MY PXM1000
Module Model	PXM1k-ETHMULTI
Module Serial Number	AN19080810
Module Hardware Version	v1.00
Module Firmware Version	v0.46
Ethernet 1 MAC Address	EC:C3:8A:20:59:04
Ethernet 2 MAC Address	EC:C3:8A:20:59:05
WiFi MAC Address	00:25:CA:3C:B4:C9
Meter Channel 1 Address	1
Meter Channel 2 Address	1
Seals Status	Open
Meter Boot Version	FP00203310

10.6 Settings

10.6.1 Meter

General Setting

The basic metering configurations needed to set up the meter can be applied from the web interface by clicking on Settings and selecting the **'Meter'** tab. On the metering tab users will see the 'General' tab selected and the page presented. The general page include the following settings:

Device Description: A description for the meter can be provided in this field which will display on the Dashboard page.

Voltage Wiring: Select the type of wiring that the meter will be monitoring from the modes in the drop down list.

Current Wiring: Select the number of CT's that will be connected to the meter to measure the current.

PT1: Enter the rated input of the potential transformer that is connected to the meter. Possible range is from 50 to 1,000,000V. By default, PT1 is a ratio of 400.

PT2: Enter the rated output of the potential transformer. Possible range is from 50 to 400 V. The default PT2 setting is 400.

Note: If the voltage input is connected directly to the meter and there are no PTs (Potential Transformers) stepping down the voltage to the meters voltage input then the PT1 and PT2 settings can be left as the default of 400.

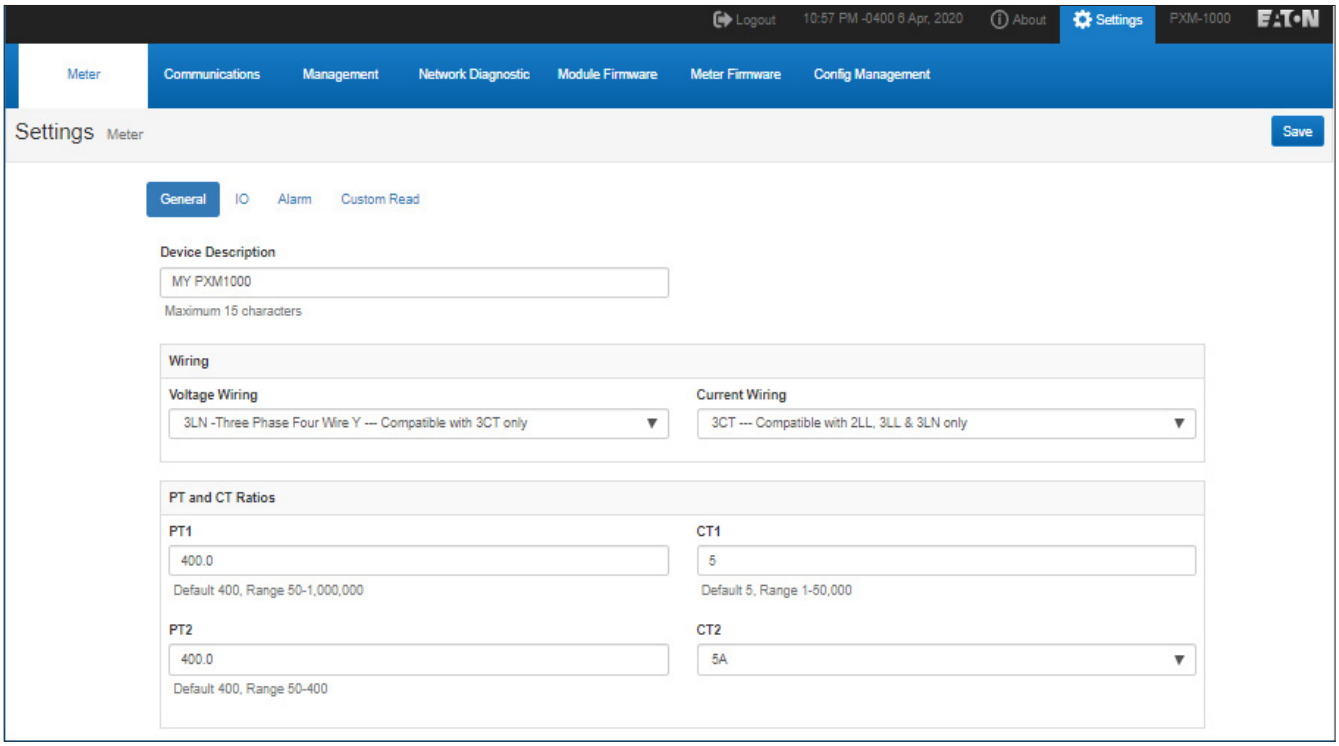
CT1: Enter the rated input of the current transformer that is used with the meter. Possible ranges for the CT1 are from 1 to 50000A.

The default settings for CT1 are dependent on the current input type of the PXM1000 meter. The following table displays the default CT1 values for the different Current input options.

PXM1000 Current Input	Default CT1 Value
1A	1
5A	5
333mV	1
RCT	1000
80mA/100mA/200mA	1

CT2: Select the rated output of the current transformer from the drop down list. By default this setting is already configured.

Note: CT2 is configurable only for 5A and mA current inputs, 5A can be either 5A or 1A and mA can be either 80mA, 100mA or 200mA.



Real time Reading: Select the mode of the readings for the meter when it is polled through Modbus. By default the meter is in Secondary mode which will require some parameters to be scaled by a relationship. Configuring the meter in Primary mode does not require any scaling.

I A Direction: Represents the flow of direction for the Phase A current being measured, configure this setting to troubleshoot issues related to incorrect polarity of readings such as real power, Power Factor and etc.

I B Direction: Represents the flow of direction for the Phase B current being measured, configure this setting to troubleshoot issues related to incorrect polarity of readings such as real power, Power Factor and etc.

I C Direction: Represents the flow of direction for the Phase C current being measured, configure this setting to troubleshoot issues related to incorrect polarity of readings such as real power, Power Factor and etc.

Demand Settings

Sliding Window: This setting refers to the demand type. There are 4 demand types that are supported by the PXM1000 series meter, users can choose one of the following:

1. **Fixed Window** - The demand is calculated based on selecting the calculation period between 1-30min. The meter will calculate and update the demand values at the end of each calculation period.
2. **Sliding Window** - The demand is calculated by selecting the calculation period between 1-30 min. The meter will average the energy accumulated within this period of time and the demand value is updated every minute.
3. **Thermal** - The demand is calculated based on thermal response, used in thermal demand meters. This method uses a sliding window to update the demand value at the end of **each calculation period.**

4. **Rolling Window** - The demand is based on selecting a calculation period between 1-30min, a subinterval(Demand Calculation Slip Time) and the demand value is **updated at each subinterval. The subinterval must be a factor of the calculation period. For example, with a calculation period of 15min, the subinterval can be configured as 5min.**

Sub Interval: The sub interval setting is only relevant if the Rolling Window Method is selected, this method requires a sub-interval time that must be a factor of the demand calculation period. the range for this setting is from 1-30 minutes.

Average Interval: The average interval window is the calculation period of the demand, and can be set from 1-30 minutes.

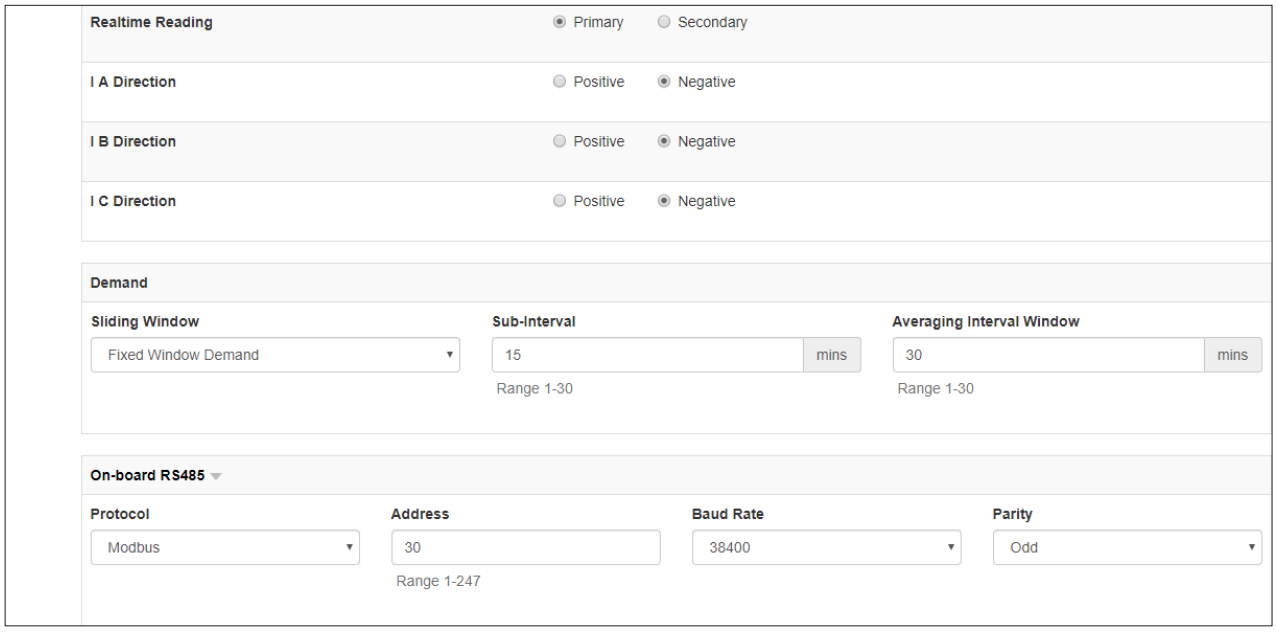
On-board RS485

Protocol: Select the protocol for the PXM1000 RS485 port, the protocol can be set to Modbus.

Address: Select the RS485 address for the meter, the range for this setting is from 1-247.

Baud Rate: The baud rate is the communication speed of the RS485 data transfer, this ranges from 1200-38400.

Parity: Select the parity bit setting for the communication.



Display

Current Password: This password relates to the four digit password used to access the meter settings from the display of the meter. By default it is 0000, the range for this setting is 0000-9999.

New Password: Enter in a new four digit password.

Repeat Password: Repeat the new password configured in the previous setting

Backlight: This setting refers to how long back light on the meters display is on for, the range is from 0-120 minutes, where 0 would disable to back light from turning off.

Rated Load: The rated load can be represented in terms of either power or current.

- If current is selected the rated current that is used would be the CT1 setting value in the PT and CT ratio settings section of the web page. For example if CT1 is set for 1000A, and the average current the meter is monitoring is 500A, the load percentage would be 50% (500/1000A).
- If power is selected the rated primary power would be used in the load percentage calculation. In a wye connected system, the max primary power can be calculated as follows:

  Max Primary Power without using PTs = 3 * (480) * (CT1)

  Max Primary Power using PTs = 3 * (PT1) * (CT1)

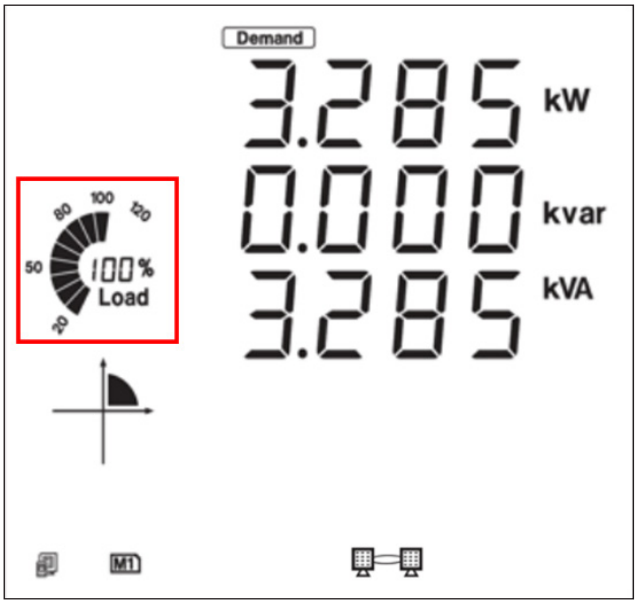
  The max primary power would be the power that is entered in this setting.

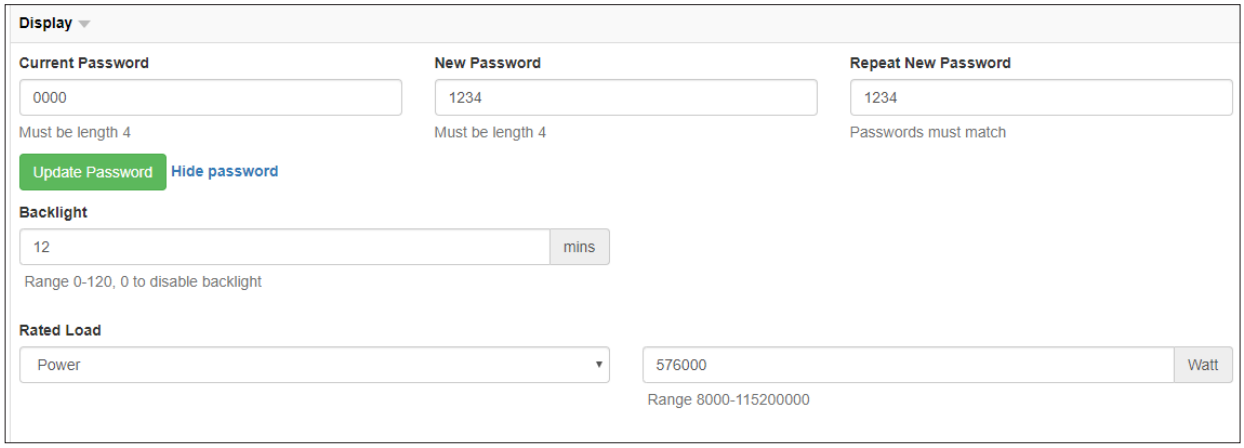
  The load percentage is displayed on the front of the PXM1000 meter display. The load percentage is calculated based on the following equation:

  Load Percentage = (Active System Power / ((5A or 1A) * User Setting))) * 100% , where the meter will have either a 5A or 1A current input. If users have PXM1000 meters with Rogowski Coil (RCT), 333mV and mA type Current Inputs then they would use 1A in this equation.

  For example if the max primary power of your system is 576000W (or 576kW), your system is currently using 211kW and the meters current input type is 5A, then the load percentage would be calculated as follows:

  Load Percentage = (211kW / (5 * 576kW)) * 100 = 7%





Advanced

Energy Type: The energy type can either be selected as Fundamental or Fundamental+Harmonics

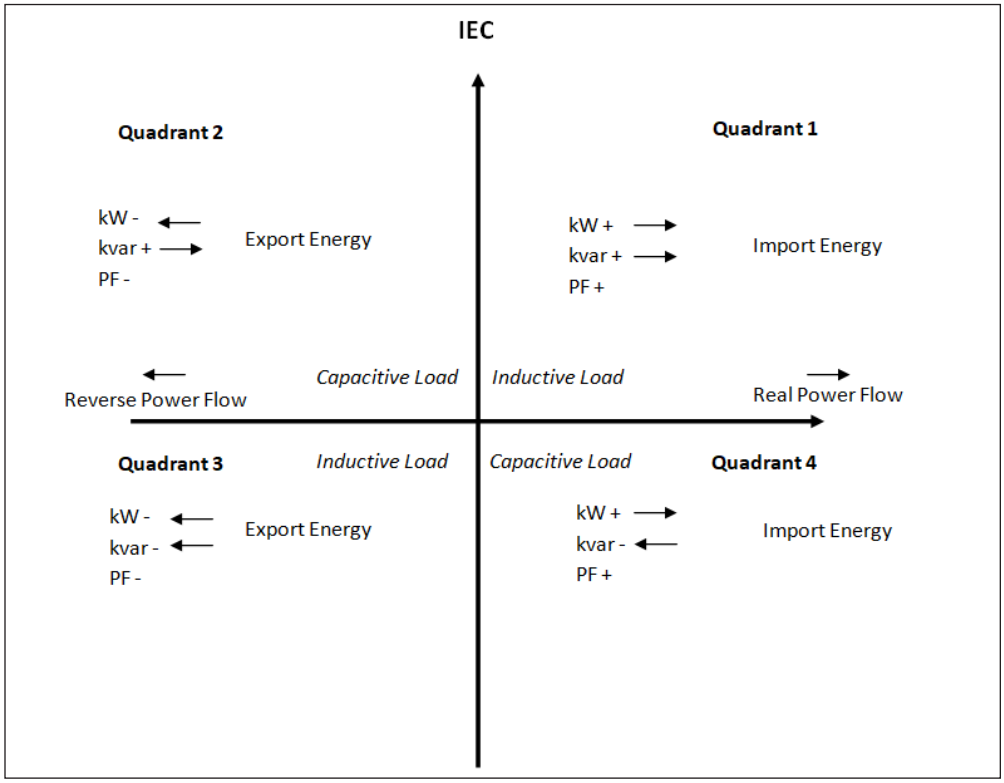
Energy Reading: This can be set to either Primary or Secondary, where Primary displays the energy accumulation in terms of the Primary and Secondary will display the energy accumulation in terms of the secondary with resolution of up to 1Wh.

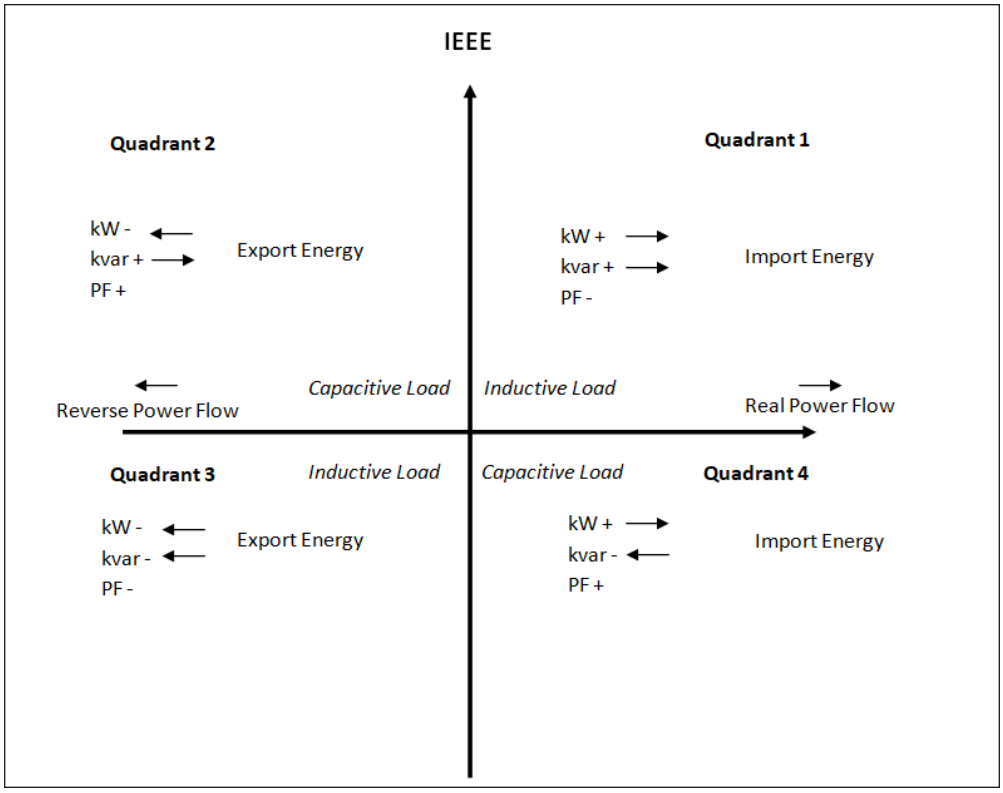
Note: When the energy reading mode is changed the energy will reset to 0 on the meter.

VAR/PF Convention: The VAR/PF convention can either be set to IEEE or IEC.

IEC the power factor is dependent on the direction of the real power flow

IEEE the power factor is dependent on the nature of the load, i.e. capacitive, inductive.





VAR Calculation Method: Can be selected as either True or Generalized

- True Method - Uses the Budeanu Concept to calculate the 'True' reactive power. This method generally takes the harmonic components to do the calculation instead of using the power vector triangle method. This method can be defined by the following by the following expression for single phase circuit:
- Generalized Method - UsesFryze's concept to calculate the 'Generalized' reactive power. This method separates instantaneous current into two components, active and reactive currents.

Active current is calculated as: $i_{\mathrm{a}}(t)=\frac{P}{V_{\mathrm{RMS}}^{2}}v(t)$

and reactive current as $i_{\mathrm{r}}(t)=i(t)-i_{\mathrm{a}}(t).$

Active and reactive powers are:

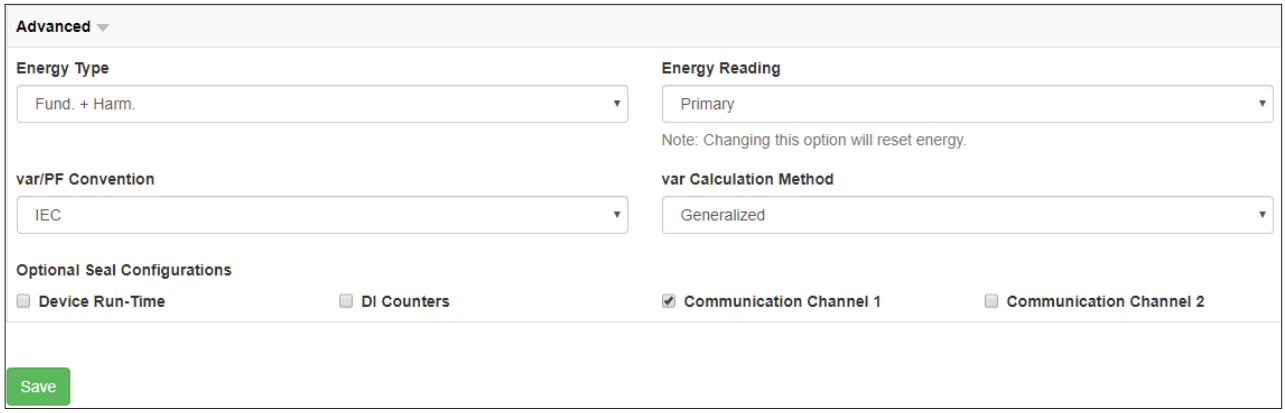
$$P=V_{\mathrm{RMS}}\cdot I_{\mathrm{a}}$$
$$Q_{\mathrm{f}}=V_{\mathrm{RMS}}\cdot I_{r}$$

where I_a and I_r represents RMS values of instantaneous active and reactive currents.

Optional Seal Configurations: Users can choose to seal the following parameters from this setting:

- Device Run Time
- DI Counters
- Communication Channel 1
- Communication Channel 2

Once all settings have been configured users can click on save and then perform a module reboot in order for the settings to be saved to the meter.



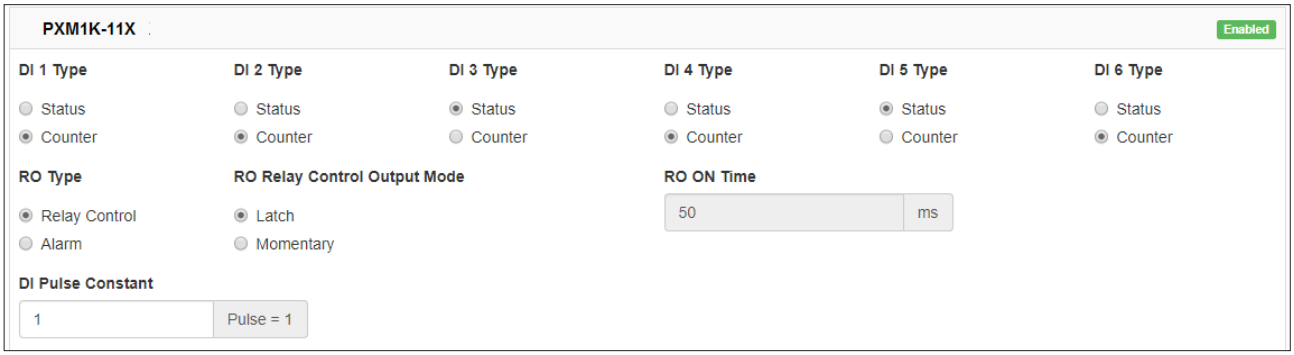
IO Settings

The PXM1000 supports 3 different I/O expansion modules that allow users add Digital Inputs, Digital Outputs, Relay Outputs, Analog Inputs, and Analog Outputs. There can be a max two of the same I/O module per meter, and will represented by the I/O logic address 1 or 2. From the PXM1K-ETHMULTI interface users can configure the settings for these I/O modules.

PXM1K-1x

The PXM1K-1 module supports 4 Digital Input channels, 2 Relay Output channels, and has a built in 24Vdc supply.

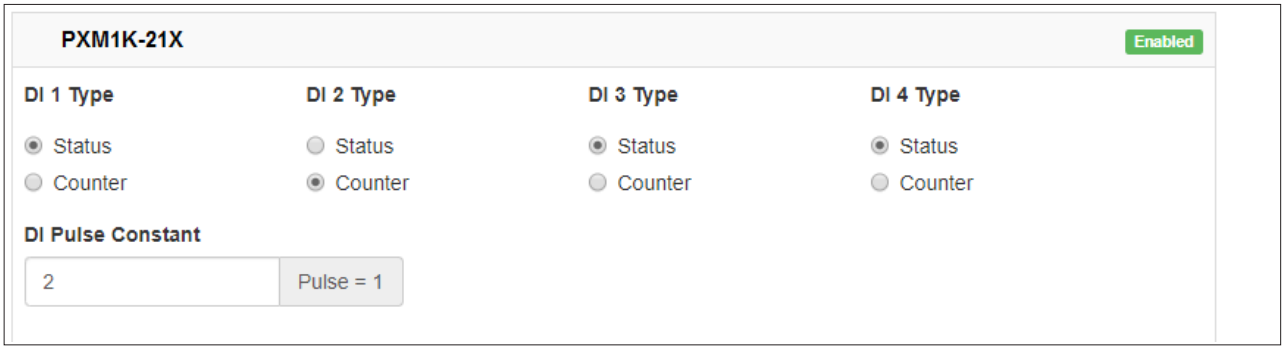
- Digital Input - Can be configured as either digital status or pulse counter.
  - DI Pulse Constant - If the DI is configured as pulse counter, then the DI pulse constant will represent how many pulses equals one count, i.e pulse constant is 2 that means that every 2 pulses the inputs reads the counter is increased by 1.
- Relay Output - Can be configured as Relay Control or Alarm Mode
  - Control Mode Output Mode - Can be configured as Latch or Momentary where users can manually turn on/off relay
    - Latch - Manually turn on/off relay
    - Momentary - Manually turn on relay for a short moment of time, the RO On time range is from 50-3000ms
  - Alarm Mode - The relay can be configured to trigger based on the alarm settings of the meter.



PXM1K-2x

The PXM1K-2 module supports 4 Digital Input channels, 2 Digital Output channels, and 2 Analog Output channels.

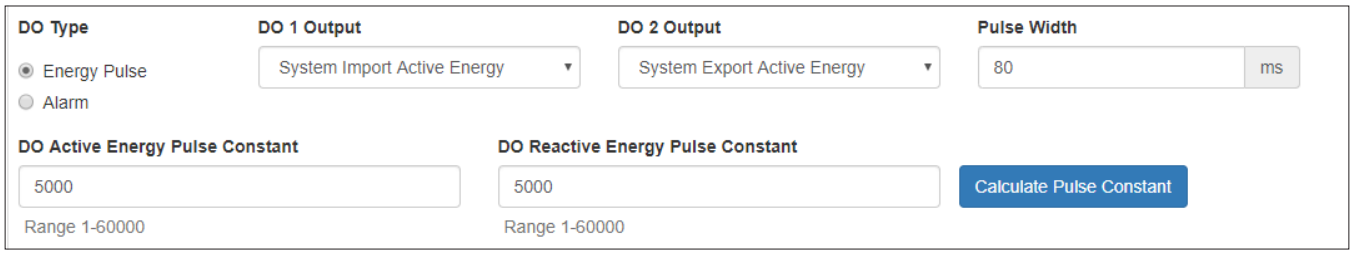
- Digital Input - Can be configured as either digital status or pulse counter.
  - DI Pulse Constant - If the DI is configured as pulse counter, then the DI pulse constant will represent how many pulses equals one count, i.e pulse constant is 2 that means that every 2 pulses the inputs reads the counter is increased by 1.



- Digital Output - Can be configured for Pulse or Alarm mode.
  - DO 1 Output - Users can select the type of energy pulse outputted from DO Channel 1
  - DO 2 Output - Users can select the type of energy pulse outputted from DO Channel 2
    - There are 4 types of energy that can be used for Energy Pulse Output:

      1. System Import Active Energy
      2. System Export Active Energy
      3. System Import Reactive Energy
      4. System Export Reactive Energy

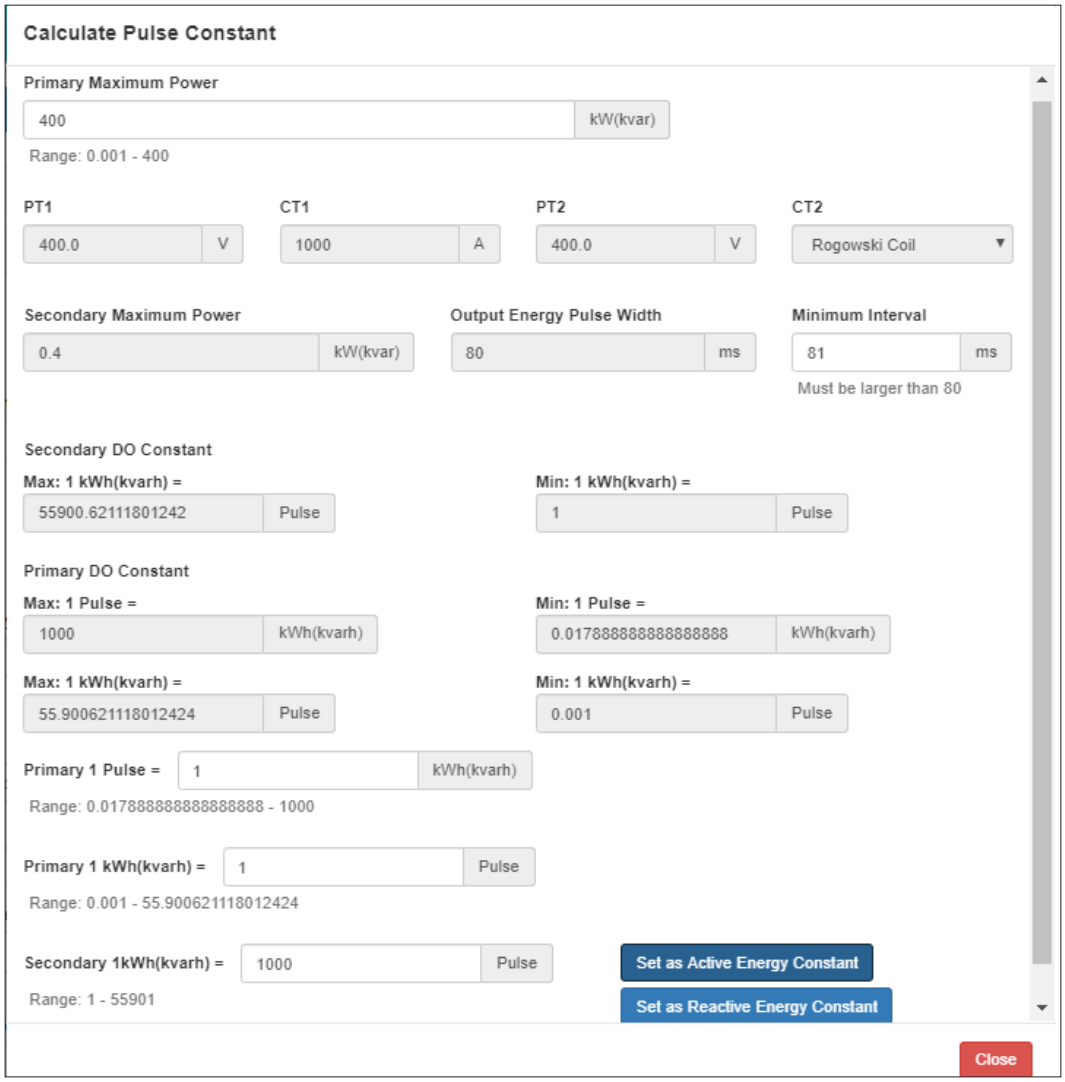
    - Pulse Width - Users can configure the Energy pulse width, the range is from 20-1000ms
    - DO Active Energy Pulse Constant - Select the Pulse Constant for Active Energy Pulse, the range is from 1-60000
    - DO Reactive Energy Pulse Constant - Select the Pulse Constant for Reactive Energy Pulse, the range is from 1-60000



Pulse Constant Calculator

The pulse constant calculator allows users to determine the pulse constant value based on the amount of energy per pulse required. Users would first need to input the maximum primary power which is the CT1 value multiplied by the PT1 value.

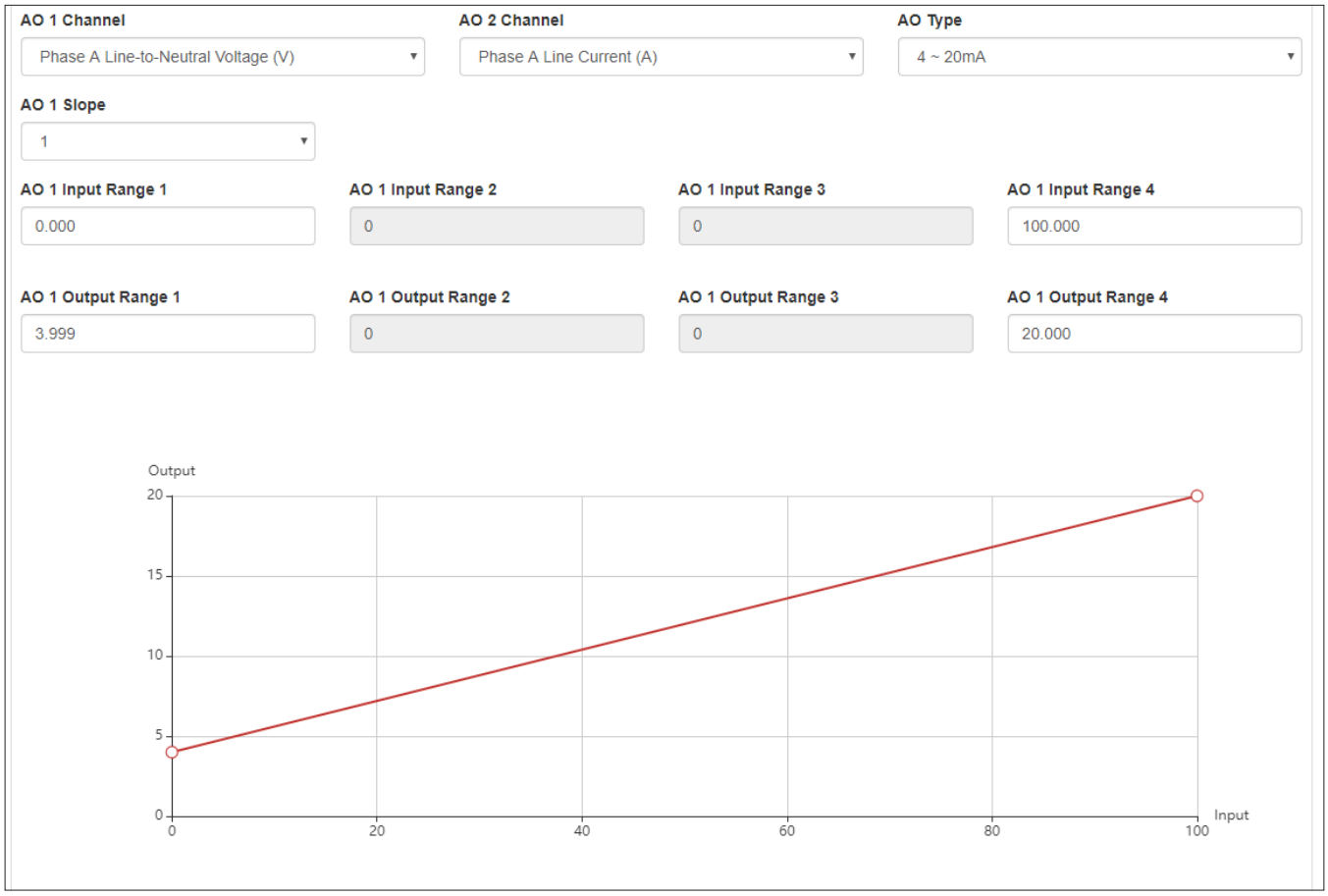
Once entered in, users can then scroll down to the 'Primary 1 Pulse =' and enter the value that one pulse would represent, i.e. 1 pulse = 1kWh. Once this is configured, a secondary pulse value will be populated which is the required pulse constant value. Users can import this value to the pulse constant setting by clicking on the 'Set as Active Energy Constant' or 'Set as Reactive Energy Constant' button.



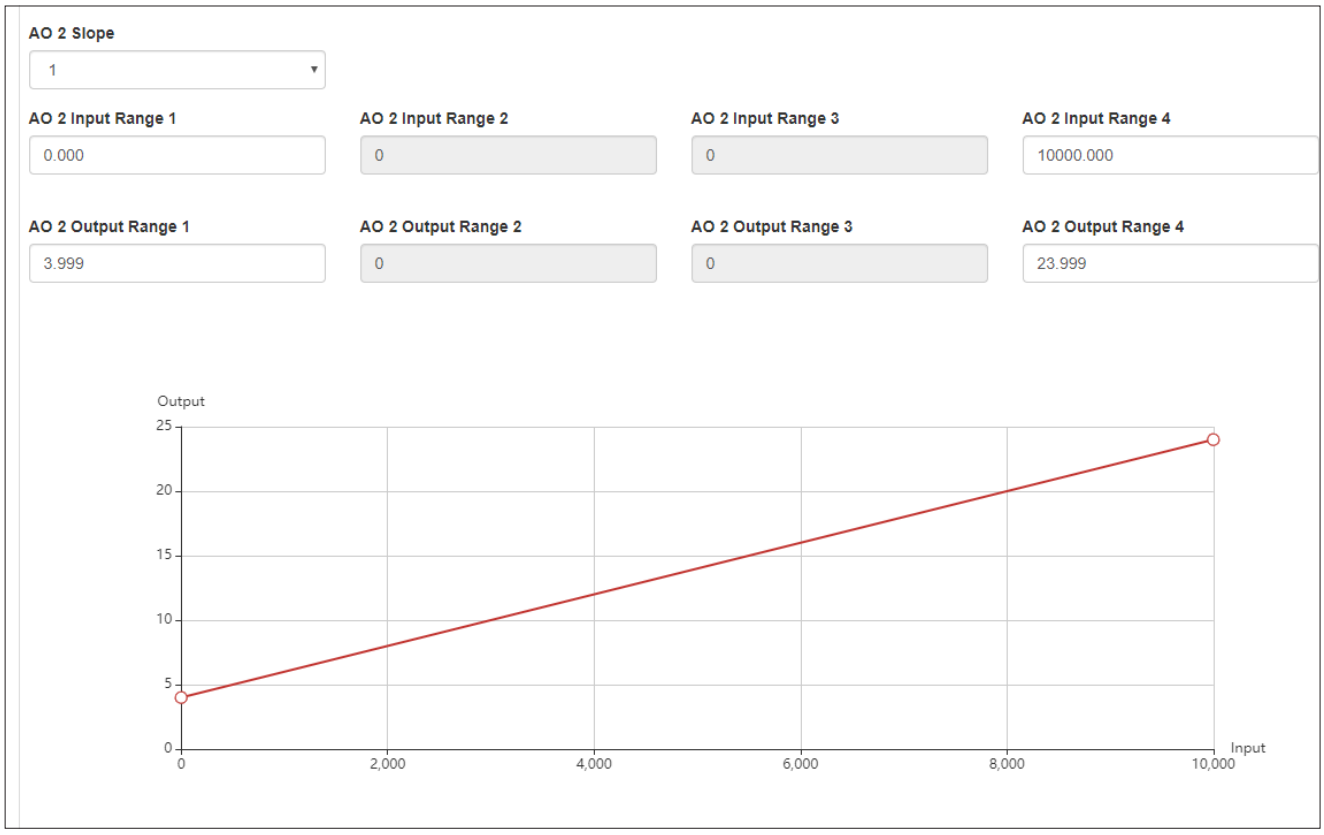
- Analog Output - There are two analog output channels that users can configure.
  - AO1 Slope - select the slope value, the range is from 1-3
  - AO 1 Input Range 1 - Enter in the starting range for the Analog Output
  - AO 1 Input Range 2/3/4 - Enter in the ending range for the Analog Output
  - AO 1 Output Range 1 - Enter in the analog output signal
  - AO 1 Output Range 2/3/4 - Enter in the ending analog output signal.

Note: The analog output range can be a 4-20mA, 0-20mA, 0-5V, 1-5V signal depending on the model of the I/O, either voltage or current model.

The interface displays a graph for the Analog output signal according to the input range that has been configured.



Similar to Analog Output channel 1, the second analog output channel can also be configured in the same manner from the web interface.

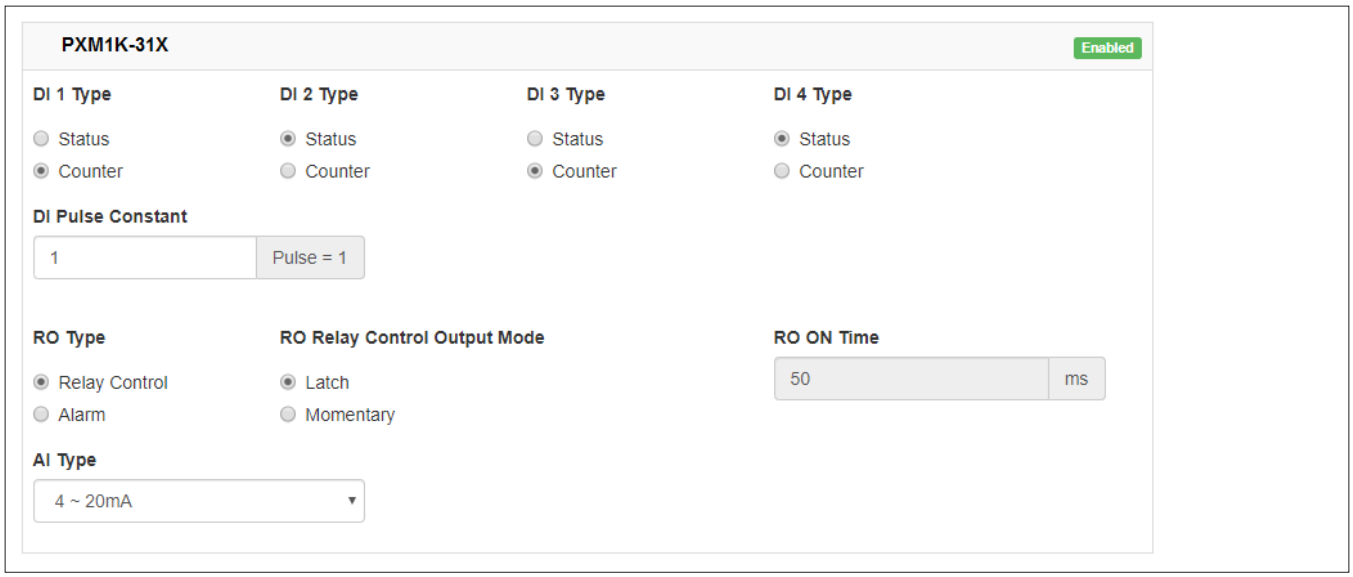


PXM1K-3x

The PXM1K-3 module supports 4 Digital Input channels, 2 Relay Output channels, and 2 Analog Input channels.

- Digital Input - Can be configured as either digital status or pulse counter.
  - DI Pulse Constant - If the DI is configured as pulse counter, then the DI pulse constant will represent how many pulses equals one count, i.e pulse constant is 2 that means that every 2 pulses the inputs reads the counter is increased by 1.
- Relay Output - Can be configured as Relay Control or Alarm Mode
  - Control Mode Output Mode - Can be configured as Latch or Momentary where users can manually turn on/off relay
    - Latch - Manually turn on/off relay
    - Momentary - Manually turn on relay for a short moment of time, the RO On time range is from 50-3000ms
  - Alarm Mode - The relay can be configured to trigger based on the alarm settings of the meter.
- Analog Input - There are two analog input channels, users can select the type based on the type of module they have.
  - The AI types are:

    1. 4-20mA
    2. 0-20mA
    3. 1-5V
    4. 0-5V

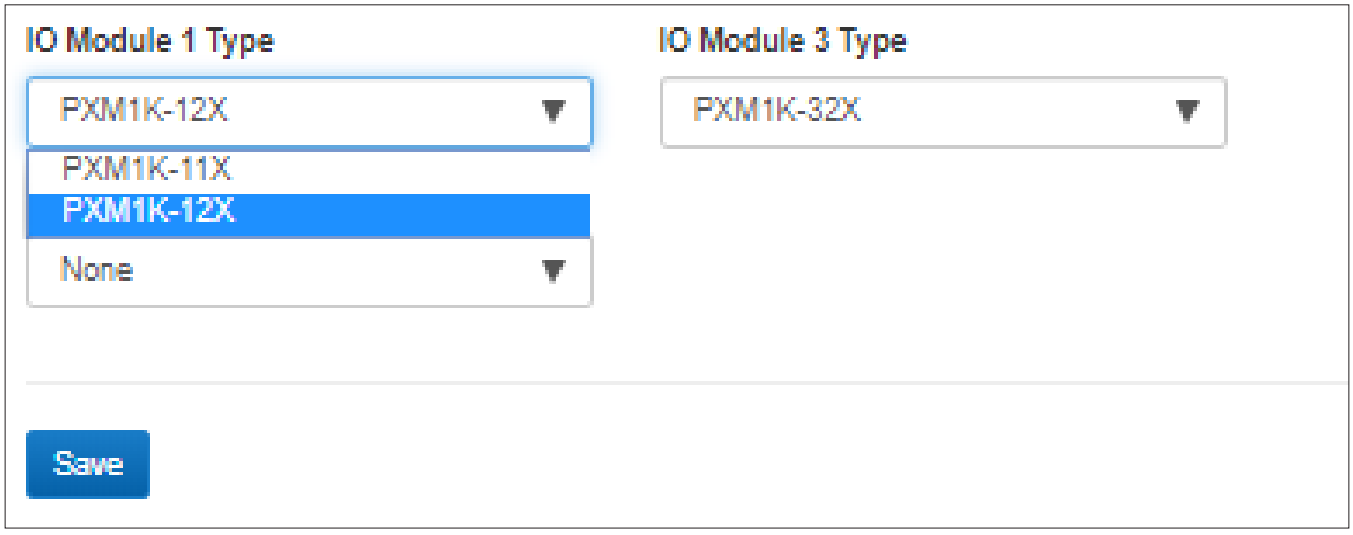


IO Logic Address

From the PXM1K-ETHMULTI interface users have the option to change the logic addresses for the PXM1K-1 and PXM1K-3 modules. This is useful when users have two modules with the same logic address. For example if the user has two PXM1K-11x and attaches them both to the meter, the meter will only recognize one of the modules. This function allows the user to change the logic number from 1 to 2 where the user would then have one PXM1K-11x module and one PXM1K-12x module.

To change the logic address the module must be attached and installed on the back of the meter after the PXM1K-ETHMULTI communications module. Simply click on the drop down menu and select the corresponding logic address.

Note: The meter be physically power cycled in order for the logic address change to take effect.



Alarm Settings

The PXM1000 meters supports over/under alarms for different metering parameters..From the PXM1K-ETHMULTI interface users can configure these alarms on the Alarm page. This page can be found by clicking on 'Settings' and selecting the 'Alarm' tab.

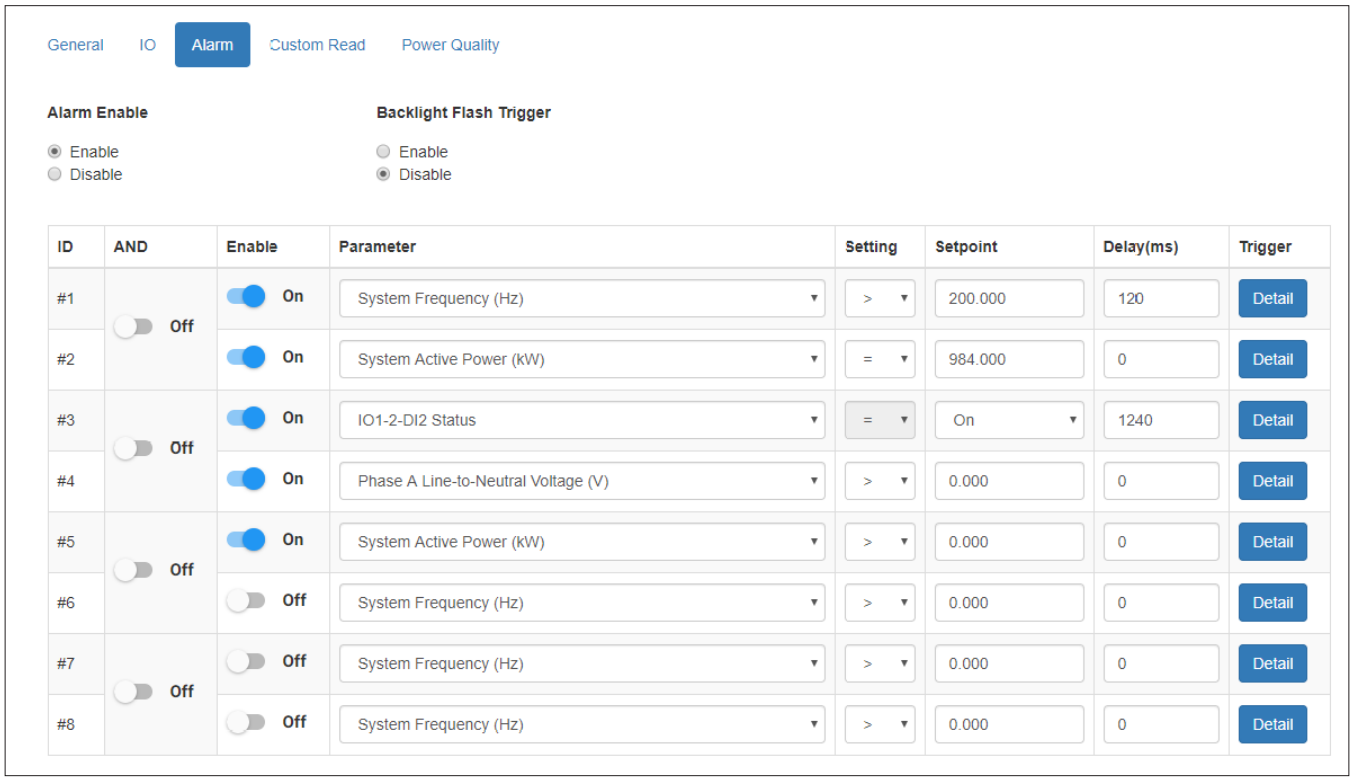
The meter supports up to 16 Alarm channels that users can monitor when the configured parameter goes over or under the alarm threshold. If users have extended I/O modules attached to the meter, digital outputs (DO) and relay outputs (RO) can be triggered upon alarm condition and used to activate external devices such as a buzzer, light, etc.

Alarm Enable: The alarm function can be enabled or disabled

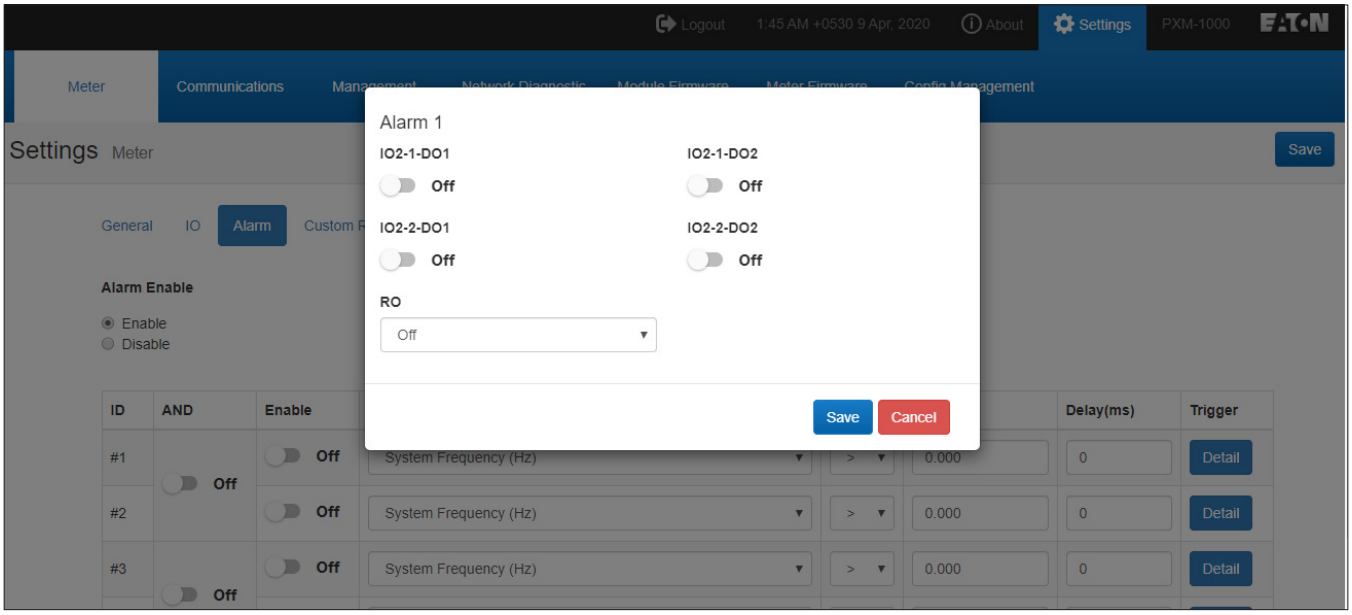
Backlight Flash Trigger: If enabled, when an alarm is triggered the backlight of the PXM1000 meter display will flash during the alarm event.

Steps to setup alarm channels:

- To configure an alarm channel, enable the preferred alarm channel(s).
- Select the required parameter for the alarm channel under the 'Parameter' column.
- Choose the required setpoint and then select the condition for the alarm, i.e greater than (>), less than (<), or equal to (=).
- Users have the ability to set a delay for the alarm trigger, the range is from 0-30000 ms.
- Users can use the 'AND' function to trigger an alarm event when two conditions have been met.



- If extended IO modules are being used, users can click on the detail tab to configure the digital output and relay outputs to be triggered when an alarm is triggered.



Once all alarm settings are configured, user must click on 'Save' and then reboot the communications module in order for the settings to be saved.

Custom Read

The PXM1300 meter supports a custom read function which allows users to customize a block of registers within the PXM1000 meter using different parameters (i.e. Basic metering, THD, Energy, etc) as well as different data types for the parameters (i.e. Int, float, etc). There is a total of 64 bytes that users have to create their customized register block.

Note: The Custom Read Function is only available in meters with firmware version 3.51 and higher.

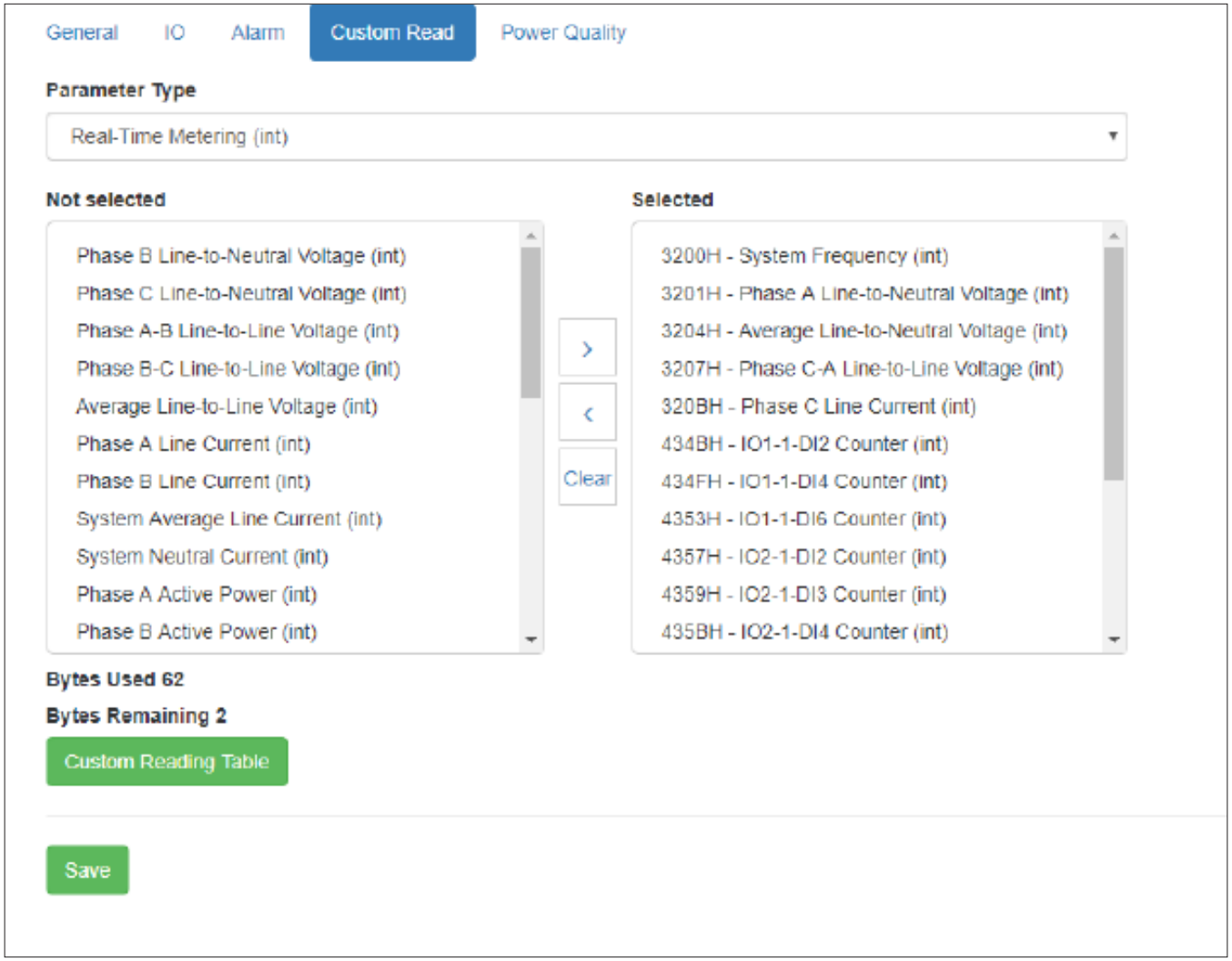
The window on the left under 'Not Selected' are the list of parameters available for the custom read block. Users can choose between different parameters by clicking on the drop down menu under 'Parameter Type'. The available parameter types include:

- Real-Time Metering
- Demand
- Energy
- THD
- Sequence
- Phase Angles
- DI Counter
- AO/AI value

Each of these parameter types are available in different data types, such as integer (int). float, and double-word. Users may select the data type for each parameter from this drop down menu.

Users can select the parameters and click on the '>' button to add the parameters to the 'Selected' window. The parameters can be removed from the register block by clicking on the '<' button, and can clear the entire block by selecting the 'Clear' button.

As users add and remove parameters, there is a 'Bytes Used' and 'Bytes Remaining' value that lets users know how much space is left in the customized register block.



Once the block is configure, users can click on 'Save'.

A copy of the custom read register block can be viewed by selecting 'Custom Reading Table'. The table can be downloaded as a csv file by clicking on 'Export Table'.

Custom Reading Table

Description	Data Type	Start Address	Reg Number
System Frequency	int	6A00H	1
Phase A Line-to-Neutral Voltage	int	6A01H	1
Average Line-to-Neutral Voltage	int	6A02H	1
Phase C-A Line-to-Line Voltage	int	6A03H	1
Phase C Line Current	int	6A04H	1
IO1-1-DI2 Counter	int	6A05H	2
IO1-1-DI4 Counter	int	6A07H	2
IO1-1-DI6 Counter	int	6A09H	2
IO2-1-DI2 Counter	int	6A0BH	2
IO2-1-DI3 Counter	int	6A0DH	2
IO2-1-DI4 Counter	int	6A0FH	2

Waveform Settings

The PXM1000W meters support a waveform caputure feature where users can capture waveforms based on power quality events such as voltage sags, voltage swells and over currents. From the PXM1K-ETHMULTI interface, users can configure these settings by clicking on the 'Settings' tab and then selecting the 'Wavform' tab.

Rated Voltage: The rated voltage of the system should be entered here, the range is from 50-400V for wye systems or 50-690V for delta systems.

Voltage Swell

- Triggering Waveform Capture - Select enable to capture voltage swell events
- Threshold - Enter in the percentage of the voltage swell to be captured, the range is from 50-140%. For example if the rated voltage is 277V, and the voltage swell threshold is set for 110%. The swell event would be captured when the voltage is 110% above 277V, which is roughly 304V.

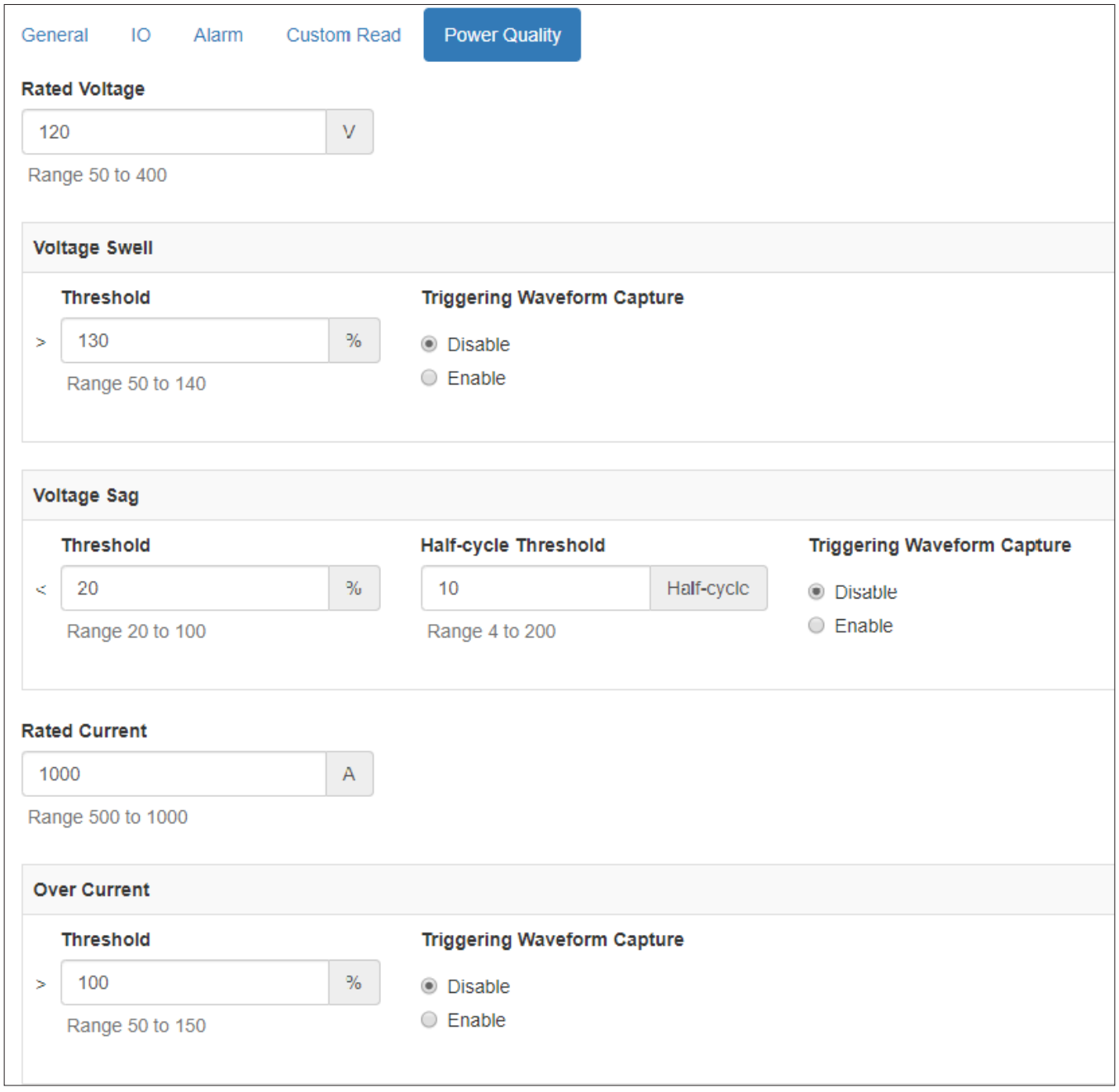
Voltage Sag

- Triggering Waveform Capture - Select enable to capture voltage sag events
- Threshold - Enter in the percentage of the voltage sag to be captured, the range is from 20-100%. For example if the rated voltage is 277V, and the voltage sag threshold is set for 50%. The sag event would be captured when the voltage drops 50% below 277V, which is roughly 138V.
- Half-cycle Threshold - Enter in the half cycle threshold for the sag event, the range is from 4-200 half cycles.

Rated Current: The rated current for the over current should be entered here, the range will be dependent on the CT1 value configured on the meter. The rated current range will be from 50-100% of the CT1 value. For example if CT1 is configured as 1000A, then the rated current range for the Power Quality event is from 500A to 1000A.

Over Current

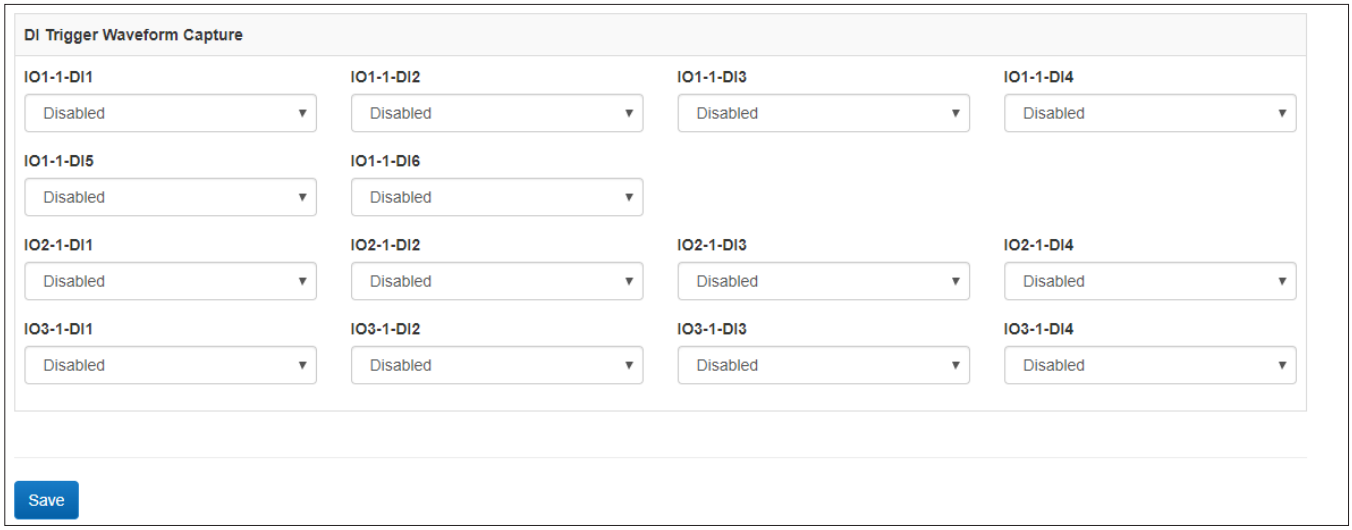
- Triggering Waveform Capture - Select enable to capture over current events
- Threshold - Enter in the percentage of the over current to be captured, the range is from 50-150%. For example if the rated current is 1000A, and the over current threshold is set for 50%. The over current event would be captured when the current is 50% of the rated current, which is 500A.



If user are using external I/O modules, the waveform capture can be triggered by the Digital Inputs Status. The DI can trigger the waveform by the following:

- From Status Off to On
- From Status On to Off
- Any Change

The DI triggering can be disabled is users do not require it to trigger the waveform capture.



10.6.2 Communications

The communication setting web page will allow the user to configure settings related to the Ethernet networks and the Wireless network. The functions and protocols that the PXM1K-ETHMULTI module supports can be configured by selecting the corresponding tab such as Emails, Time/Date , Datalog, PX-EVA Post for communicating with the PX-EVA software, BACnet-IP, SNMP.

Network

The first page the user will see after selecting the Communications option under the Settings tab is the Network page. The network settings allow users to configure all network related settings including both Ethernet 1 and Ethernet 2 as well as WiFi.

RSTP Protocol

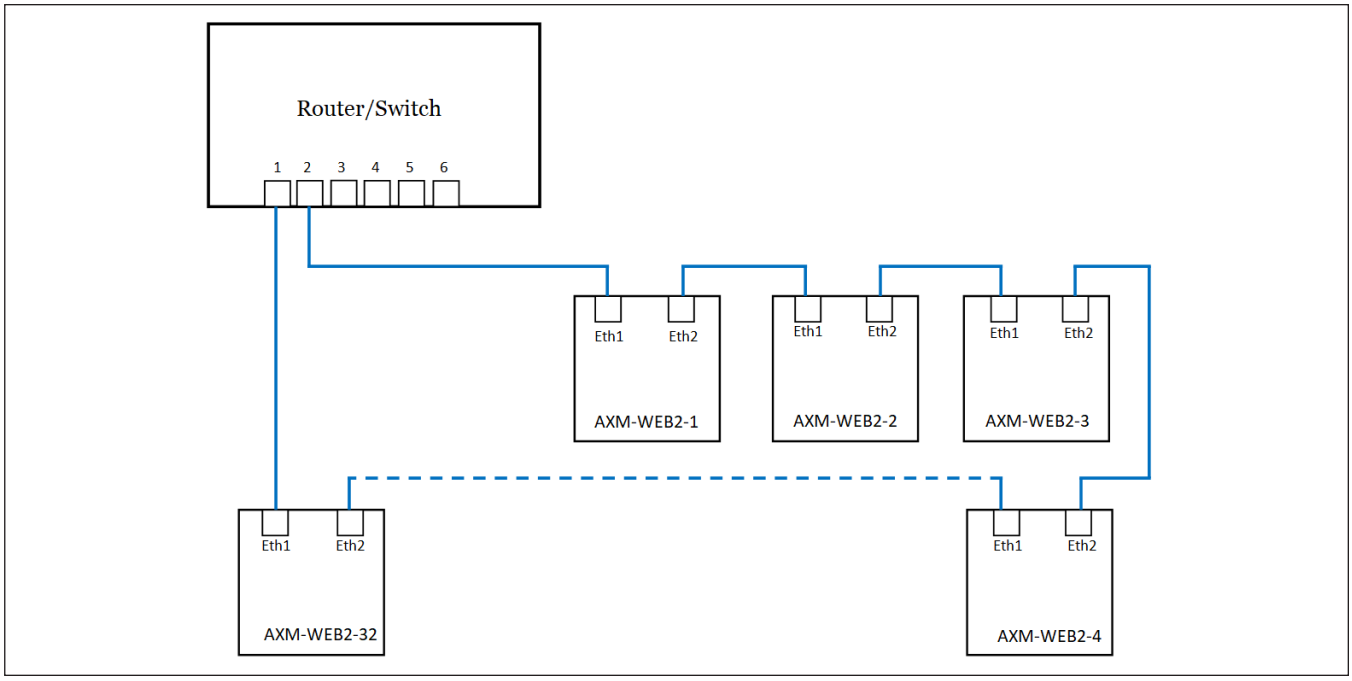
The PXM1K-ETHMULTI supports the RSTP protocol where users can daisy chain the Ethernet ports on the PXM1K-ETHMULTI module to a network switch.

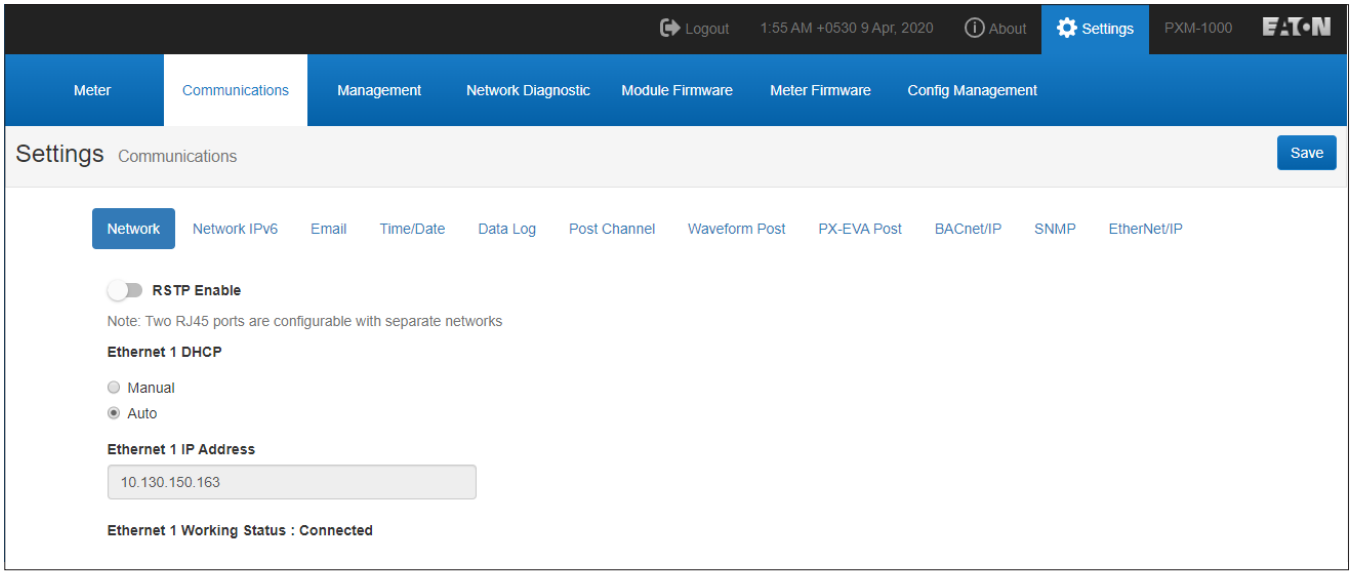
Users can configure the meters IP manually or by setting the DHCP set as Auto.

Note: NWhen the RSTP is enabled users will not be able to configure Ethernet 1 and Ethernet 2, there is only 1 IP per meter using RSTP protocol.

Network Topology

Users can can daisy chain up to 32 devices using the RSTP protocol. This can cut down the amount of network switches required in different applications and allows the use of 1 network switch/router to be used with up to 32 devices. Each device can be accessed by configuring a unique IP address or having the IP addresses assigned automatically by the network.





Network Settings

The settings for the Ethernet 1 and Ethernet 2 are as followed:

Ethernet 1 DHCP: Select 'Manual' to manually configure the IP address to access the meter. If set to 'Manual', you'll also need to set the Subnet Mask and Gateway. By default the IP address for ETH1 will be 192.168.1.254

Select 'Auto' to have the meter assigned a IP address automatically. With this selection the Subnet Mask, and Gateway will also be automatically assigned.

Note: After changing DHCP to Auto, check the display of the meter(N02 NET Settings) to obtain the new IP address that has been assigned. The new IP address will be displayed only after a module reboot is performed and completed.

IP Address: If the DHCP is configured to Manual, the IP address can be configured from this page. Default is 192.168.1.254

Subnet Mask: If the DHCP is configured to Manual, the Subnet Mask can be configured from this page. Default is 255.255.255.0

Gateway: If the DHCP is configured to Manual, the Gateway can be configured from this web page. Default is 192.168.1.1

The status of the Ethernet 1 port will display if it is connected or disconnected.

Ethernet 2 DHCP: By default the Ethernet 2 port is configured to have its DHCP set to 'Auto'. If configured to 'Manual' the default Manual IP address is 192.168.1.253. Users can configure the IP address to any IP once the DHCP is configured for 'Manual', users will also need to set the Subnet Mask and Gateway if using this method.

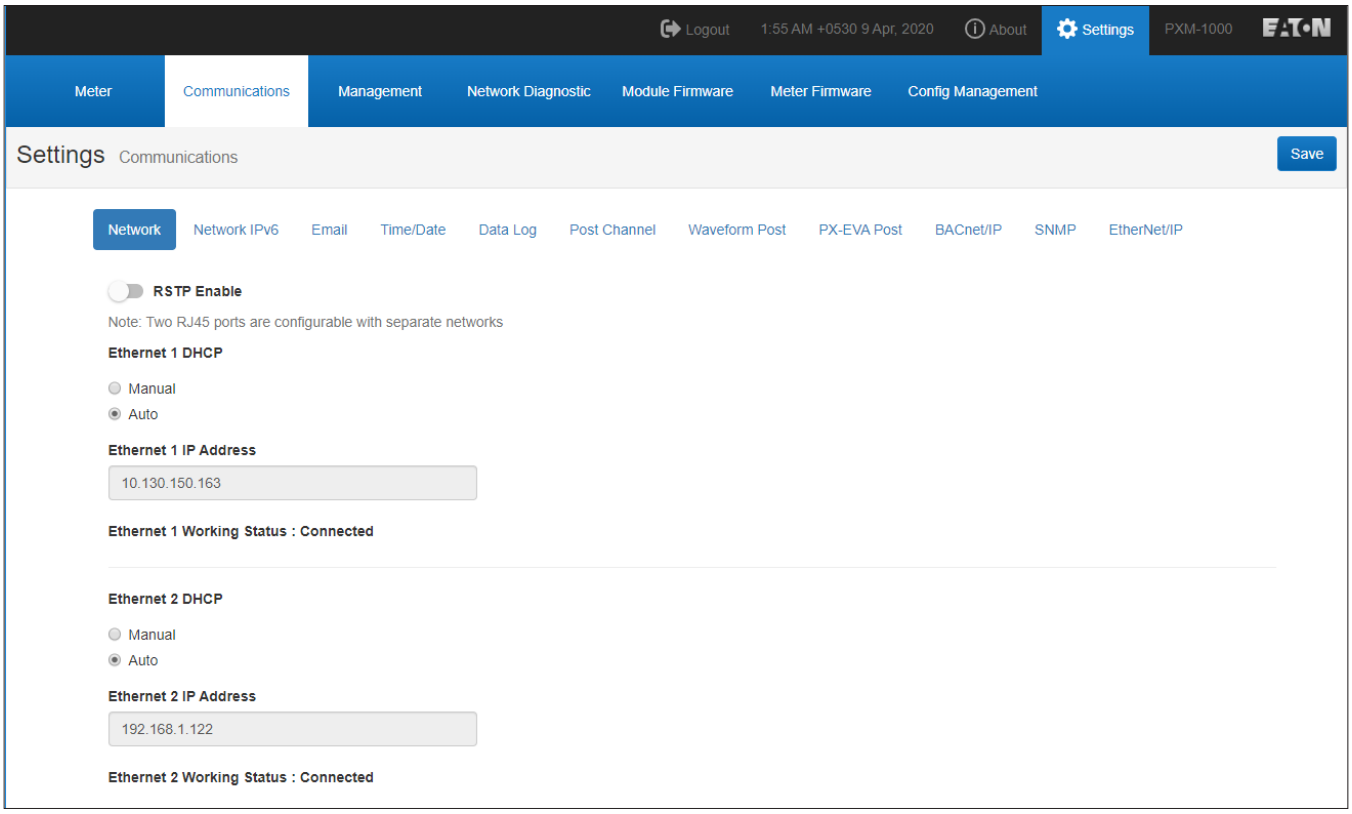
Note: The IP address of the Ethernet 2 can be found page N12 of the NET Settings. The PXM1000 protocol setting must be configured to PXM1K-ETHMULTI to view this from the meters NET settings.

IP Address: By default the IP address is configured by DHCP, this field will be grayed out. If the DHCP is configured to Manual, the IP address can be configured from this page.

Subnet Mask: If the DHCP is configured to Manual, the Subnet Mask can be configured from this page.

Gateway: If the DHCP is configured to Manual, the Gateway can be configured from this web page.

The status of the Ethernet 2 port will display if it is connected or disconnected.



WiFi Enabled: Select the Enable or Disable communication through WiFi.

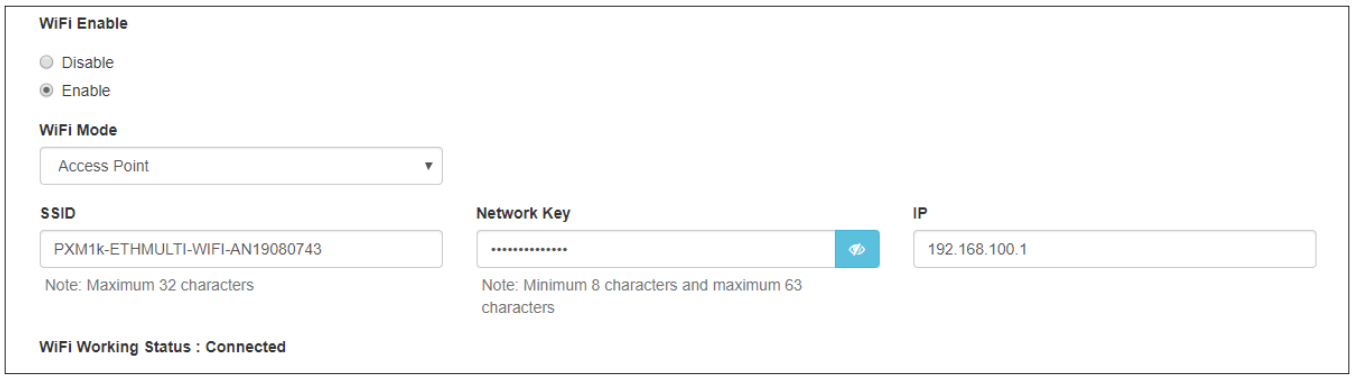
WiFi Mode: The WiFi can be configured to work in two modes just like any other WIFI device. It can be configured as either Access Point(AP) or Station mode.

Access Point: Default configuration for PXM1K-ETHMULTI. The PXM1K-ETHMULTI will act as a wireless access point and will allow other wireless devices to connect and access the PXM1K-ETHMULTI.

- In Access Point mode, users can configure the SSID, Network Key and IP of the PXM1K-ETHMULTI module as well as the DHCP DNS servers.

Station: The PXM1K-ETHMULTI will behave like a wireless client and bridge to another wireless network that is available.

- In Station mode, users can select the Wireless network to connect to under the "Connect to SSID" setting. Click on "Select from Available Networks" and enter the Network Key for the wireless network that the PXM1K-ETHMULTI will bridge to.
- If users are connecting to an open Wireless network that is not password protection, the password field can be left blank.
- The PXM1K-ETHMULTI also supports Enterprise WiFi, where users can connect using an enterprise level WiFi network which is common in many colleges/universities, hospitals, etc. When attempting to connect to an enterprise level WiFi network the interface will show options to connect to the network with a username and password.



In station Mode the DHCP can configured as either manual or auto.

- If manual, users can configure the IP, Subnet Mask and Gateway and DNS Servers.
- If auto, users can check the meter's display to get the IP address and all other network configurations assigned by the wireless network. The user can also configure the DNS servers if the DHCP is set to Auto.

Note: The WiFi IP address for the PXM1K-ETHMULTI will be in parameter N11 of the NET settings. The PXM1000 protocol setting must be configured to PXM1K-ETHMULTI to view this from the meters NET settings.

DNS Server 1: Enter the address of DNS server 1 in this field.

DNS Server 2: Enter the address of DNS server 2 in this field.

HTTPS Port: Enter the HTTPS port number of the meter. By default, this setting is configured to 443. The range can be from 6000 to 9999.

Note: NOTE: This setting should never be configured to 80. Enable the HTTP Enable configuration to access the web interface at port 80.

HTTP Enable: Enable HTTP so the the PXM1K-ETHMULTI cab be accessed through the HTTP protocol, by default the HTTP port is 80 but it can be configured from 6000-9999.

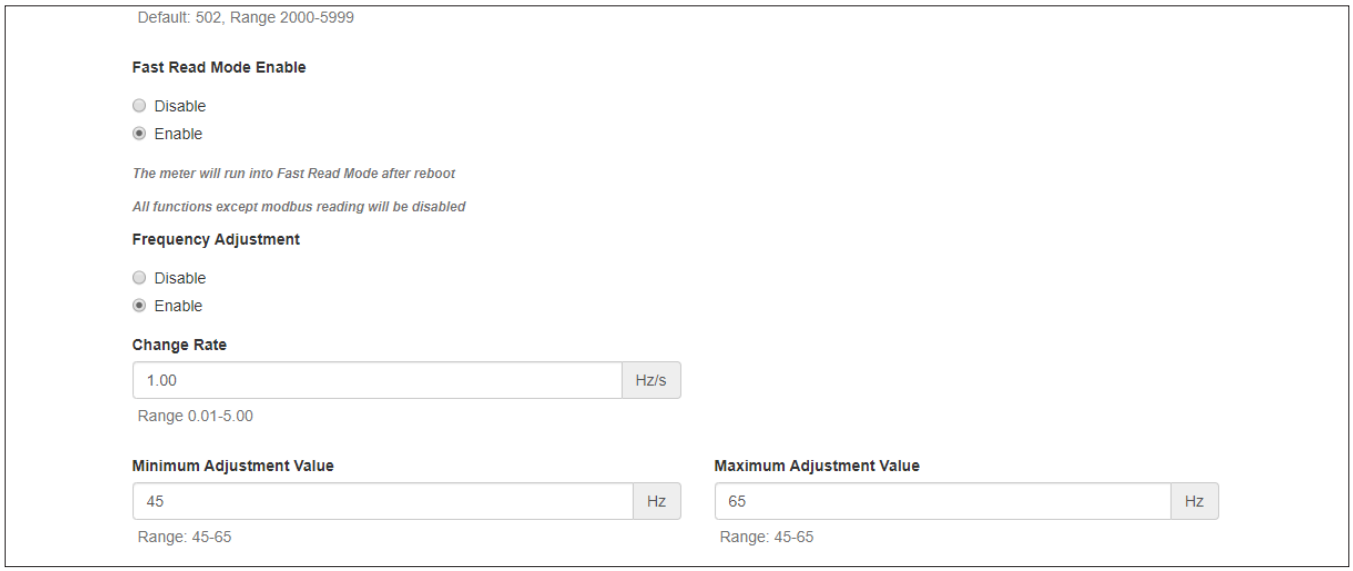
Modbus TCP Port: Enter the Modbus port number of the meter. By default, this setting is configured to 502. The range can be from 2000 to 5999.

Fast Read Mode Enable: Selecting Enable allows the user to read the real time parameters at 100ms.

- Frequency Adjustment: Allows users to control the rate of change of frequency. The range can be set from 1.00-5.00 Hz/s, by default this is set for 1 Hz/s.
  - The minimum range is 45Hz and the max range is 65Hz, any frequency outside of the range will not have the frequency adjustment applied.

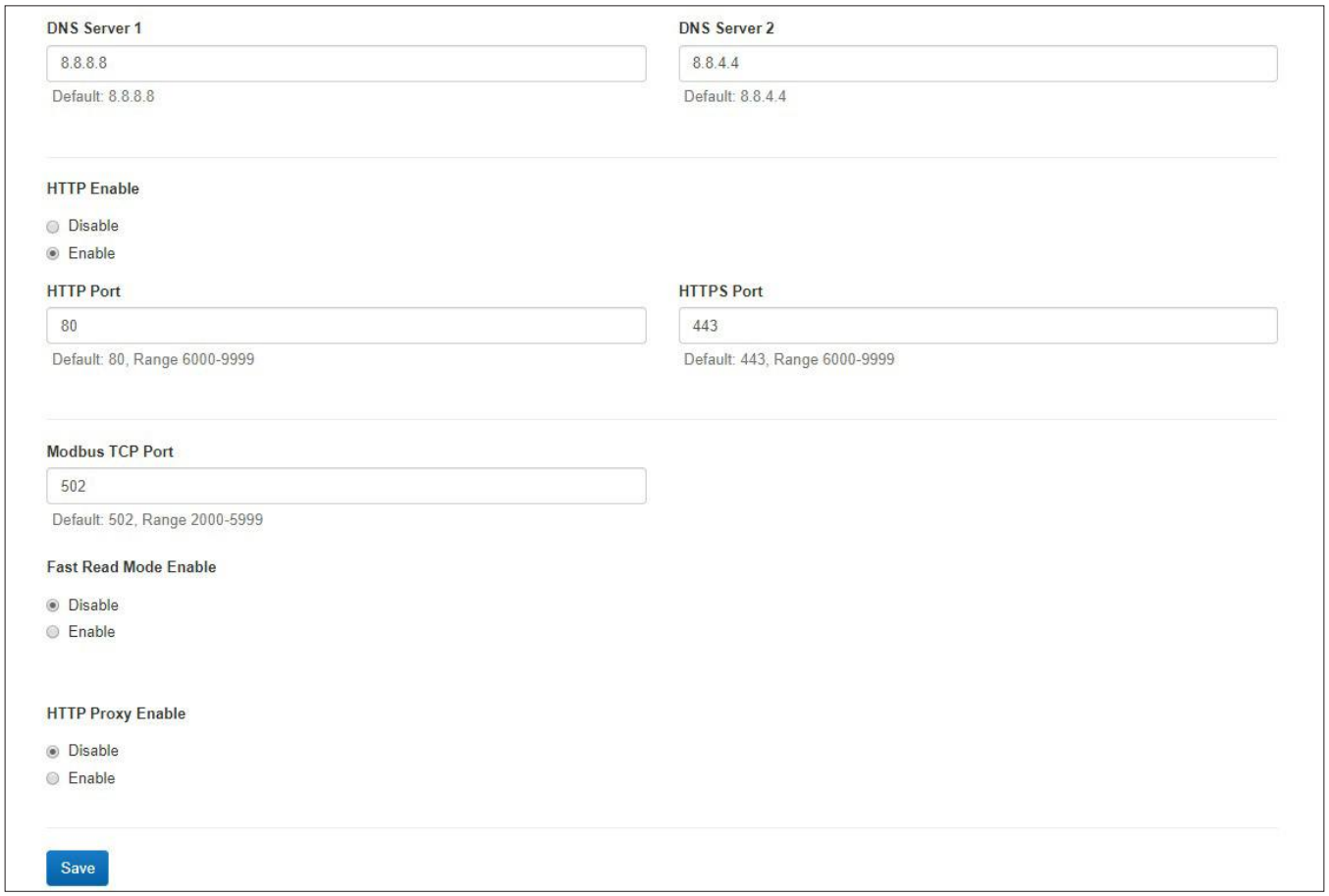
Note: Frequency Adjustment is only available when the meter is in Fast Read Mode.

Note: When Fast Read mode is enabled, all functions except Modbus and the Web server are disabled.



Proxy Server Enable: Select enable to allow for forwarding of data log files to pass through the Proxy server first and then the data post server. IE. PX-EVA.

After making any changes on the network settings page, click 'Save'. Users will be prompted to reboot the PXM1K-ETHMULTI immediately or later. If later is chosen the PXM1K-ETHMULTI will need to be rebooted from the 'Management' page in order for the settings to take effect.



IPv6

The PXM1K-ETHMULTI module supports IPv6 communication where users can use IPv6 to access the web interface as well as connect via SNMP protocol. The settings for IPv6 can be accessed by clicking on Settings and selecting the Communications tab. On the Communications page select the IPv6 tab to configure the settings.

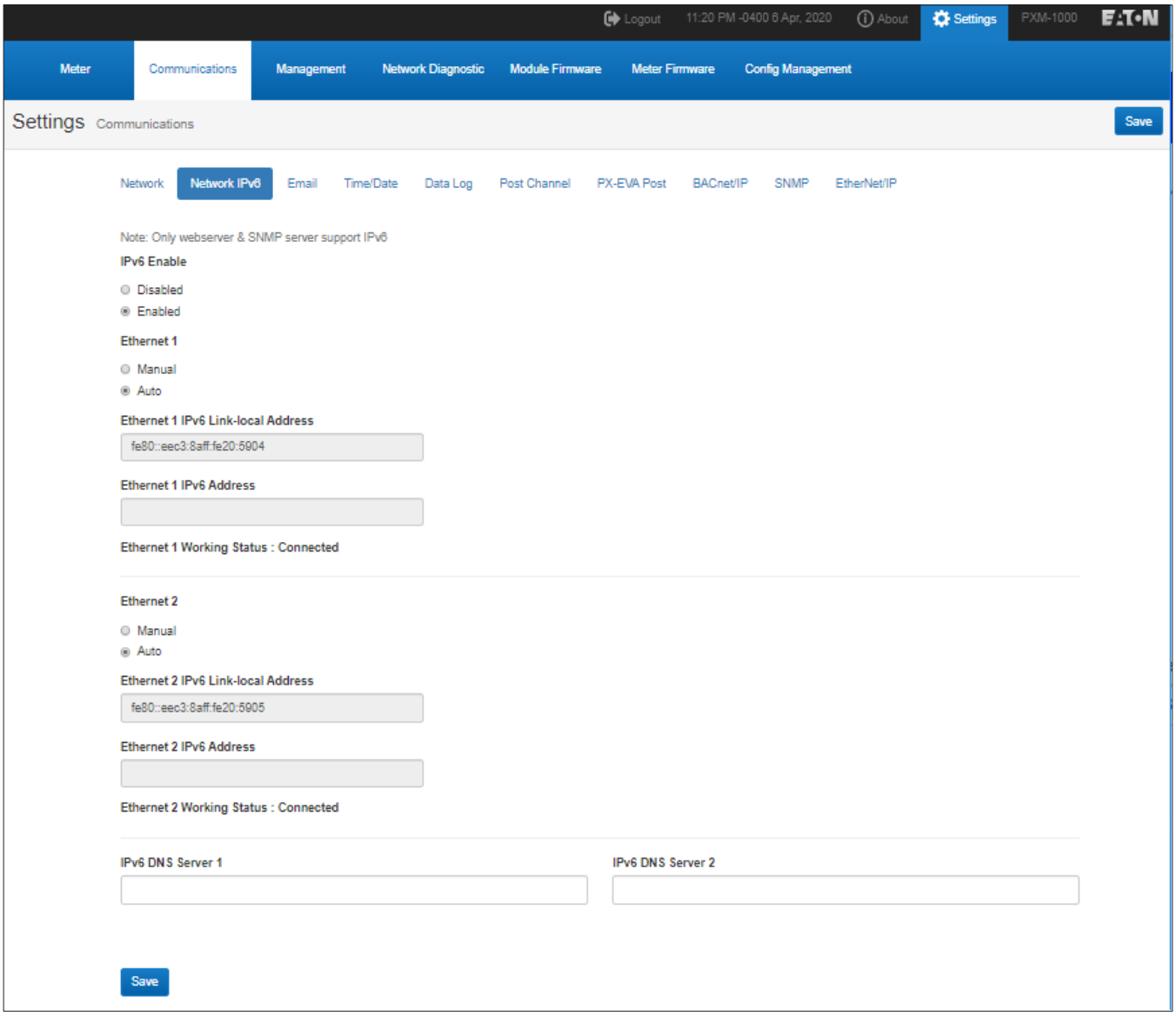
IPv6 Enable: Enable to access the settings for IPv6

Ethernet DHCP: This can be set to manual or auto.

- When set to Manual, users must configure the IPv6 address, the Subnet Prefix Length, and the Gateway.
- When set to Auto, the network will assign an IPv6 address automatically.

Ethernet Link Local Address: Is an IPv6 address that is automatically configured on the device with prefix 'fe80' followed by the MAC address of the module.

Ethernet Status: Displays whether there is an Ethernet cable connected or disconnected.



Email

The PXM1K-ETHMULTI supports the SMTP protocol where users can setup the email function to enable the meter to send emails based on a specific time interval or whenever there is an alarm or SOE event or a combination of both. The Email configuration page can be accessed by clicking on the **'Email'** tab on the settings page. Users must know their SMTP server provider and details regarding their SMTP server, which can be provided by users' IT personnel.

There are three modes available for sending emails that the user can enable.

The first mode is **'Triggered Sending'** where emails are sent immediately when there is a new alarm, SOE, or waveform event.

The second mode is **'Timed Sending'** where users can receive emails at a certain period of time based on the time interval configured. The email will include the data that is selected to be sent.

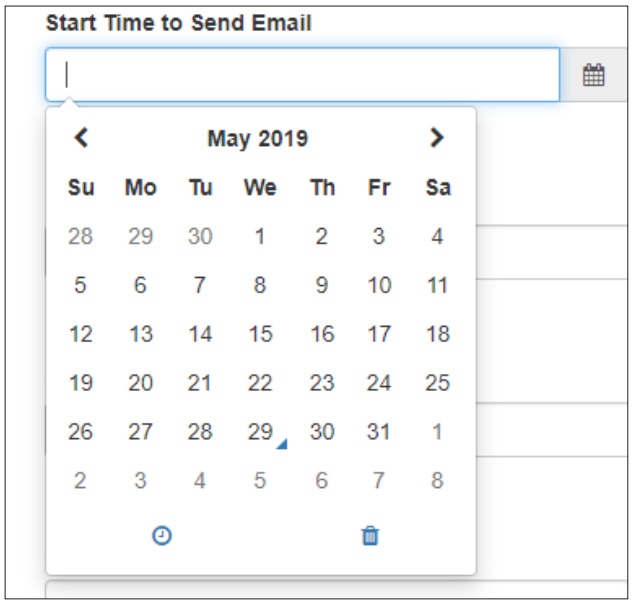
The third mode is when both of the above are enabled.

To use this function the following settings need to be configured:

SMTP Enabled: Select 'Enable' to enable and to further configure the settings related to the SMTP function.

Start Time to Send Email: Select the date and time for when the emails should begin to send.

- Click on the icon to configure the time and date.
- Click on the icon in the bottom right to clear the time and date.



SMTP Server: Enter the URL of a valid SMTP server. I.E. mail.eatoncom or smtp.gmail.com

SMTP Port: Enter the port number associated with the SMTP server.

SMTP From: Enter a name or phrase which will appear to let you know who the mail is from. I.E. 'Technical Support'

SMTP Subject: Enter a subject line for the emails

Authentication: Users can have email authentication on or off. If authentication is on users will need to provide the SMTP username and password.

- **SMTP Username:** Enter the SMTP user name for the SMTP server set above.
- **SMTP Password:** Enter the SMTP user password for the username set above.
- **TLS/SSL:** Users have the option to send emails using TLS/SSL protocols

SMTP To Address 1;2;3: Enter up to three recipients to which you wish to have the email sent.

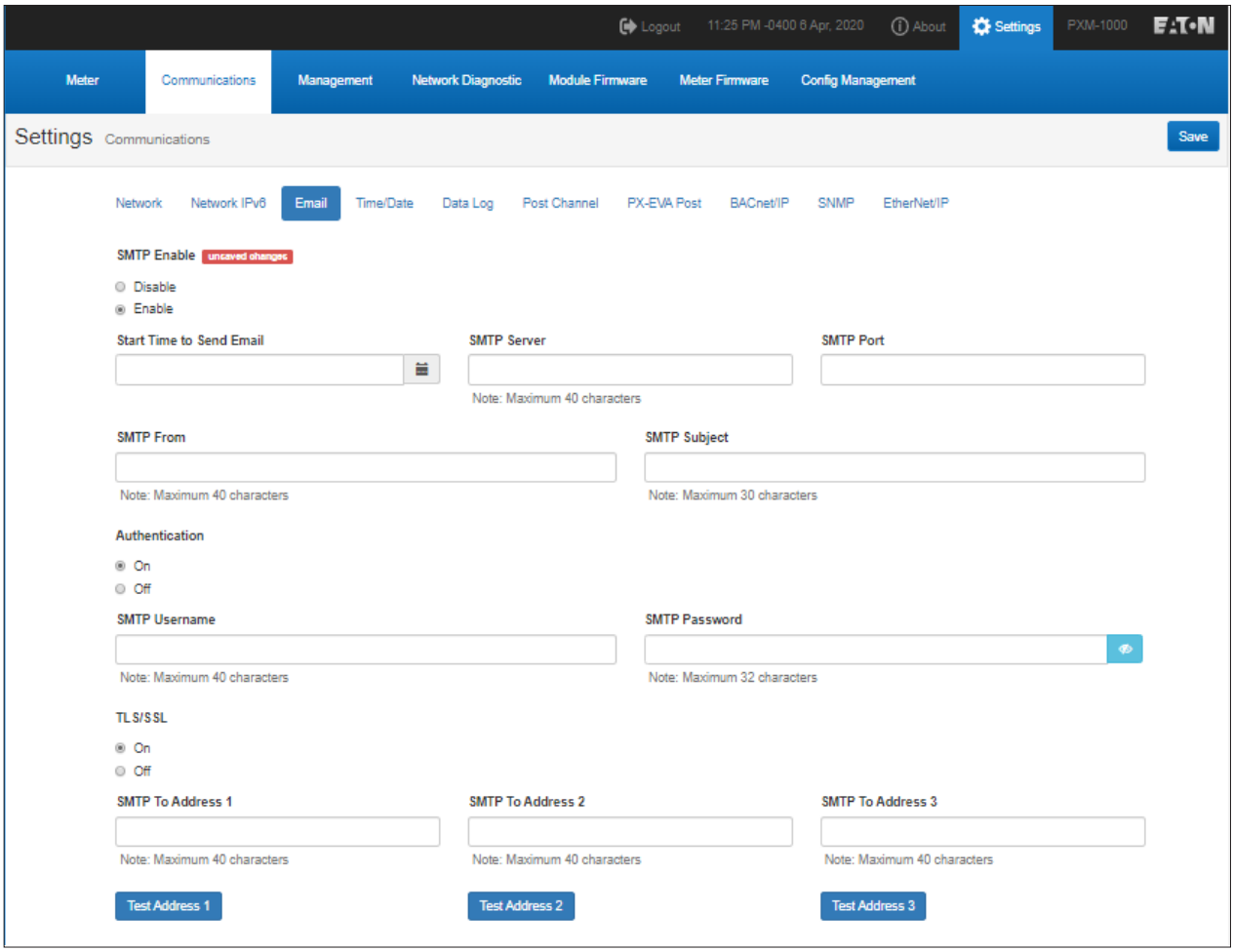


SMTP To Address 1', 'SMTP To Address 2' and 'SMTP To Address 3'.

Test Address 1,2,3: Test the if the email can be sent to 'SMTP To Address 1', 'SMTP To Address 2', 'SMTP To Address 3'.

Note: If the test address function fails, users can view the email post failure by clicking on the 'Details' option from the test post screen.

After configuring the above settings, the next step is to select the content for the emails.



The content of the emails can either be time based triggered or event based triggered.

For receiving emails on a time based under Enable Periodic Email Reporting:

Enter a time between 5-1440 mins in the Set time interval

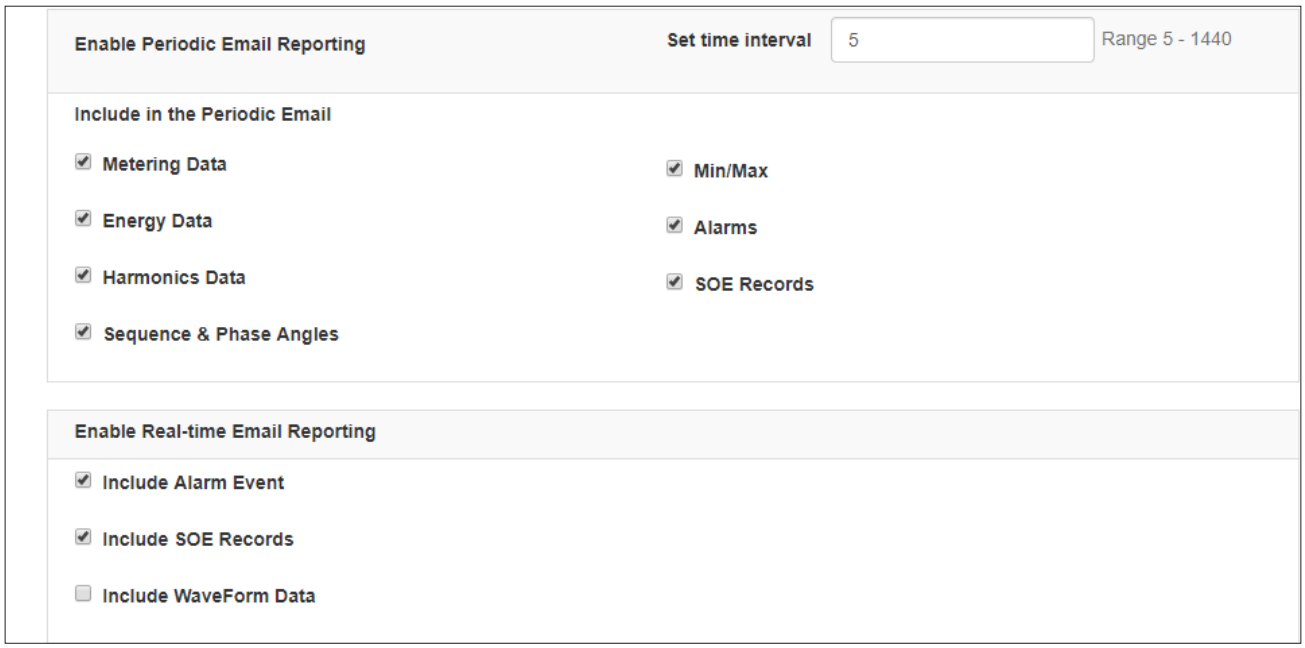
- Check off the box beside the parameters for the content the user should receive.
  - **Metering Data:** Report on Real-time voltage, current, power and etc.
  - **Energy Data:** Report on energy parameters.
  - **Harmonics Data:** Report on the voltage and current harmonics from 2nd to 63rd.
  - Sequence & Phase Angles: Report on the positive, negative and zero components of the voltage and current waveform.
  - **Min/Max:** Report on the maximum and minimum statistics that the meter has recorded since the lifetime of the meter or from the last reset of the min/max statistics.
  - **Alarm:** Report of the alarm log.
  - **SOE Record:** Report of the SOE log.
  - Waveform: Report of the waveform log.

The user will receive an email with csv file attachment.

For receiving emails on a event based select either Alarm Event, SOE Record or Waveform Data under the Enable Real-time Email Reporting.

Note: Waveform Data is only available for PXM1300 model.

The user will receive an email with csv file attachment corresponding to the triggered event selected.



Time/Date

The device clock of the PXM1000 series meter can be set through the web interface of the PXM1K-ETHMULTI module. The PXM1K-ETHMULTI module also supports the NTP (Network Time Protocol) protocol so that the module can update the meter's device clock by synchronizing with a time server.

The module can sync with up to 3 time servers. If a time server is down, the module will synchronize with the second or third time server if they are configured.

The settings for the time and date can be found by clicking on the 'Settings' and selecting the 'Communications' tab. Users can select 'Time/Date' to configure the time settings.

The following must be configured to set the time/date and NTP settings:

NTP Enabled: Select enable to further configure the settings related to the NTP (Network Time Protocol) function

Device Clock: Configure the date and time on the meter

- Click on the icon to configure the date and time.
- Click on the  icon in the bottom right to clear the time and date.

Sync Time: Click on Force Update to have the PXM1K-ETHMULTI sync its time with the NTP server

NTP Server 1;2;3: Enter up to 3 NTP servers in the "NTP Server 1", "NTP Server 2" and "NTP Server 3" fields.

Examples of North American SNTP servers are:

0.us.pool.ntp.org

1.us.pool.ntp.org

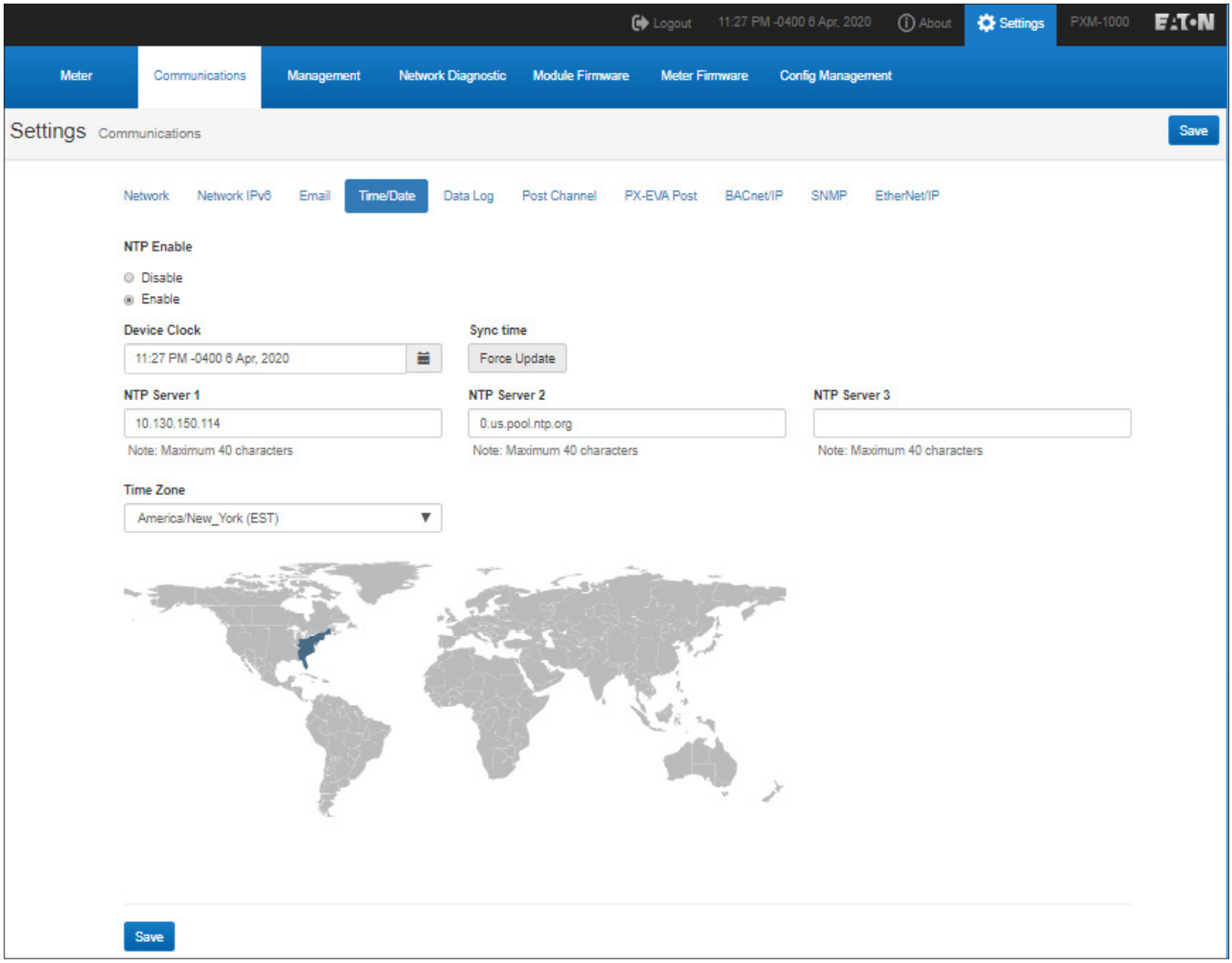
2.us.pool.ntp.org

3.us.pool.ntp.org

For more NTP servers based on region, visit the following site:
http://www.pool.ntp.org/en/

Time Zone: Select the time zone the meter is in or the time zone in which you would like the meter's time to be synchronized to from the drop down list. Users can also select the timezone by clicking on the region in the map.

Click 'Save' after configuring the time settings. Users will be prompted to reboot the PXM1K-ETHMULTI immediately or later. If later is chosen the PXM1K-ETHMULTI must be rebooted from the 'Management' page in order for the settings to take effect.



Data Log

The PXM1K-ETHMULTI supports logging data onto its on board memory.

The module supports three loggers for different parameters and requirements.

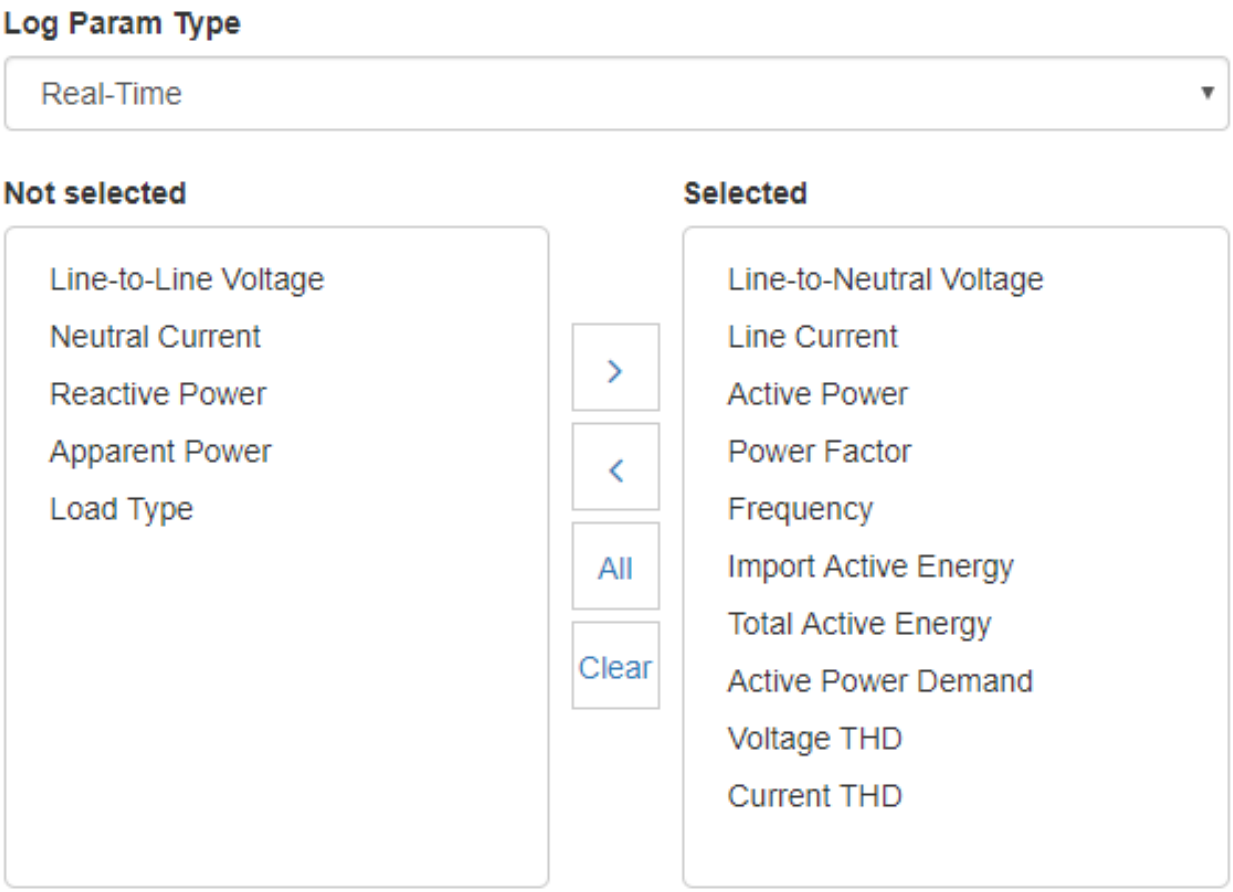
The data can be downloaded as a .csv file from the datalog page in the logs section or by using a HTTP/FTP client.

Logger Enable: To use the data log function to log the data onto the module, select 'Enable' to view and configure the settings that are applicable.

Post Channel: Select the channel to push the datalog to. Only an enabled post channel can be selected here. A post channel can be enabled in the 'Post Channel' tab on the settings page.

Log Param Type: Users can select the type of parameters they wish to log into logger.

Users can use the '>' button to add selected parameters into the data log, and use the '<' button to remove selected parameters from the data log. Users can also use the 'All' or 'Clear' buttons to add all or clear all parameters to and from the data log. The supported parameter types include real-time readings, energy readings, demand readings, power quality readings and I/O readings.



Log Param Type Detail: This setting allows users to modify what values they see in the data log. Users can select the following parameter details:

- Instantaneous
- Minimum
- Maximum
- Average

Only the 'Real-time' and 'Demand' parameters support the minimum, maximum and average parameter type details. All other parameter types such as Energy, Power Quality, and IO only support the instantaneous values displayed in the the data log.

The image below describes how these parameter details function in the data log. The image of the csv file below depicts a sample of a data log where the data is logged every 5 minutes. The first few columns are highlighted showing the Phase A Line-Neutral voltage.

- Column B highlighted in yellow shows the instantaneous value for Phase A voltage at every 5 minute interval.
- Column C highlighted in green shows the minimum value for Phase A voltage between each 5 minute interval.
- Column D highlighted in red shows the maximum value for Phase A voltage between each 5 minute interval.
- Column E highlighted in orange shows the average value for Phase A between each 5 minute interval.

A1 Time

	A	B	C	D	E	F	G	H	I	J	K	L
1	Time	V1	V1_MIN	V1_MAX	V1_AVG	V2	V2_MIN	V2_MAX	V2_AVG	V3	V3_MIN	V3_MAX
2	2019-06-20 9:55	120.067	120.067	120.077	120.0714667	120.084	120.078	120.087	120.084	120.091	120.09	120.096
3	2019-06-20 10:00	120.072	120.032	120.076	120.0728667	120.079	120.077	120.087	120.081	120.091	120.09	120.094
4	2019-06-20 10:05	120.068	120.01	120.076	120.0720667	120.084	120.08	120.086	120.083	120.093	120.091	120.094
5	2019-06-20 10:10	120.073	120.065	120.073	120.0707143	120.083	120.079	120.088	120.083	120.092	120.091	120.094
6												
7												
8												
9												
10												

Timestamp Format: Select the format of the timestamp for the data that is logged. The format for the timestamp can be based on the Local Time, UTC Seconds or based on ISO8601 Format.

Log File Name Format: Select the format of the log file name for the data that is logged. The format for the log file name can be based on the UTC timestamp or based on Time Interval Format.

Log Interval: Select how frequently the meter will log data to the file that will be uploaded to the server from the drop down list. The logging interval can be from 1 second to 1 month. The minimum time interval option is according to the selected parameter.

- The Real-time & IO's min Log Interval is 1 sec
- The Energy's min Log Interval is 15 sec
- The Demand & Power Quality's min Log Interval is 1 min

Note: If selected parameters are Real-time and I/O, the min log interval is 1 sec. If selected parameters are Real-time and Energy, the min log interval is 15 sec.

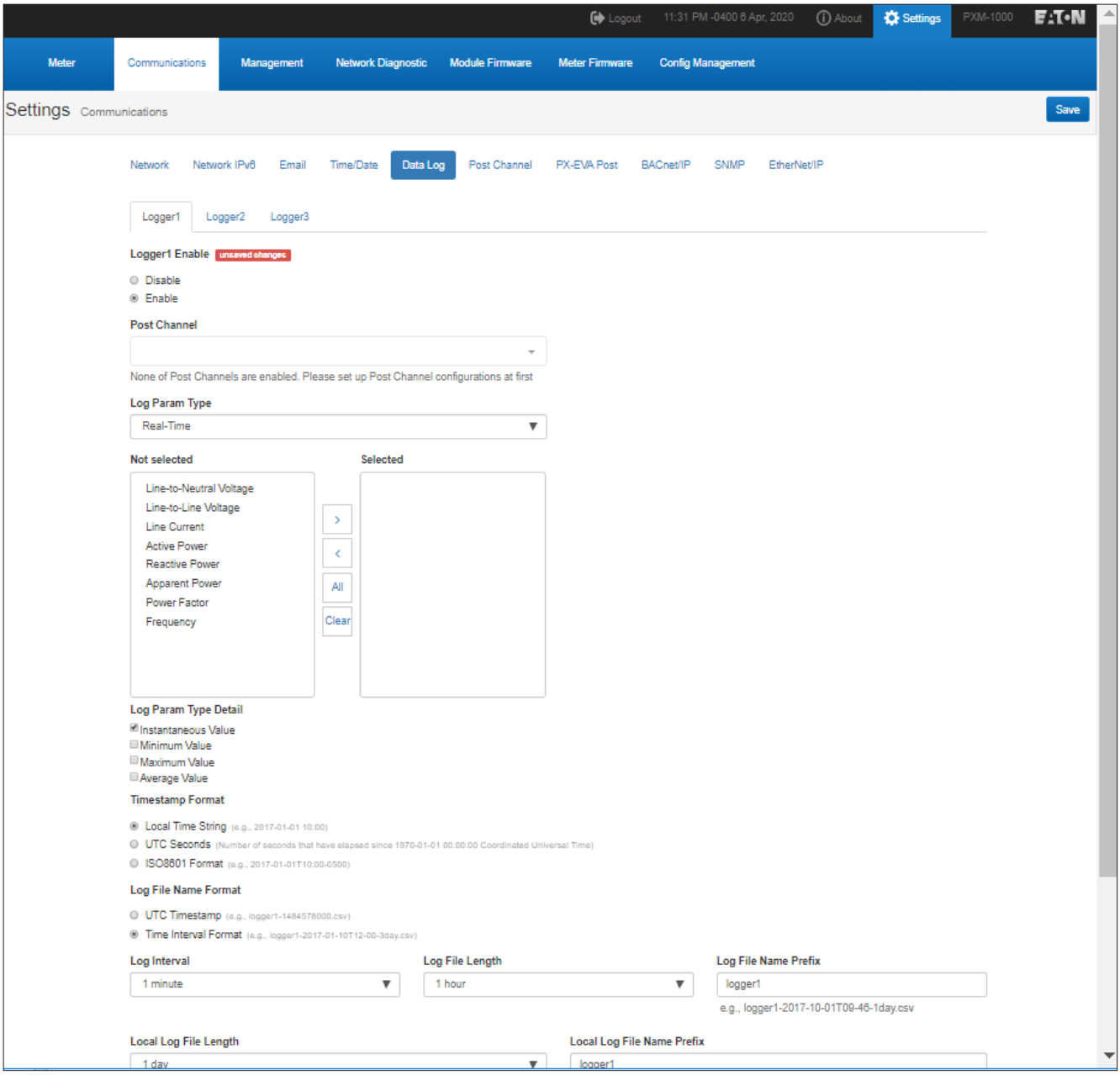
Post File Length: Select how frequently the log file will be uploaded to the server from the drop down list The log file length can be from 1 minute to 1 month.

Log File Name Prefix: Provide a name for the log file posted to post channel which will be appended to the beginning of the log file if "Time Interval Format" is selected as the Post File Name Format. By default "logger1" will be appended to the beginning of the log file.

Local Log File Length: Select the length of the local log file as 1 hour, 1 day, 7 days or 1 month of data from the drop down list.

Local Log File Name Prefix: Provide a name for the local log file which will be appended to the beginning of the log file if "Time Interval Format" is selected as the Post File Name Format. By default "logger1" will be appended to the beginning of the log file.

Note: The Post File Length and Local Log File Length must be less than or equal to the log interval selected.


Logout
11:31 PM -0400 6 Apr, 2020
About
Settings
PXM-1000
Meter
Communications
Management
Network Diagnostic
Module Firmware
Meter Firmware
Config Management
Settings Communications
Save
Network
Network IPv6
Email
Time/Date
Data Log
Post Channel
PX-EVA Post
BACnet/IP
SNMP
EtherNet/IP
Logger1
Logger2
Logger3
Logger1 Enable
unsaved changes
Disable
Enable
Post Channel
None of Post Channels are enabled. Please set up Post Channel configurations at first
Log Param Type
Real-Time
Not selected
Selected
Line-to-Neutral Voltage
Line-to-Line Voltage
Line Current
Active Power
Reactive Power
Apparent Power
Power Factor
Frequency
All
Clear
Log Param Type Detail
Instantaneous Value
Minimum Value
Maximum Value
Average Value
Timestamp Format
Local Time String (e.g., 2017-01-01 10:00)
UTC Seconds (Number of seconds that have elapsed since 1970-01-01 00:00:00 Coordinated Universal Time)
ISO8601 Format (e.g., 2017-01-01T10:00-0500)
Log File Name Format
UTC Timestamp (e.g., logger1-1484578000.csv)
Time Interval Format (e.g., logger1-2017-01-10T12-00-3day.csv)
Log Interval
1 minute
Log File Length
1 hour
Log File Name Prefix
logger1
e.g., logger1-2017-10-01T09-46-1day.csv
Local Log File Length
1 day
Local Log File Name Prefix
logger1

SFTP Enable: To download the logged data from the module using a FTP client, select Enable. The log file will then be available to be downloaded using a FTP client using the following credentials:

Host: *sftp://IPaddressofthemeter*

Username: *sftpuser*

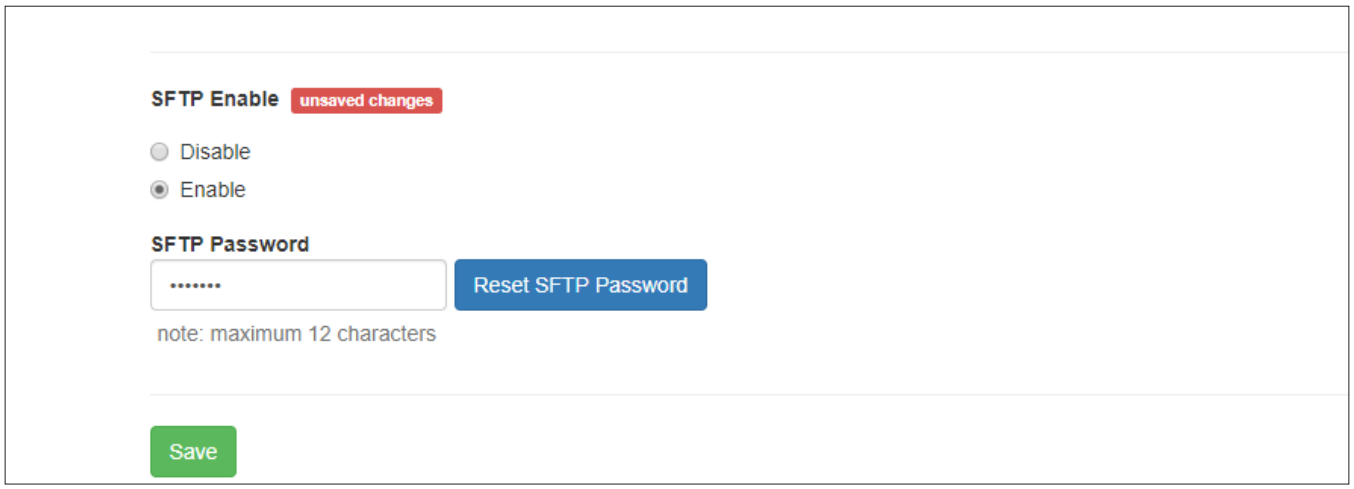
SFTP Password: *pxm1k-ethmulti*

Port: 22

By default the password for retrieving the backup log files is **pxm1k-ethmulti**. The user can configure any password or can reset to the default of pxm1k-ethmulti by clicking on the "Reset SFTP Password".

Note: Affter enabling the SFTP function the user must reboot the communication module in order to access the data logs with the default password of 'pxm1k-ethmulti'.

Click 'Save' after changing any settings. Users will be prompted to reboot the PXM1K-ETHMULTI immediately or later. If later is chosen the PXM1K-ETHMULTI must be rebooted from the 'Management' page in order for the changes to take effect.



Post Channel

The PXM1K-ETHMULTI supports the HTTP and FTP Post functions to send data from the meter to a HTTP/FTP server. The PXM1K-ETHMULTI can post .csv files to three different HTTP or FTP servers using HTTP Post or FTP Post.

In the case when there is no connection to the server, the PXM1K-ETHMULTI will store the posts and send it out after the connection is restored. A maximum of 3000 files will be buffered on module. The Clear Post Channel Logs button will allow users to clear the buffered files on meter.

The PXM1K-ETHMULTI can post data to a server at intervals of time ranging from 1 minute to 1 month.

The settings for configuring the post channels to post the data can be found by clicking on 'Settings' and then selecting 'Communications' tab. Click "Post Channels" to configure any of the three post channels.

Post Channel 1 Enable: Enable the Post Channel 1 in order to configure the settings needed to post data via the HTTP(S)/FTP post functions

Post Method: Select the method for posting the files, the user can choose HTTP/HTTPS or FTP

Post Name Fixed: This configuration needs to be enabled in order for user to control the name of the file that will be posted. Otherwise file name will be based on the Log File Name Format configuration from the Data Log settings

Post File Name: Users can enter a name for the file that will be posted as if 'Post Name Fixed' is enabled

If the HTTP/HTTPs post method is selected:

HTTP/HTTPS URL: Enter the URL for the HTTP/HTTPS server. The URL needs to begin with the prefix http:// or (https://)

HTTP/HTTPS Port: Enter the port number the server will be listening on

HTTP/HTTPS Meter ID: Enter a name or description for the meter to be identified on the server

If the FTP post method is selected:

FTP URL: Enter the URL for the FTP server. The URL needs to begin with the prefix ftp://

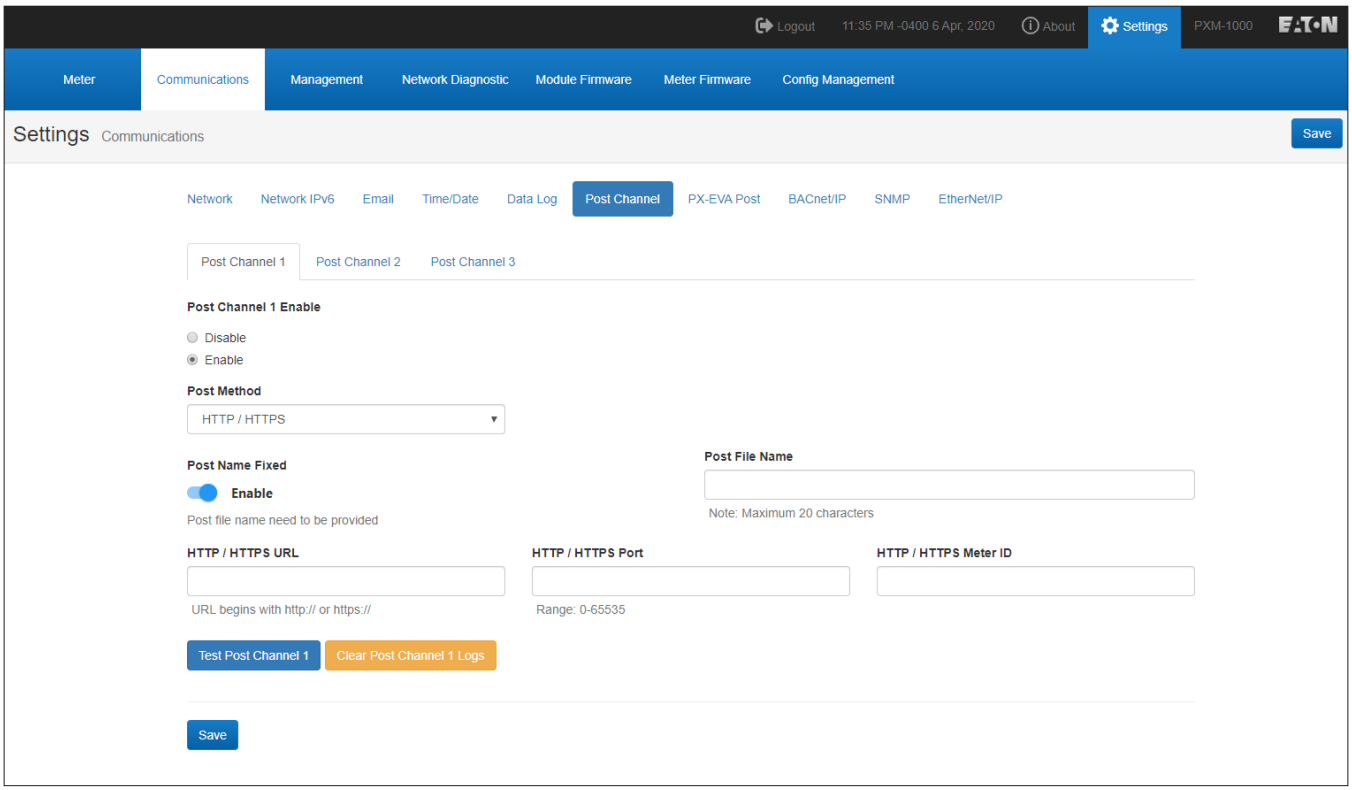
FTP Port: Enter the port number the server will be listening on

FTP Username: Enter the username required to log into the FTP server

FTP Password: Enter the password required to log into the FTP server

Note: The 'TEST Post Channel' button should only be utilized after clicking the 'Save' button otherwise a fail response will be observed. If a fail response occurs after clicking 'Save' confirm the network settings or credentials for the server.

Click 'Save' after changing any settings. Users will be prompted to reboot the PXM1K-ETHMULTI immediately or later. If later is chosen the PXM1K-ETHMULTI must be rebooted from the 'Management' page in order for the settings to take effect.



Select the Post Channel 2 tab to configure the settings for post to a second server.

Post Channel 2 Enable: Enable the Post Channel 2 in order to configure the settings needed to post data via the HTTP(S)/FTP post functions

Post Method: Select the method for posting the files, the user can choose HTTP/HTTPS or FTP

Post Name Fixed: This configuration needs to be enabled in order for user to control the name of the file that will be posted. Otherwise file name will be based on the Log File Name Format configuration from the Data Log settings

Post File Name: User can enter a name for the file that will be posted as if Post Name Fixed is enabled

If the HTTP/HTTPs post method is selected:

HTTP/HTTPS URL: Enter the URL for the HTTP/HTTPS server. The URL needs to begin with the prefix http:// or (https://)

HTTP/HTTPS Port: Enter the port number the server will be listening on

HTTP/HTTPS Meter ID: Enter a name or description for the meter to be identified on the server

If the FTP post method is selected:

FTP URL: Enter the URL for the FTP server. The URL needs to begin with the prefix ftp://

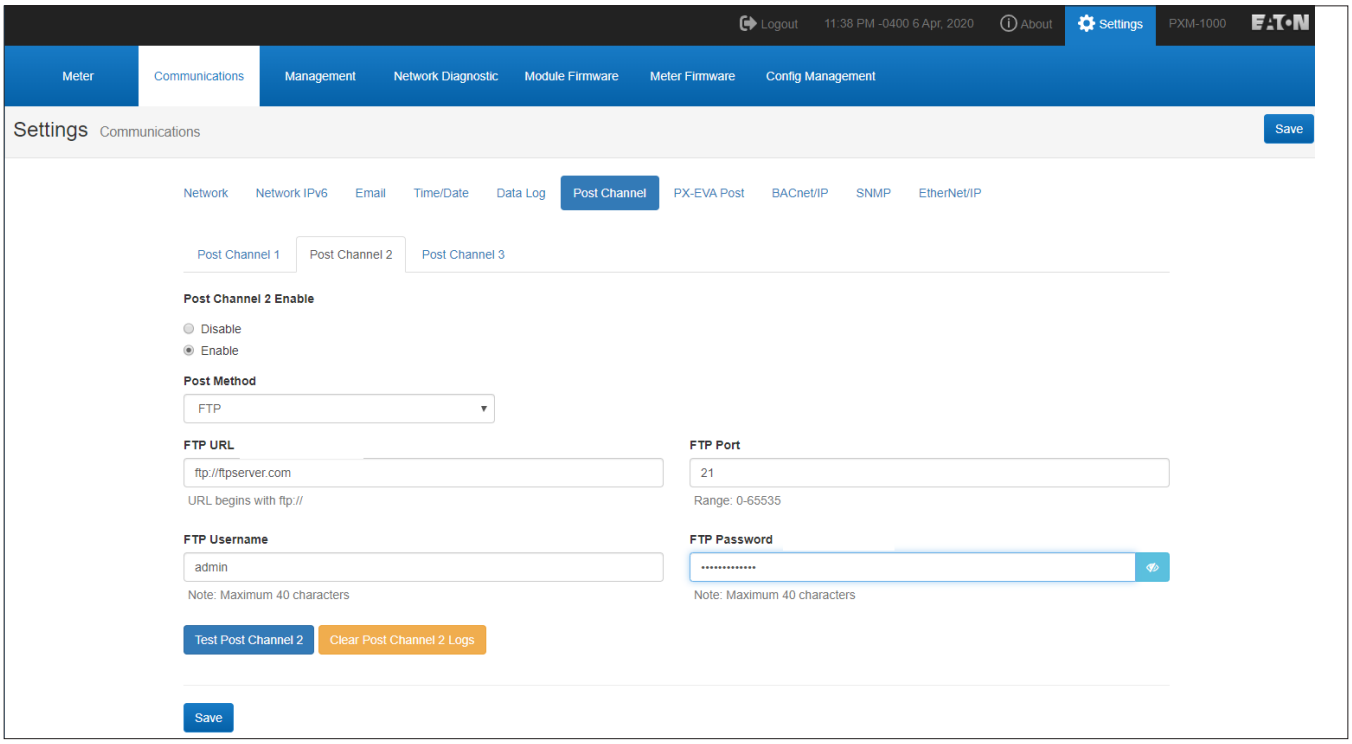
FTP Port: Enter the port number the server will be listening on

FTP Username: Enter the username required to log into the FTP server

FTP Password: Enter the password required to log into the FTP server

Note: The "TEST Post Channel 2" button should only be utilized after clicking the 'Save' button otherwise a fail response will be observed. If the test post fails users can view the test post details by clicking on the 'Details' option from the test post screen.

Click 'Save' after changing any settings. Users will be prompted to reboot the PXM1K-ETHMULTI immediately or later. If later is chosen the PXM1K-ETHMULTI must be rebooted from the 'Management' page in order for the settings to take effect.



Select the Post Channel 3 tab to configure the settings for post to a second server.

Post Channel 3 Enable: Enable the Post Channel 3 in order to configure the settings needed to post data via the HTTP(S)/FTP post functions

Post Method: Select the method for posting the files, the user can choose HTTP/HTTPS or FTP

Post Name Fixed: This configuration needs to be enabled in order for user to control the name of the file that will be posted. Otherwise file name will be based on the Log File Name Format configuration from the Data Log settings

Post File Name: User can enter a name for the file that will be posted as if Post Name Fixed is enabled

If the HTTP/HTTPs post method is selected:

HTTP/HTTPS URL: Enter the URL for the HTTP/HTTPS server. The URL needs to begin with the prefix http:// or (https://)

HTTP/HTTPS Port: Enter the port number the server will be listening on

HTTP/HTTPS Meter ID: Enter a name or description for the meter to be identified on the server

If the FTP post method is selected:

FTP URL: Enter the URL for the FTP server. The URL needs to begin with the prefix ftp://

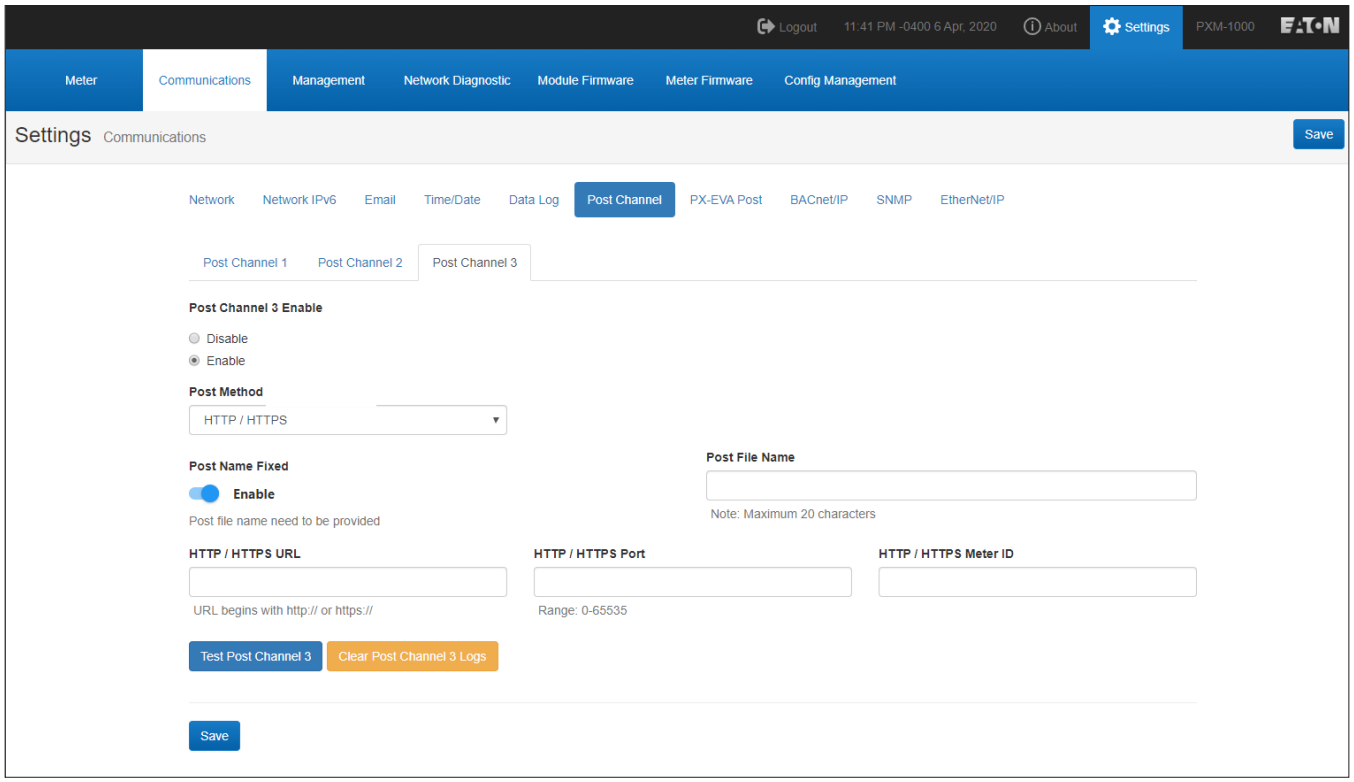
FTP Port: Enter the port number the server will be listening on

FTP Username: Enter the username required to log into the FTP server

FTP Password: Enter the password required to log into the FTP server

Note: NOTE: The "TEST Post Channel 3" button should only be utilized after clicking the 'Save' button otherwise a fail response will be observed. If the test post fails users can view the test post details by clicking on the 'Details' option from the test post screen.

Click 'Save' after changing any settings. Users will be prompted to reboot the PXM1K-ETHMULTI immediately or later. If later is chosen the PXM1K-ETHMULTI must be rebooted from the 'Management' page in order for the settings to take effect.



Waveform Post

The PXM1K-ETHMULTI supports the HTTP and FTP Post functions to send the meters waveform data from the meter to a HTTP/FTP server. The PXM1K-ETHMULTI can post the Comtrade (.cfg and .dat) files to either an HTTP or FTP servers using the Waveform Post. The settings for the Waveform Post can be found by clicking on the 'Settings' tab and selecting 'Communications', from the Communications page click on the 'Waveform Post' tab.

Note: This function is only available on PXM1300 models which support the Waveform Capture Function, all other models will not have this feature available.

Waveform Post Enable: Select Enable to enable the waveform post and configure the settings further.

Scan Interval: Users can configure a scan interval where the PXM1K-ETHMULTI module scans at the PXM1300 meter to check whether there are any power quality events that occurred during the selected scan interval. For example if the scan interval is set for 15 minutes the module will scan for all the power quality events that occurred within 15 minutes and post it to the server.

The scan interval ranges from 15 seconds to 1 month.

File Name Prefix: Users can configure the file name prefix for the Comtrade file that is sent to the server.

Post Method: From the drop down menu select either FTP or HTTP/HTTPs

For FTP configure the following:

FTP URL: Enter the URL for the FTP server. The URL needs to begin with the prefix ftp://

FTP Port: Enter the port number the server will be listening on

FTP Username: Enter the username required to log into the FTP server

FTP Password: Enter the password required to log into the FTP server

For HTTP/HTTPs configure the following:

HTTP/HTTPS URL: Enter the URL for the HTTP/HTTPS server. The URL needs to begin with the prefix http:// or (https://)

HTTP/HTTPS Port: Enter the port number the server will be listening on

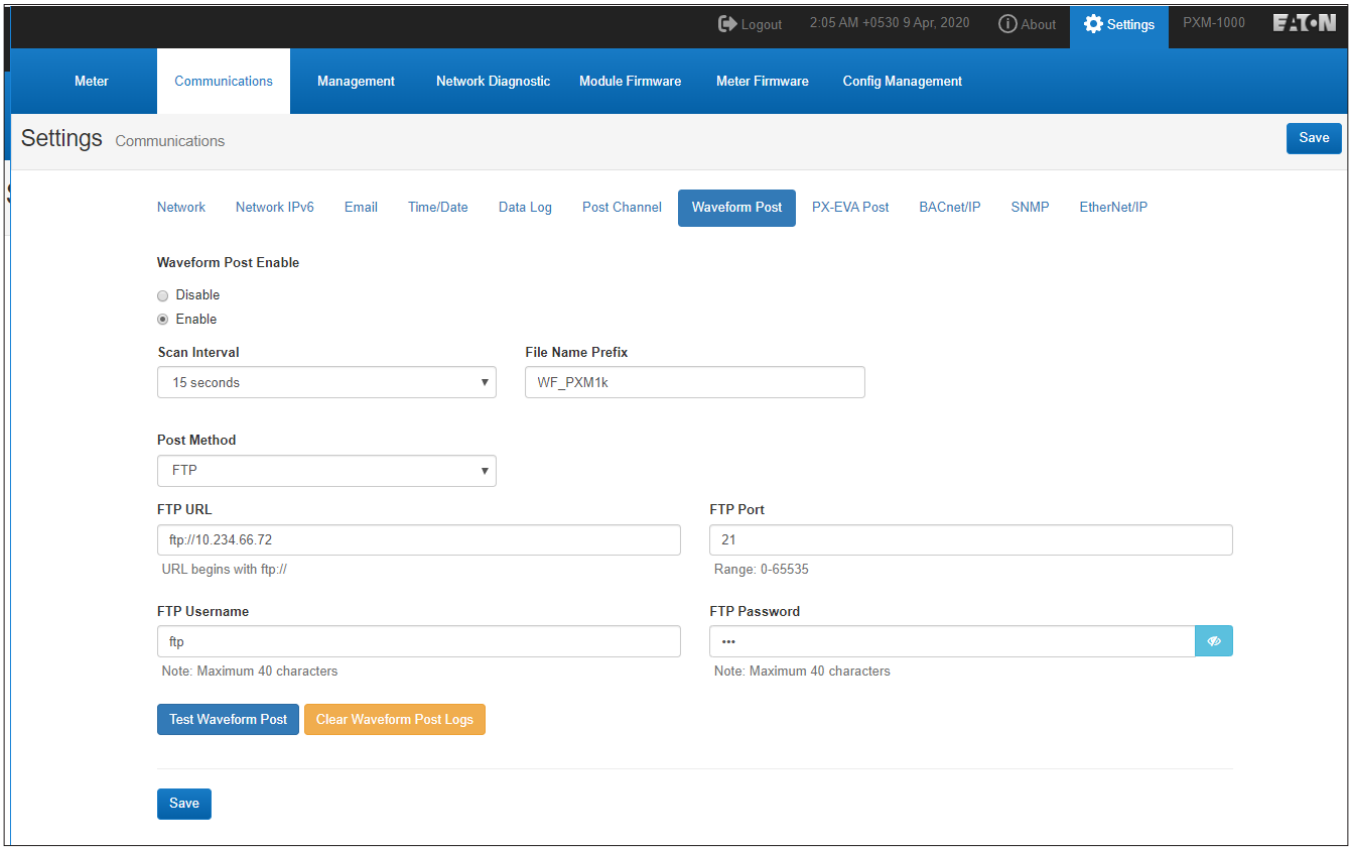
HTTP/HTTPS Meter ID: Enter a name or description for the meter to be identified on the server

Note: The 'Test Waveform Post' button should only be utilized after clicking the 'Save' button otherwise a fail response will be observed. If the test post fails users can view the test post details by clicking on the 'Details' option from the test post screen.

Similar to the Post Channel function discussed earlier, in the case when there is no connection to the server the PXM1K-ETHMULTI will store the posts and send it out after the connection is restored. A maximum of 3000 files will be buffered on module. The 'Clear Waveform Post Channel Logs' button will allow users to clear the buffered waveform files on meter.

There is no interval setting for sending the waveform data using the waveform post, the data will post directly to the FTP/HTTP server when a power quality event has occurred.

Note: All waveform capture settings must be configured on the PXMview software.



PX-EVA

The PXM1K-ETHMULTI module can directly interface with the Eaton Cloud software PX-EVA. The PXM1K-ETHMULTI will post data to the cloud software every five minutes.

PX-EVA will require the serial number of the PXM1K-ETHMULTI module which will then provide a token that will be used to configure the PXM1K-ETHMULTI so it can send its data to PX-EVA.

The settings for the PX-EVA post function can be found by clicking on the 'Settings' tab and selecting 'Communications'. Select '**PX-EVA'** to access the settings to configure the PXM1K-ETHMULTI to send data to the cloud.

PX-EVA Enable: Select 'Enable' to enable the function and to further configure the settings related to PX-EVA.

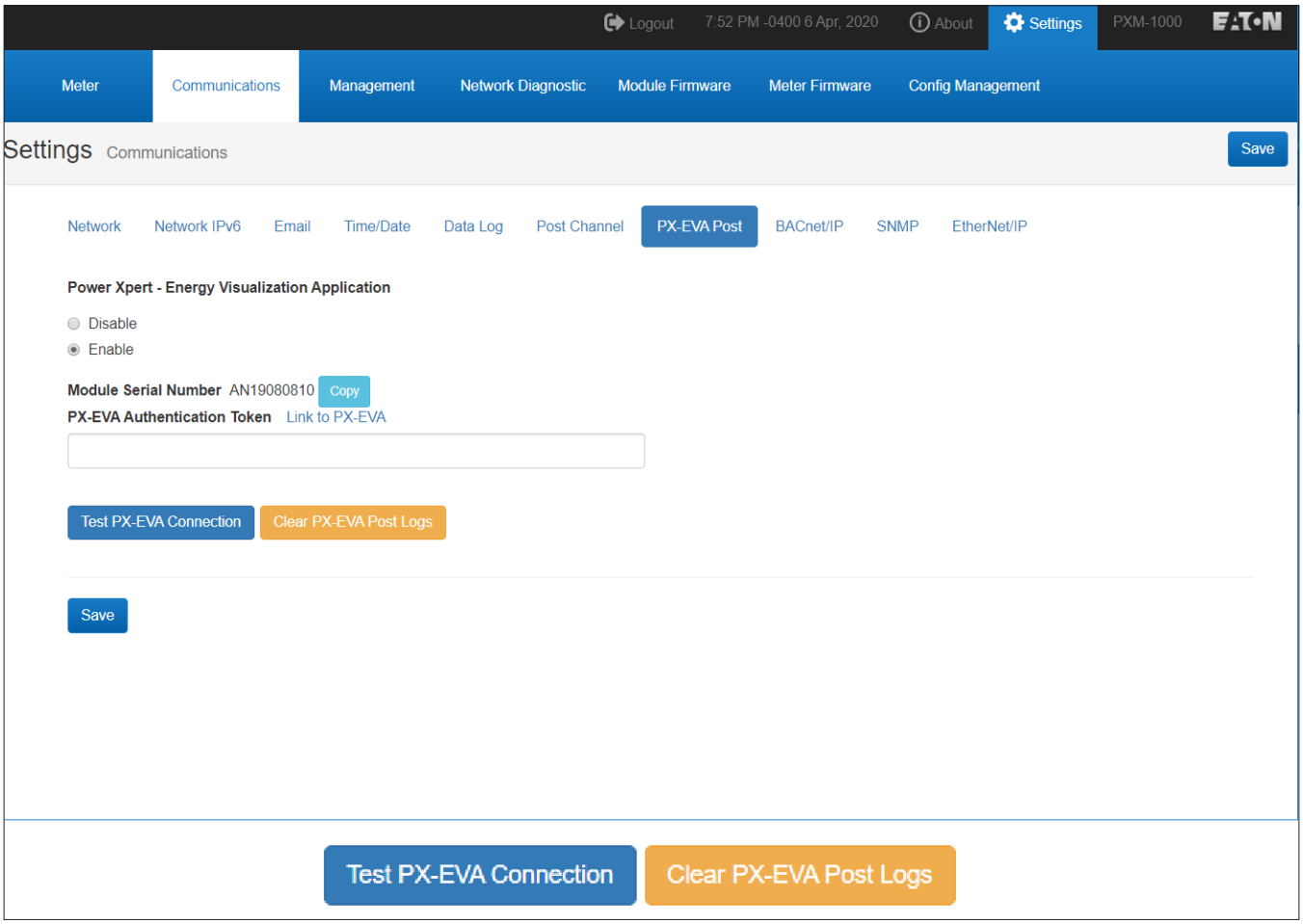
PX-EVA Token: Copy and paste the token provided by the PX-EVA software into this field.

Note: The "TEST PX-EVA" button should only be utilized after clicking the 'Save' button otherwise a fail response will be observed. If a fail response occurs after clicking 'Save', please double check the serial number entered in PX-EVA, the token pasted in the web page as well viewing the test post details by clicking on the 'Details' option.

Users can use the 'Link to PX-EVA' to access the cloud software and configure the required settings on that platform. Users must have sufficient access to add devices on their account in order to correctly configure the meter on the software. For inquiries on creating your PX-EVA account please contact MRSupport@eaton.com.

Click 'Save' after changing any settings. Users will be prompted to reboot the PXM1K-ETHMULTI immediately or later. If later is chosen the PXM1K-ETHMULTI must be rebooted from the 'Management' page for the settings to take effect.

The PXM1K-ETHMULTI will post the data continuously every 5 minutes after the reboot.



BACnet/IP

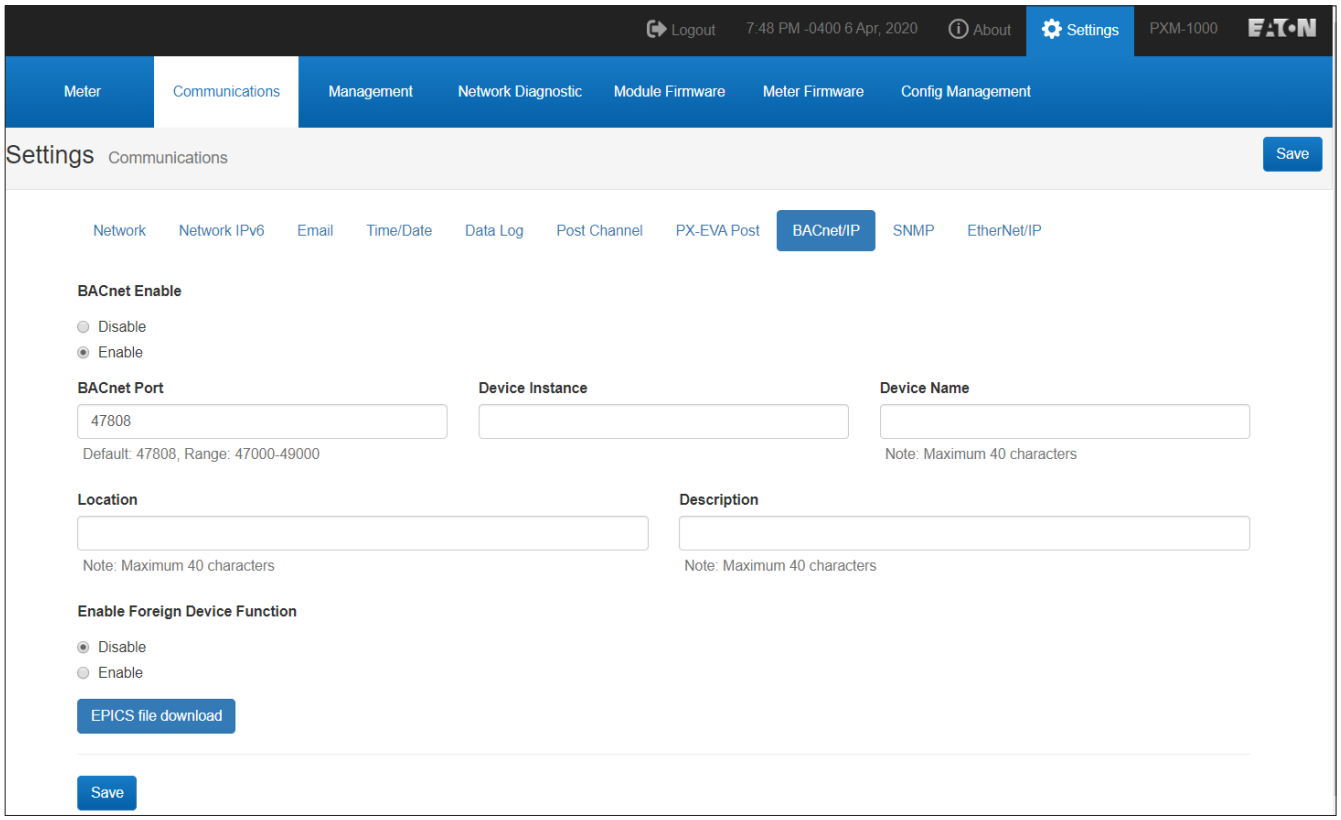
The PXM1K-ETHMULTI module supports the BACnet/IP protocol. The settings for the BACnet/IP protocol can be found on the web by clicking on the 'Settings' tab and selecting 'Communications'. Once on the communications page select "BACnet/IP" to access the settings to configure the PXM1K-ETHMULTI to communicate with a BACnet client.

BacNet Enabled: Select Enable to enable the BACnet protocol.

BACnet Port: Enter the BACnet or UDP port number. Default port is 47808.

Device Instance: Enter the instance number for the device in the BACnet system. It must be unique within the system.

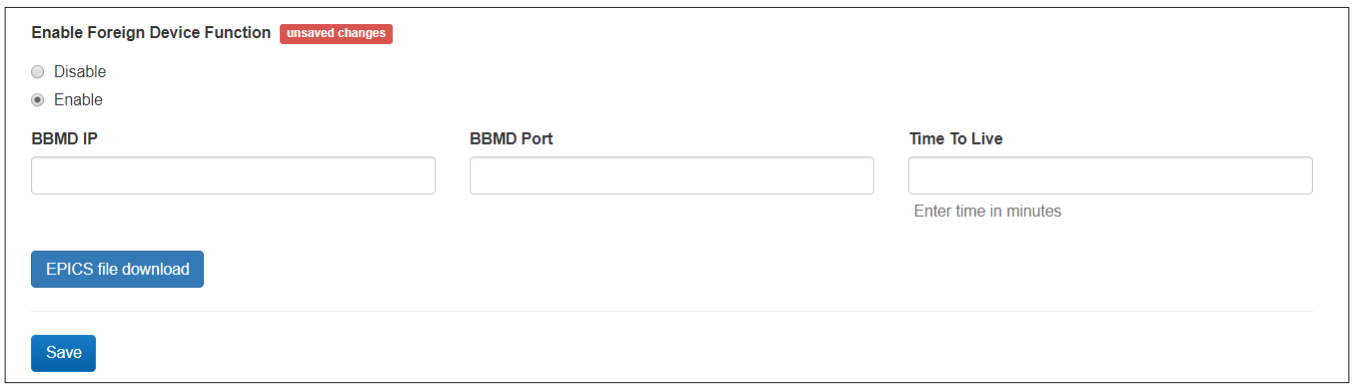
Device Name: Enter a name for the device to distinguish it from other devices within the network.



Under the "Enable Foreign Device Function", select 'Enable' to communicate with a BACnet device from another subnet.

- Enter the IP of the BACnet Broadcast Management Device(BBMD) under the 'BBMD IP' field for the device which will receive broadcast messages on one subnet and forward them to another subnet.
- Enter BACnet Port of the BBMD in "BBMD Port"
- Enter a value between 5-1440 min in the "Time To Live" for how often the foreign device will register in the BBMD's foreign device table.

Click 'Save' after changing any settings. Users will be prompted to reboot the PXM1K-ETHMULTI immediately or later. If later is chosen the PXM1K-ETHMULTI must be rebooted from the 'Management' page in order for the settings to take effect.



SNMP

The PXM1K-ETHMULTI module supports the Simple Network Management Protocol(SNMP) protocol for reporting the metering data to the management station. The PXM1K-ETHMULTI uses a public community string for read-only access. By default the module will communicate using SNMP port 161. The PXM1K-ETHMULTI also supports 'traps' to send unsolicited messages to up to four management stations.

The settings for the SNMP protocol can be found by clicking on the 'Settings' tab and selecting 'Communications'. From the communications page select the 'SNMP' tab to access the settings to configure the PXM1K-ETHMULTI for communication with a SNMP management station.

SNMP Enable: Select 'Enable' to enable the function and to further configure the settings related to the SNMP protocol.

SNMP Port: By default the SNMP Port is configured to 161. The SNMP Port can be any value from ranging from 16100 to 16199.

Read Only Community: By default the community string is Public, this configuration is similar to a password which allows only authorized users to access the meters data.

Trap Enable: Select 'Enable' so that the meter will send a message to the management station when an event is triggered. The event could be a change in Digital Input Status. The notification can then be sent to up to 4 stations.

Trap Target 1: Enter the IP address and port number of station number 1 that should be notified when there is an event.

Trap Target 2: Enter the IP address and port number of station number 2 that should be notified when there is an event.

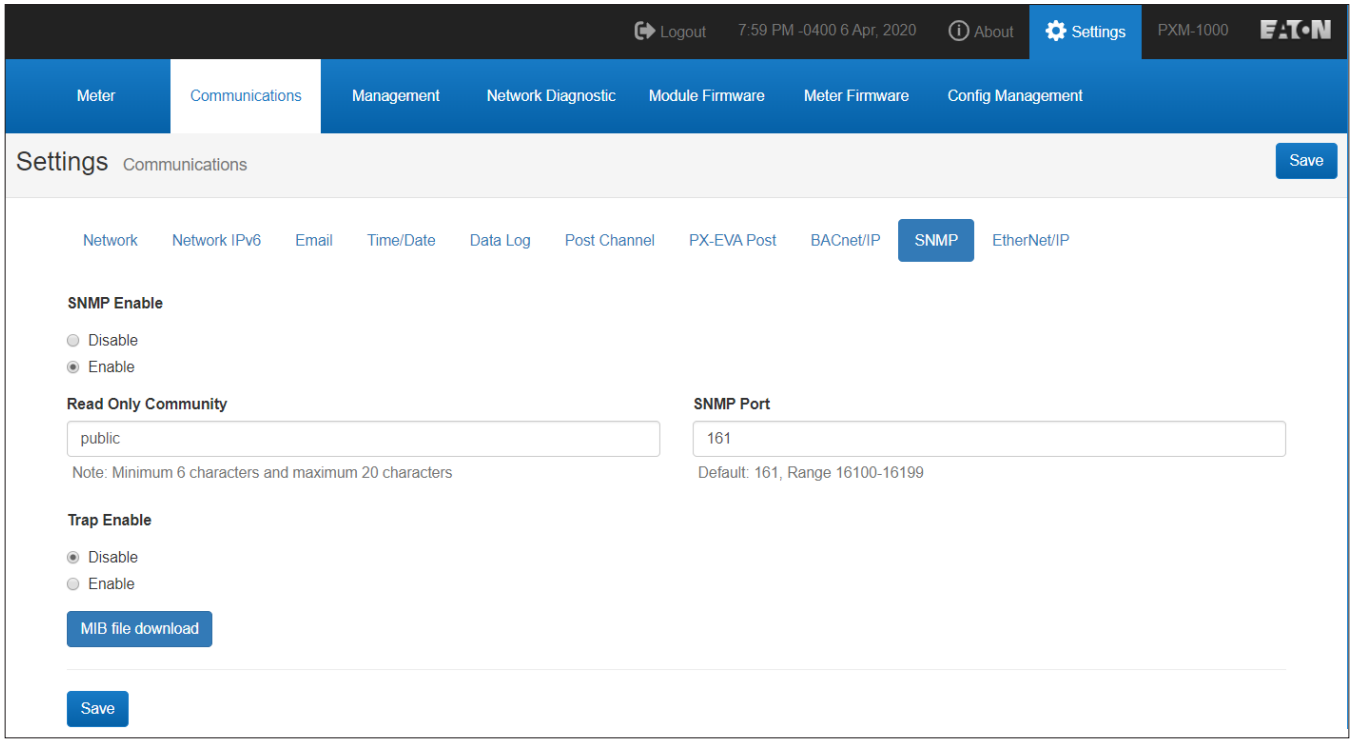
Trap Target 3: Enter the IP address and port number of station number 3 that should be notified when there is an event.

Trap Target 4: Enter the IP address and port number of station number 4 that should be notified when there is an event.

Report Buffer Size: Enter the size of the buffer for the amount of notifications will be stored before being sent to the management station. A maximum of 30 notifications can be stored.

Report Hold Time: Enter the time in seconds for how long the notification will be in queue before it gets sent to the management station. By default, this setting is configured to 0 so the notification will be sent immediately after an event occurs. This setting could be configured from 0-30 seconds.

Click 'Save' after changing any settings. Users will be prompted to reboot the PXM1K-ETHMULTI immediately or later. If later is chosen the PXM1K-ETHMULTI must be rebooted from the 'Management' page in order for the settings to take effect.



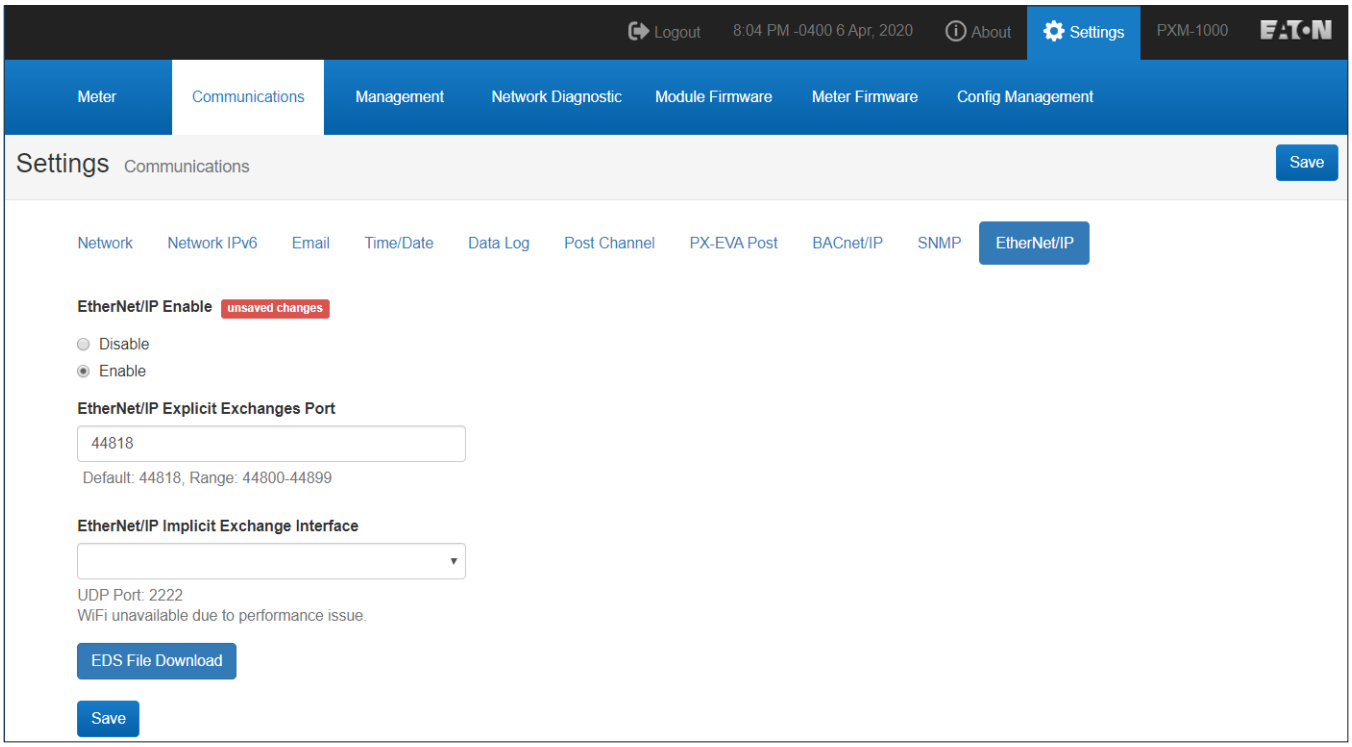
EtherNet/IP

The PXM1K-ETHMULTI module supports the Ethernet/IP protocol which is an industrial based network protocol that uses standard Ethernet and TCP/IP technology.

EtherNet/IP Enable: Select Enable to enable the EtherNet/IP protocol

EtherNet/IP Explicit Exchanges Port: Users can configure the EtherNet/IP port, the default port is 44818 and the port number ranges from 44800-44899.

EtherNet/IP Implicit Exchange Interface: Users can select Ethernet 1 or Ethernet 2 for communication.



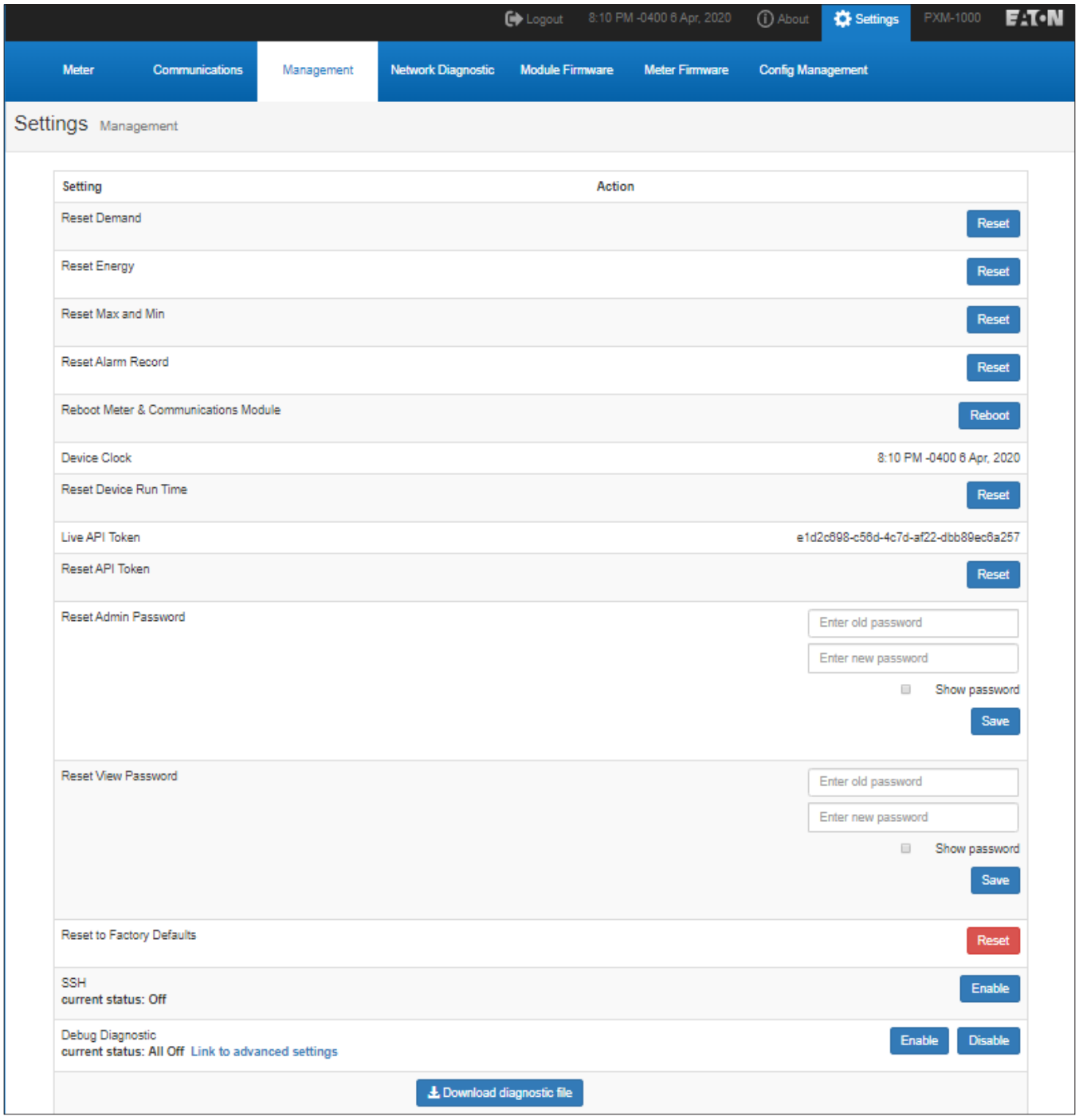
10.6.3 Management

Parameter Reset

The Management web page allows the user to clear and reset certain parameters in the meter. The following parameters can be reset from the Management page:

- Demand
- Energy
- Max and Min
- Alarm Record
- Device Run Time

10. Web Interface Readings and Parameter Settings



Reboot Meter & Communications Module

Users can also reboot the web module and meter which is required after any communication or meter setting is changed, if a module reboot is not performed the settings will not be saved to the meter and will go back to its default settings. This not only resets the communication module, it also performs a soft reboot on the PXM1000 meter.



Change Password

The access level passwords can be changed from the Management page as well, all new passwords must be 6 characters or more.



Reset to Factory

The PXM1K-ETHMULTI supports a reset to factory function where if reset the module settings would be configured back to its default factory settings. This impacts all configurations and logs stored on the module.

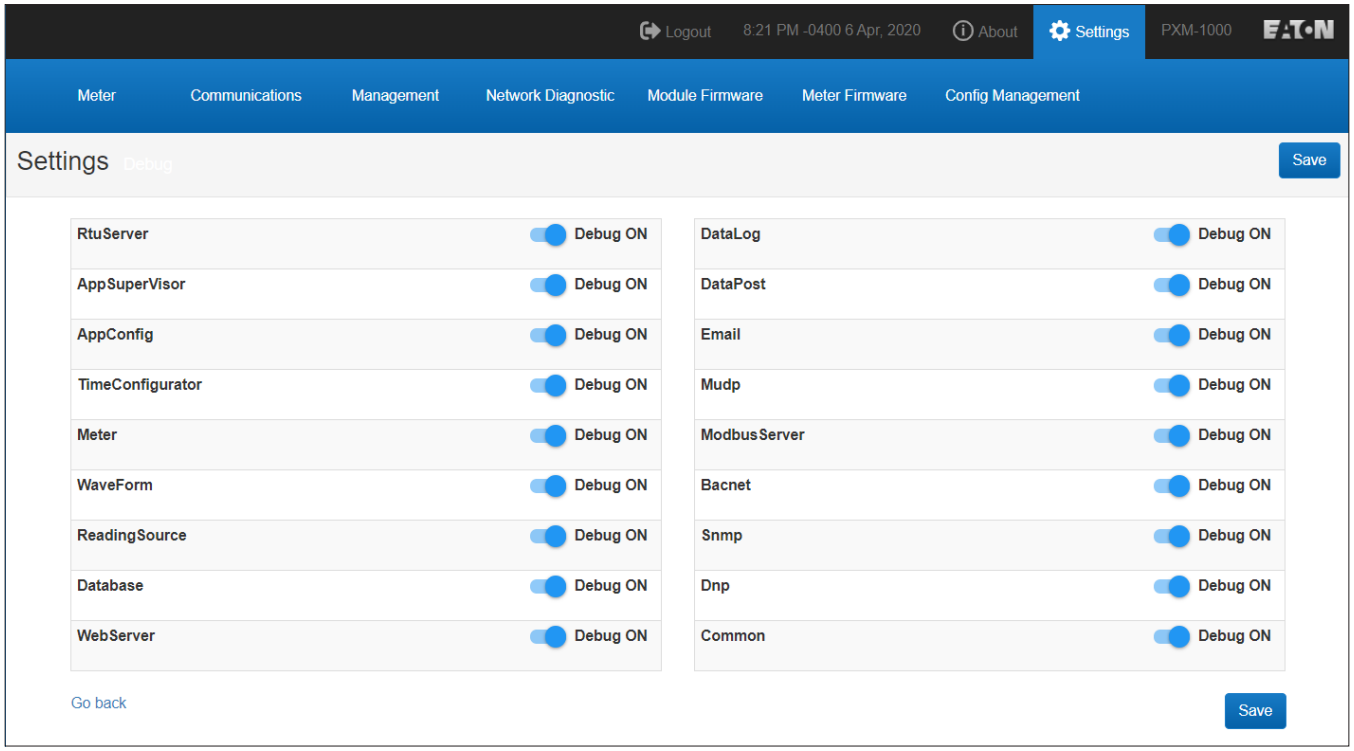
Note: This setting is permanent and cannot be undone!

Debug Diagnostic

The debug diagnostic allows the user to enable or disable the debug logs. The current status will say 'All off' if disabled, 'All On' if enabled.



Users can click on the advanced link, to turn on or off specific debug logs. If certain debug logs are enabled the current status will show 'Partial On'.

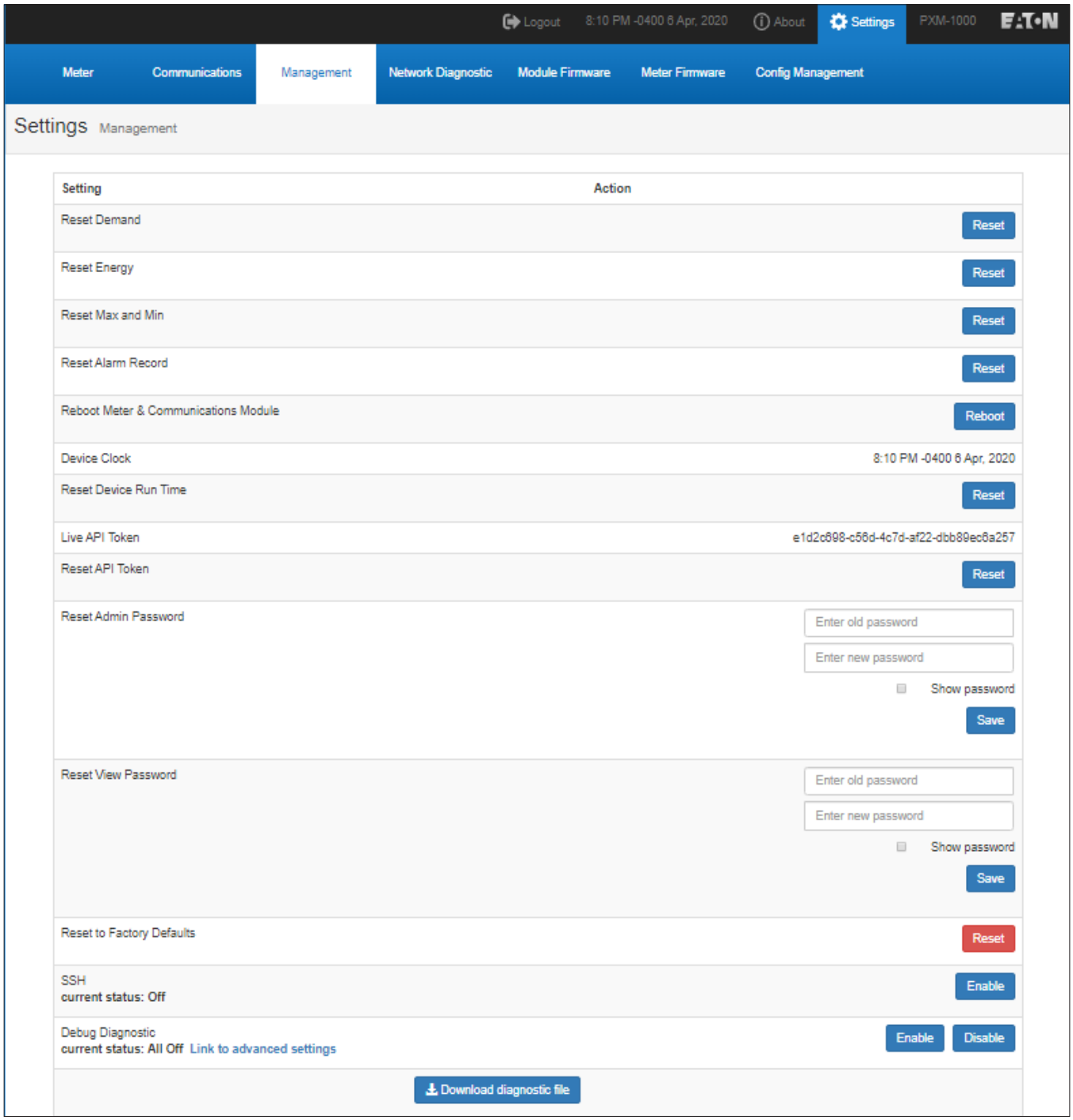


Note: The system performance may be affected by enabling the debug logs.

Diagnostic File

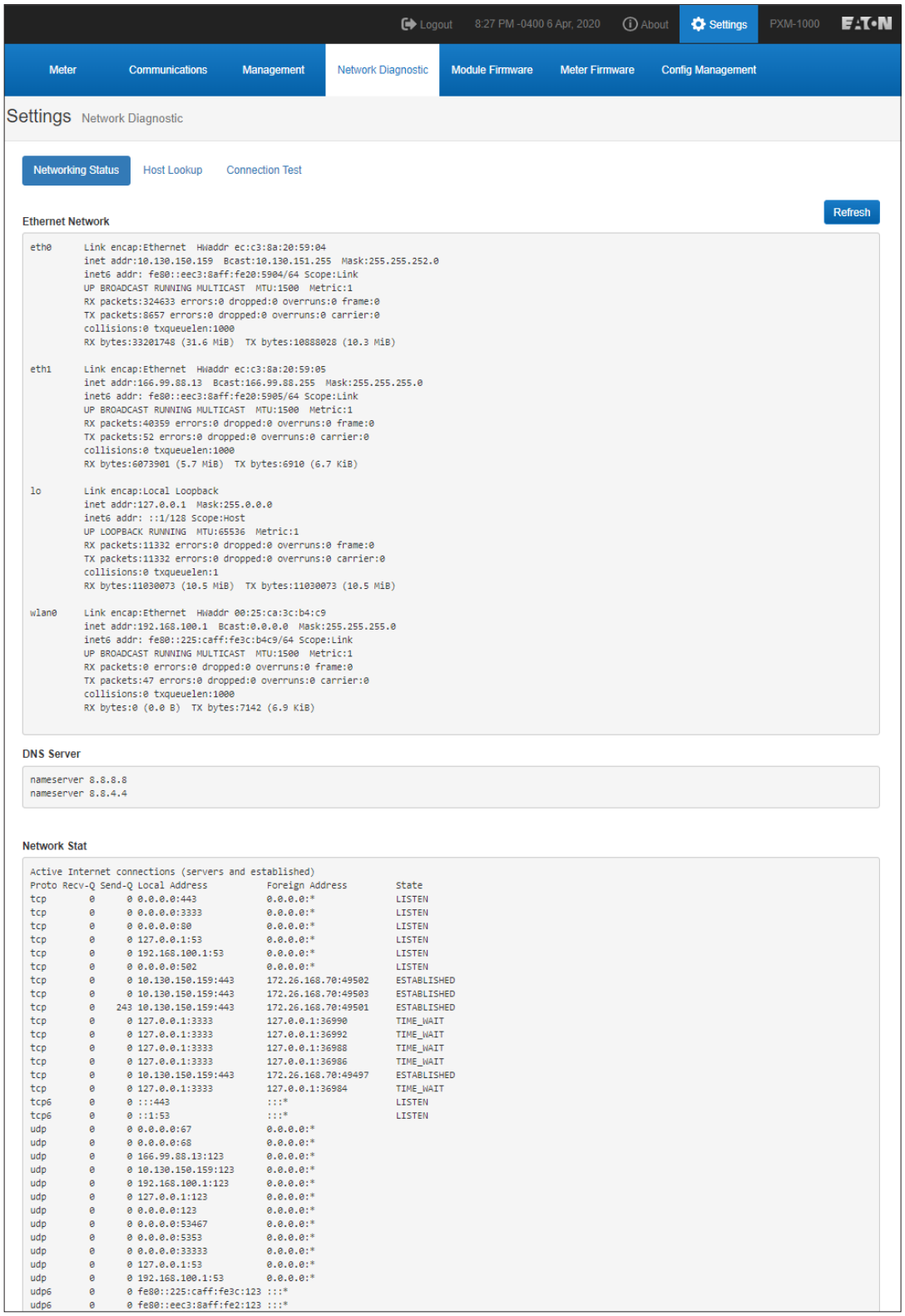
The is a diagnostic file on the PXM1K-ETHMULTI module that users can download which can be used to analyze the modules diagnostics.

Note: Please send the diagnostic file to Eaton Technical support (MRSupport@eaton.com) for analysis.

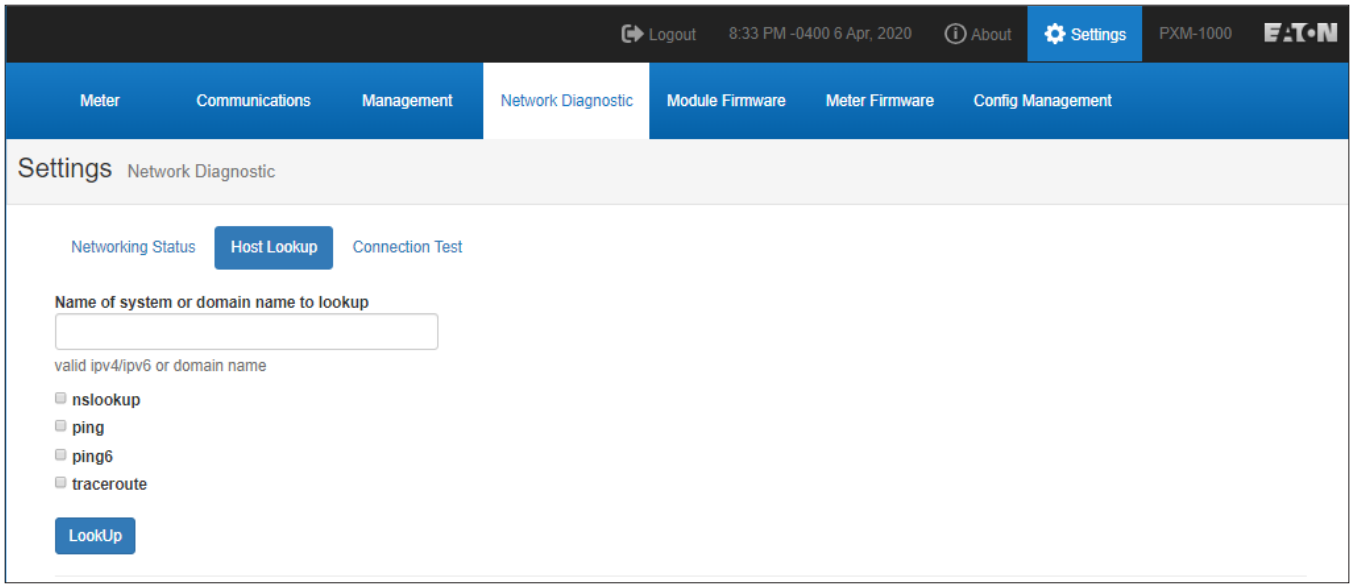

Logout
8:10 PM -0400 6 Apr, 2020
About
Settings
PXM-1000
Meter
Communications
Management
Network Diagnostic
Module Firmware
Meter Firmware
Config Management
Settings Management
Setting
Action
Reset Demand
Reset
Reset Energy
Reset
Reset Max and Min
Reset
Reset Alarm Record
Reset
Reboot Meter & Communications Module
Reboot
Device Clock
8:10 PM -0400 6 Apr, 2020
Reset Device Run Time
Reset
Live API Token
e1d2c698-c56d-4c7d-af22-dbb89ec8a257
Reset API Token
Reset
Reset Admin Password
Enter old password
Enter new password
Show password
Save
Reset View Password
Enter old password
Enter new password
Show password
Save
Reset to Factory Defaults
Reset
SSH
current status: Off
Enable
Debug Diagnostic
current status: All Off
Link to advanced settings
Enable
Disable
Download diagnostic file

10.6.4 Network Diagnostic

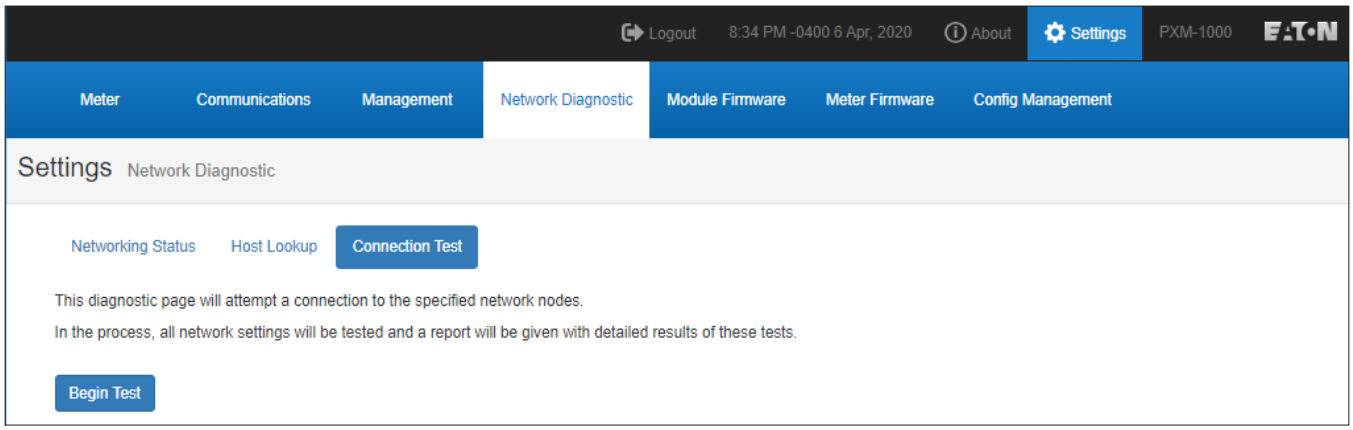
The Network Diagnostic page can be used to monitor the network status of the module.



In the 'Host Lookup' tab users can utilize the 'ping' function to test the reach-ability to other networks. Users can also use the 'ping6' function to ping an IPv6 address.



User can also use the 'Connection Test' function to test the local network that the module connected to. The test result will show SUCCESS and PASS if there is no issues found.

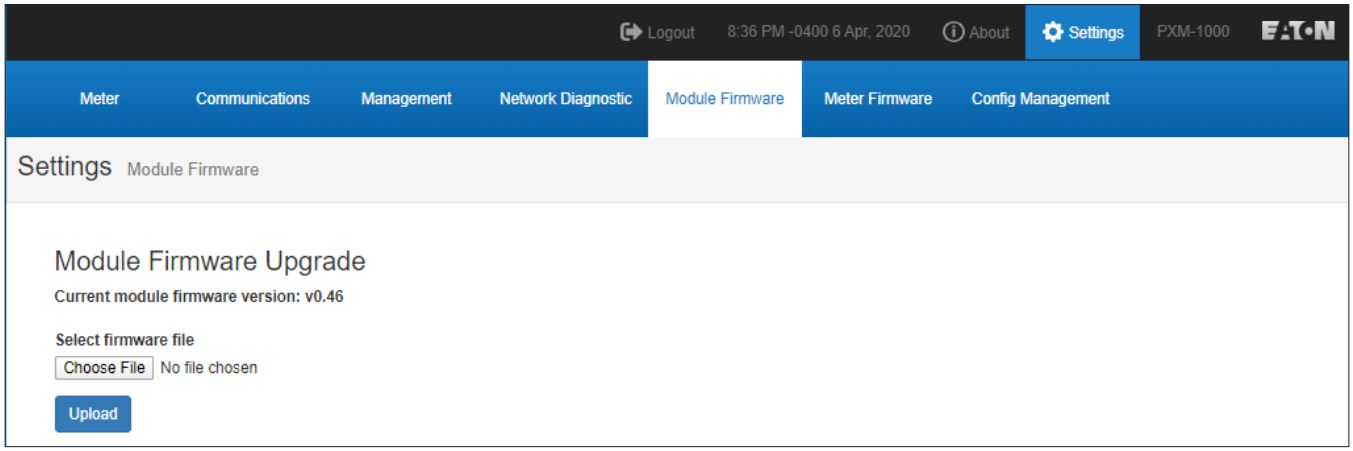


10.6.5 Module Firmware Update

The Module Firmware web page is used for updating the firmware version on the PXM1K-ETHMULTI module. Users can manually update the firmware by uploading the firmware file. The current version of the firmware will be displayed on the Module firmware update page and can also be viewed on the 'Device Information' page of the web interface.

There is an Auto Firmware Update feature available also, this allows users to update the module automatically without manually going into the web server and performing the update.

Note: Users can also contact Eaton technical support (MRsupport@eaton.com) for latest firmware.



Auto Firmware Update

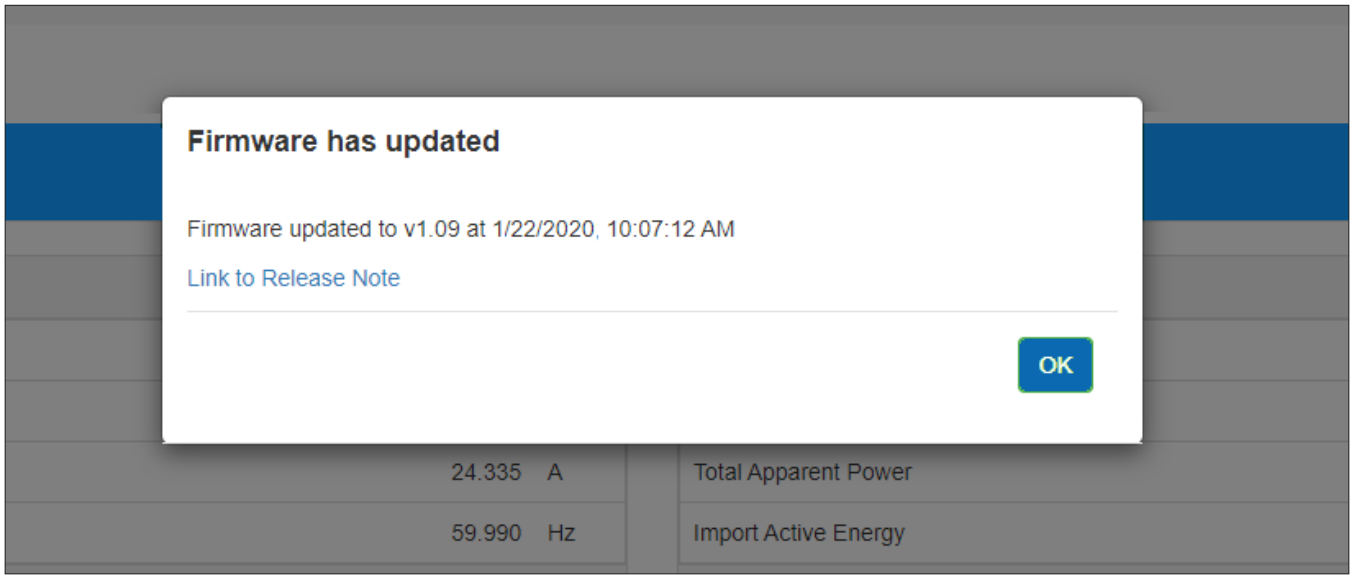
The auto firmware update allows users to select three different updating options.

- Disable - Disables the auto firmware update function
- Critical Update Only - Updates the module to the latest critical firmware
- Automatically keep firmware to Latest - Updates the module to the latest firmware

If users select critical or latest firmware update options, the time for the update can be configured. By default the update time is set for 3am-4am.

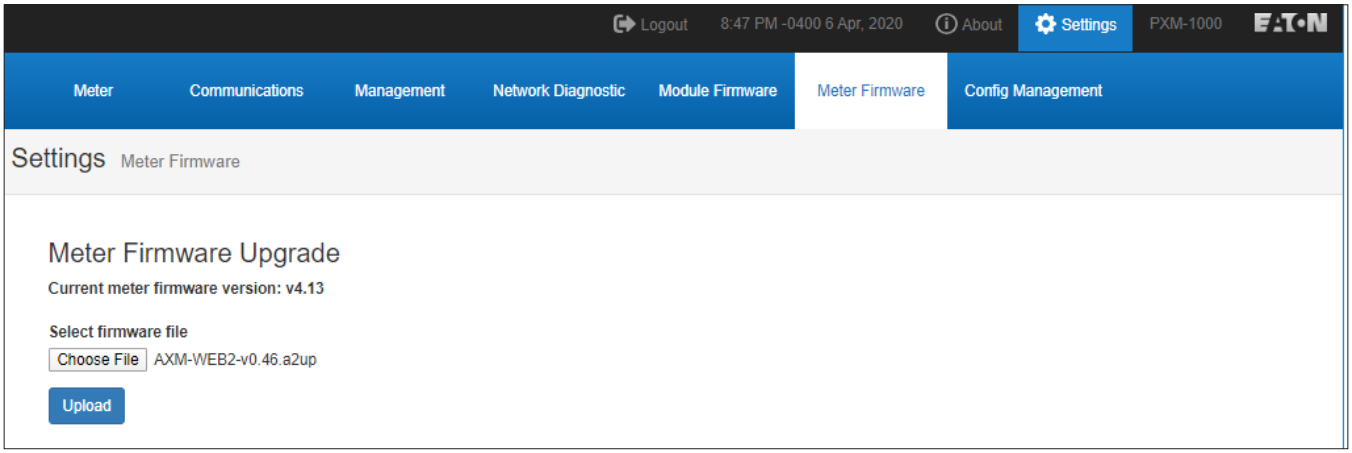
Note: The one hour time block means that the update will occur anytime within the hour.

After the automatic firmware update, when users log in to the web interface for the first time after the update they will see a message displayed, which shows the time and date to which the module was updated as well as the firmware version to which iit was updated.

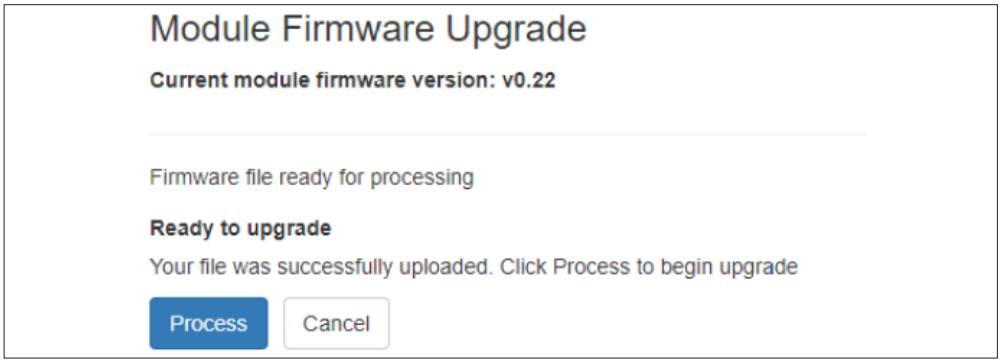


Manual Update

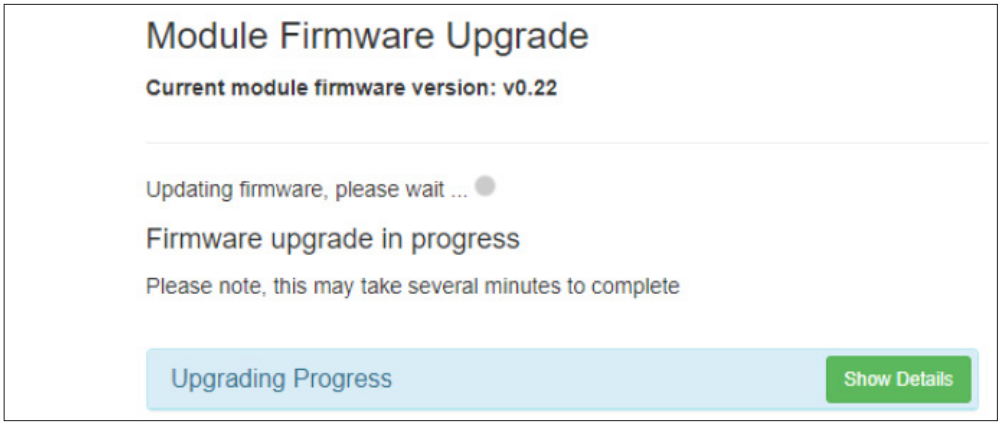
Select and upload the PXM1K-ETHMULTI firmware file, it is a .aup file extension.



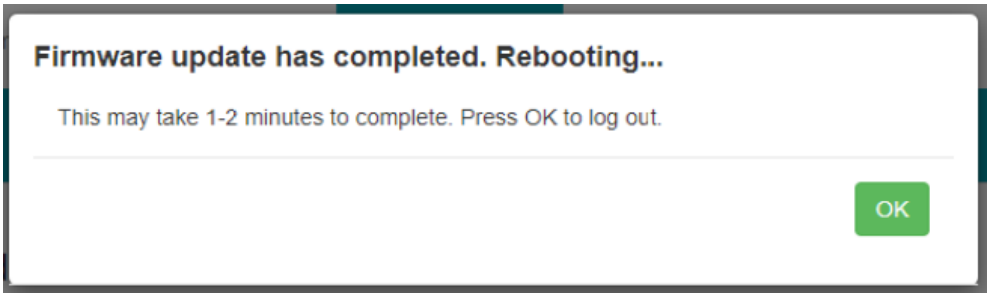
Once the upload was successfully uploaded you will see the following page confirming that the file was uploaded.



Click 'Process' to begin the update.



Click 'OK' to log out the web interface and wait for 1-2 minutes to complete the reboot.

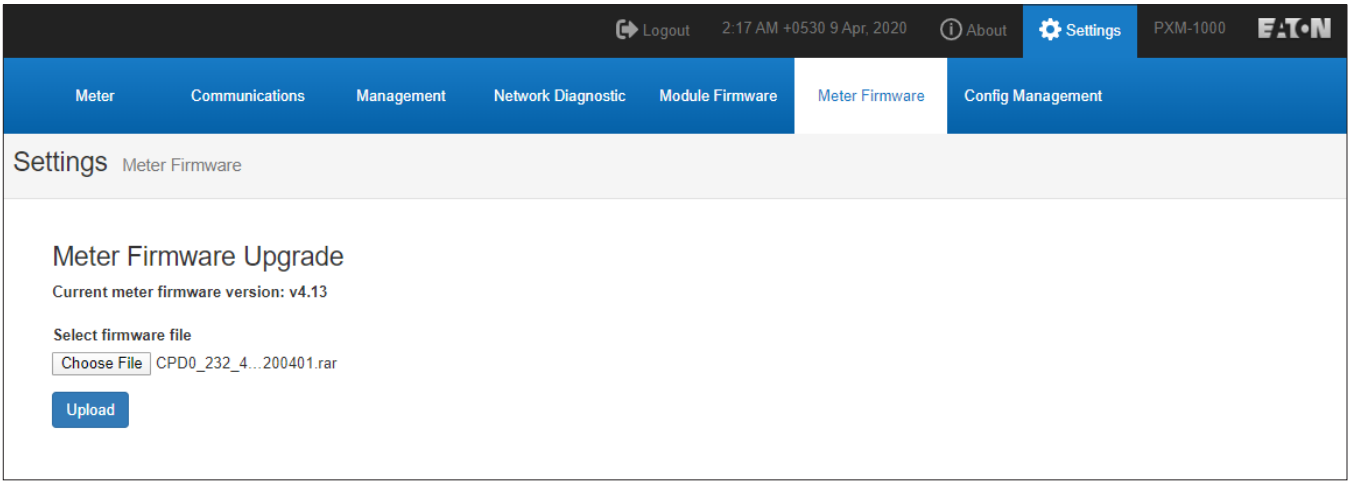


Login to the web interface of PXM1K-ETHMULTI after the reboot is complete, and go to the 'About' page to check if the module firmware version is updated.

10.6.6 Meter Firmware Update

The PXM1K-ETHMULTI allows users to update the meters firmware from its web server. The meters firmware can be updated manually by uploading the required firmware file.

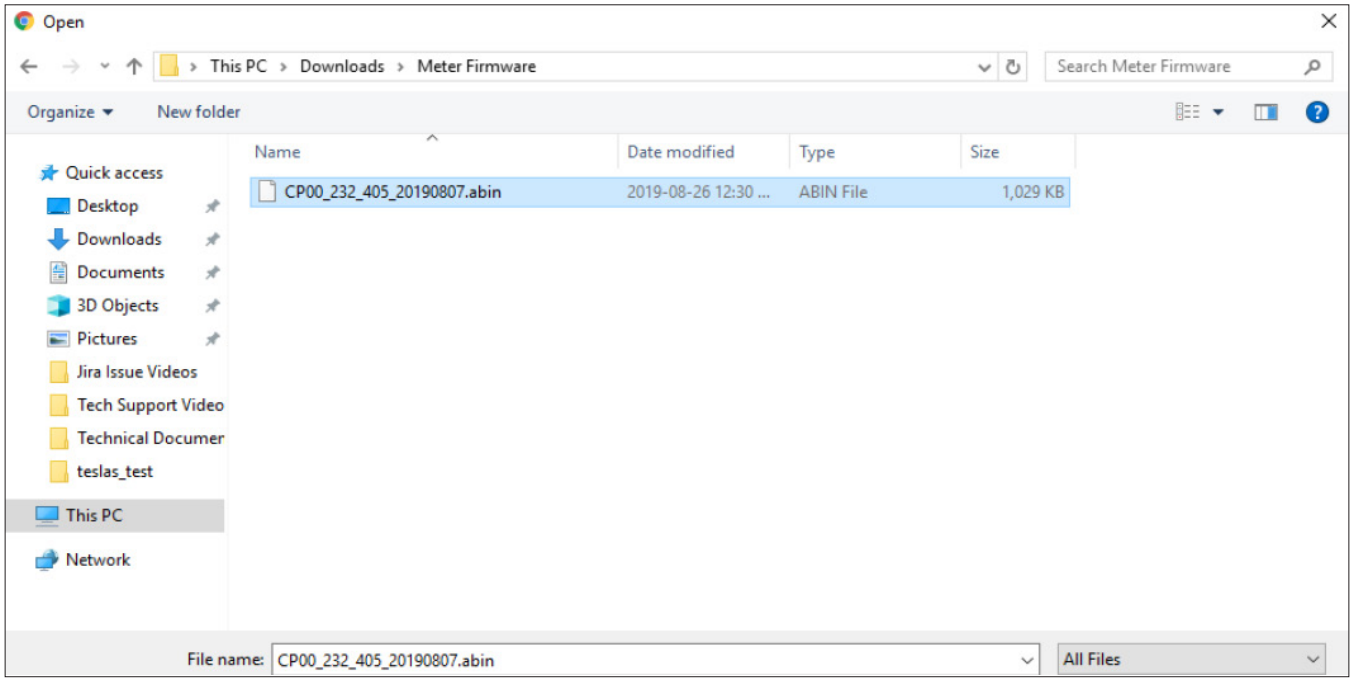
Note: NOTE: The meter firmware update option is only available on meters that are firmware version 3.69 and above.



Manual Update

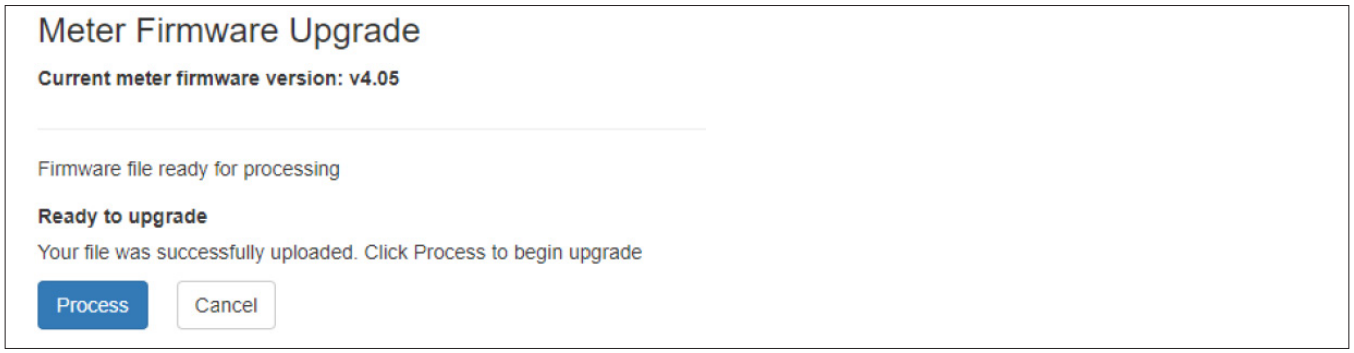
Users can update the meter firmware manually by loading the firmware file to the web server.

Click on 'Choose File' and upload the correct file. All firmware files are .abin type files.



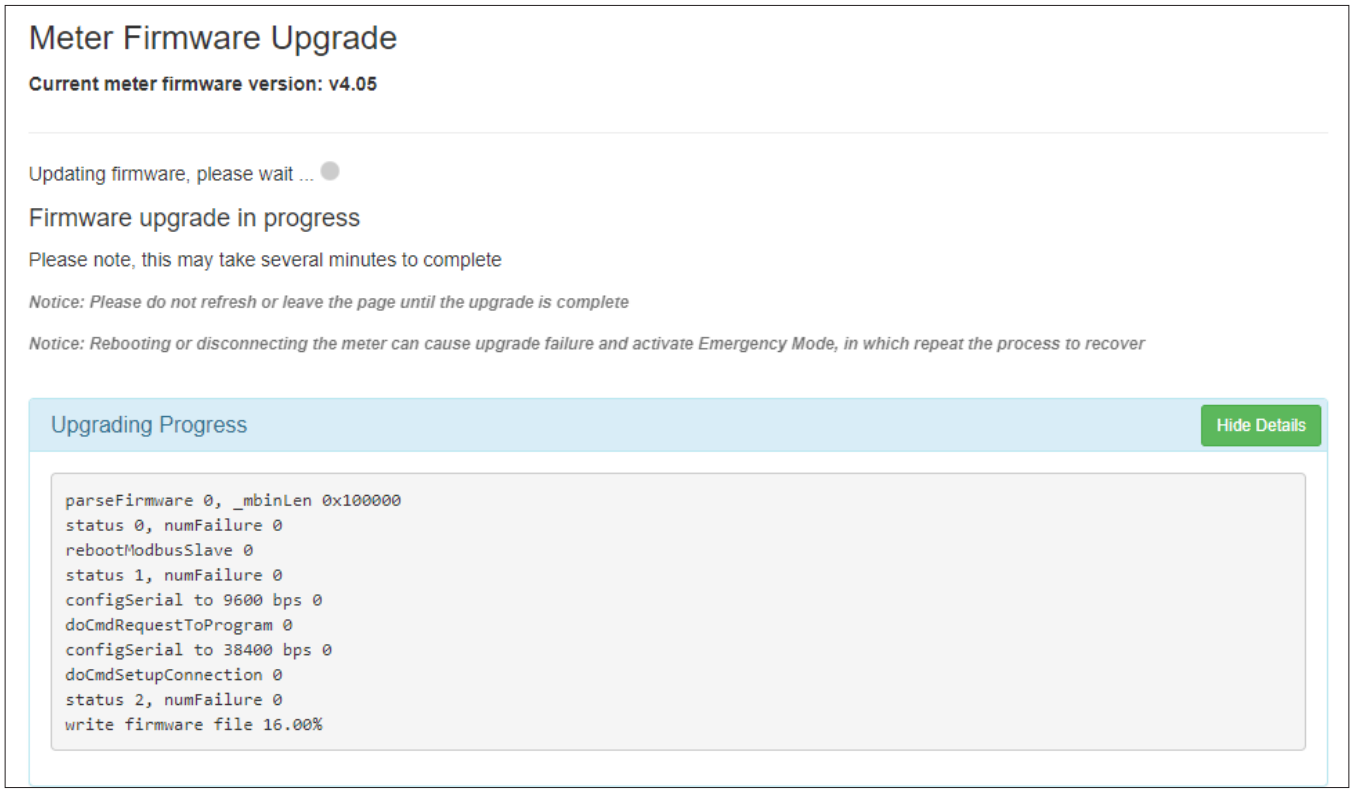


Once the file is selected click on 'Upload'.

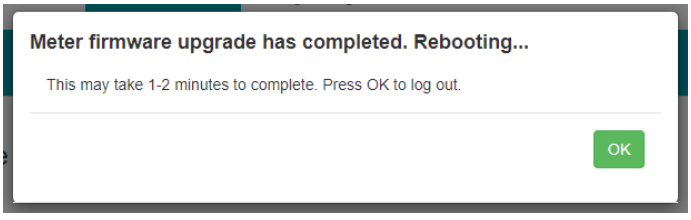


After the firmware has uploaded to the web server, click on 'Process'.

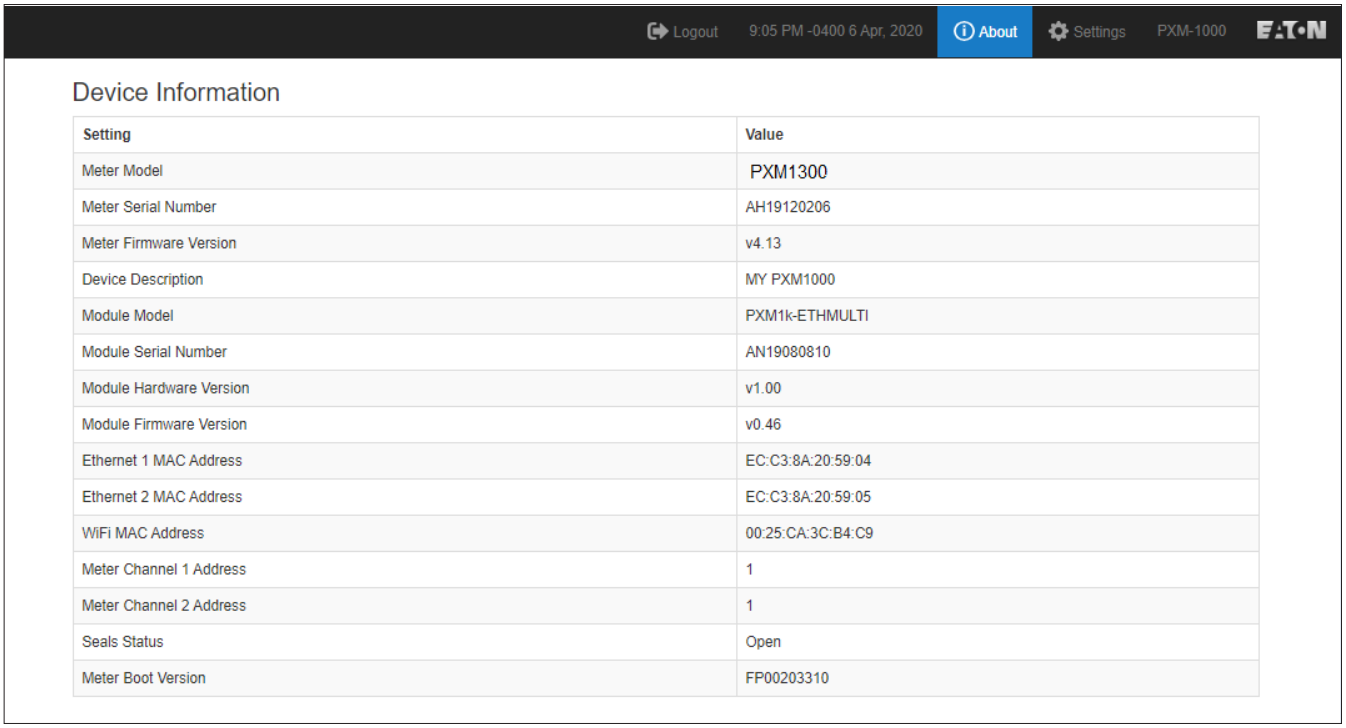
The firmware update will take approximately 5-10 minutes to complete.



When the firmware update is complete, the module will reboot. The rebooting process will take 1-2 minutes to complete.



After the meter reboots, users will be able to log back into the web interface. When logged in click on the 'About' tab located on the top right corner of the web page to view the 'Device Information' page. From the Device Information page users can ensure that the meter was updated correctly to the right firmware version.



Device Information

Setting	Value
Meter Model	PXM1300
Meter Serial Number	AH19120206
Meter Firmware Version	v4.13
Device Description	MY PXM1000
Module Model	PXM1k-ETHMULTI
Module Serial Number	AN19080810
Module Hardware Version	v1.00
Module Firmware Version	v0.46
Ethernet 1 MAC Address	EC:C3:8A:20:59:04
Ethernet 2 MAC Address	EC:C3:8A:20:59:05
WiFi MAC Address	00:25:CA:3C:B4:C9
Meter Channel 1 Address	1
Meter Channel 2 Address	1
Seals Status	Open
Meter Boot Version	FP00203310

10.6.7 Emergency Mode

During the meter firmware update process if the meter loses power or connectivity to the web interface the meter will run in emergency mode. In this mode the next time the user logs into the web interface instead of seeing the dashboard the following screen will be displayed.

It lets users know that the meter firmware update process has failed and to try updating the meter firmware again. The meter will stay in this mode until the meters firmware has been updated successfully.

Users can update the meters firmware manually or through the remote server while the meter is in Emergency mode. After the update is complete the meter will come out of emergency mode and users will be able to see the web server in its normal mode.

Note: The meter will only go into emergency mode if the Meter firmware update fails, emergency mode is not applicable if the Module firmware update fails.



Meter Channel 1 Address	1
Meter Channel 2 Address	1
Seals Status	Open
Meter Boot Version	FP00203310

Hint: Firmware Upgrade Failed

Hint: Go to "Settings->Meter Firmware" and re-upload a fimware file

10.6.8 Config Management

The PXM1K-ETHMULTI module has a configuration management page that allows users to save all web settings with the exception of a certain settings into a configuration file. This is useful if users have more than one meter that needs to be programmed with the same settings, and eliminates any error when trying to configure another PXM1K-ETHMULTI module.

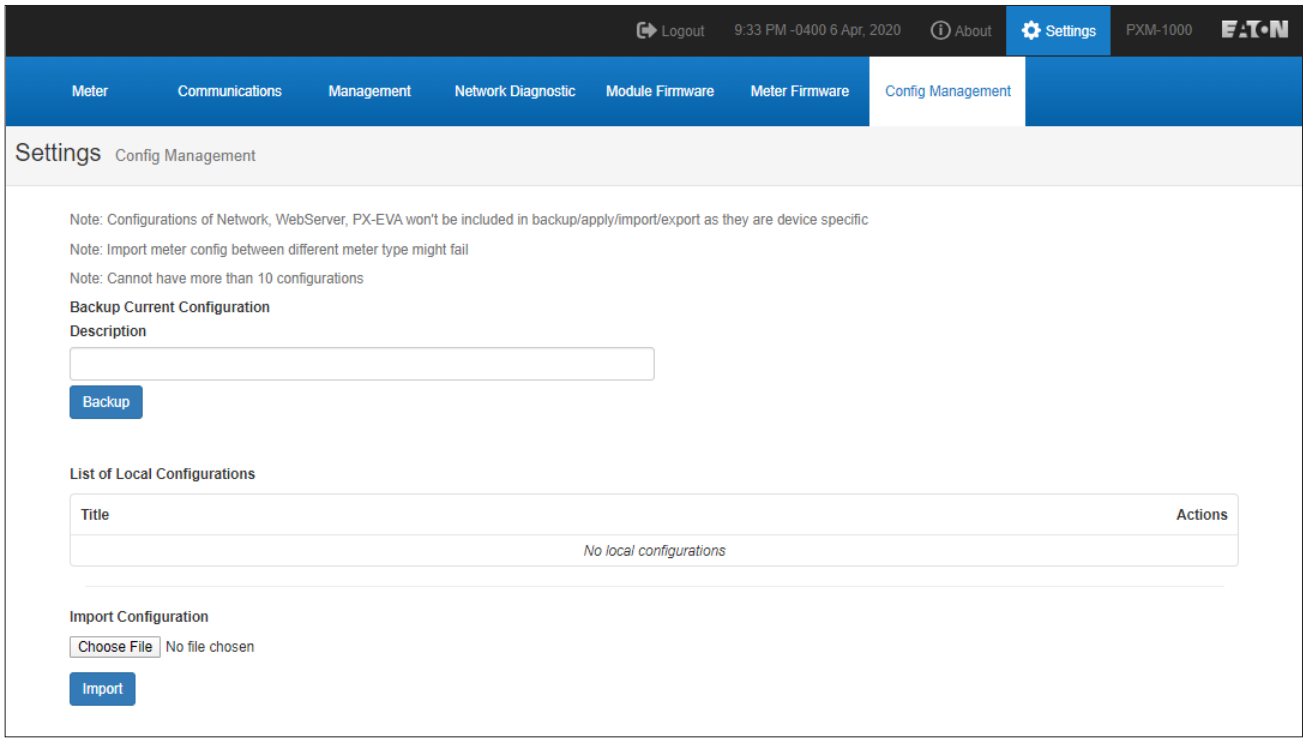
The following settings are saved in the configuration file:

- All Meter settings (General, IO, Alarm, Custom Read, Waveform)
- Network settings (Only DNS1, DNS2, Modbus TCP Port, HTTP Proxy)
- All Email settings
- All Time/Date settings
- All Data Log settings
- All Post Channel settings
- All BACnet settings
- All SNMP settings
- Management settings (the View and Admin Access Level passwords, SSH, and Debug Configuration)

The settings that are not included or effected by the Config Management file is:

- Most Network settings (RSTP, DHCP, IP, Submask, Gateway, HTTP Port for both Ethernet 1 and 2. All WiFi settings, Fast Read Mode, HTTP enable, and HTTPS port are not changed)
- IPv6
- PX-EVA

The Config Management page can be accessed by clicking on the 'Settings' tab and selecting 'Config Management'.



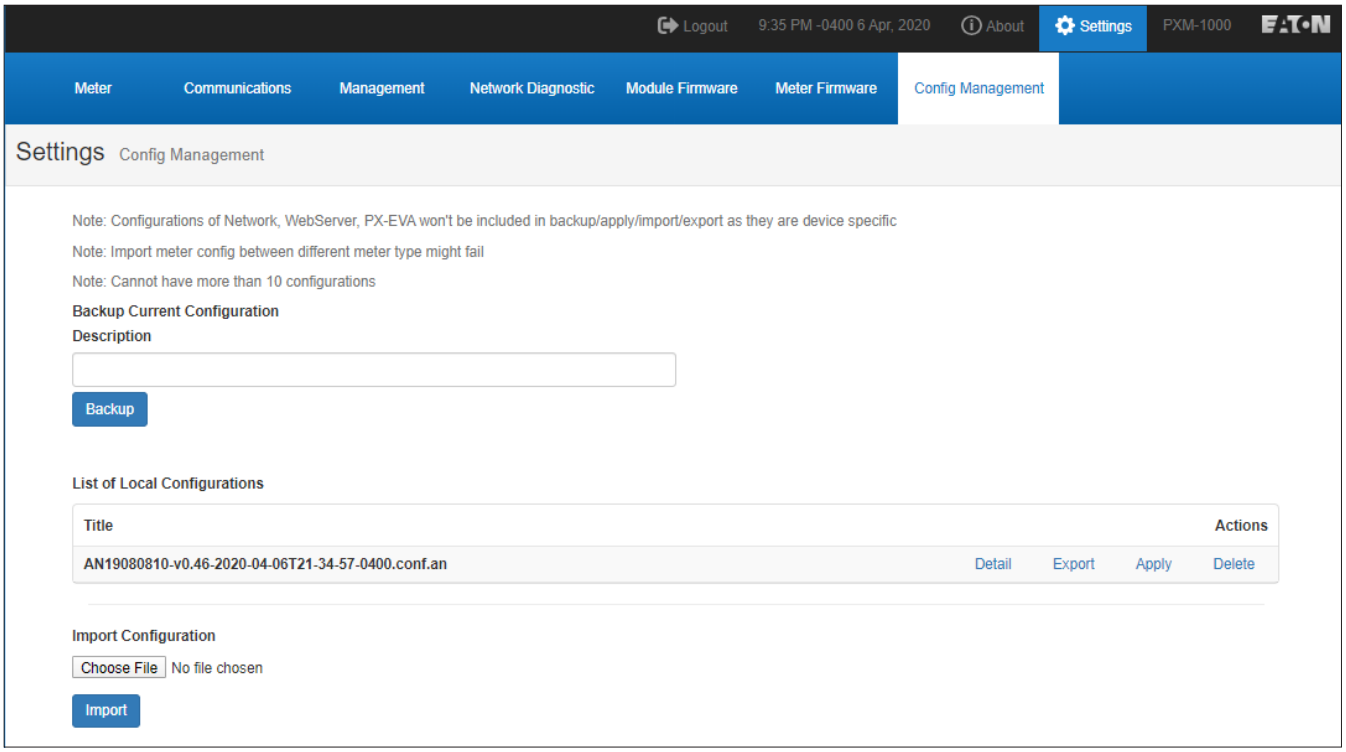
Users can create a backup of the current configurations on the PXM1K-ETHMULTI interface.

Backup Current Configuration Description: Enter in a description for the backup configuration file.

Once the description is entered in click on the 'Backup' button.

The backup is displayed in the List of Local Configurations. The file has a file format that includes the module serial number, module firmware version, and time stamp that the file was created.

Note: NOTE: Users cannot have more than 10 configurations in the List of Local Configurations.

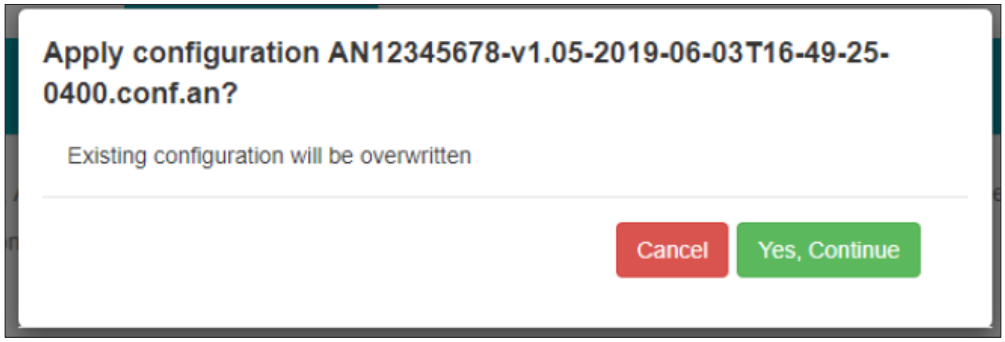


Users can click on 'Detail' to view the description of the configuration file. The details include the Model name, serial number, time created. firmware version and the description entered when the backup was created.

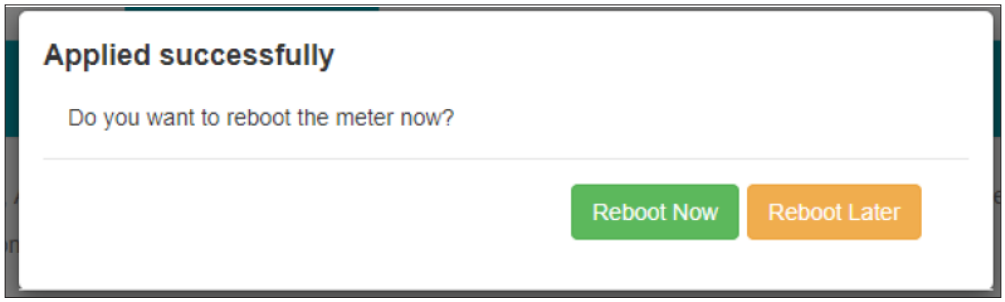


Users can export the configuration file and use it on other PXM1K-ETHMULTI modules. The file is downloaded as a .an file.

To implement the configuration file click on 'Apply'. A prompt warning the user that the existing .conf.an file will be overwritten is shown. Click 'Yes' to continue.



A module reboot is required for the configuration to take effect. if users decided to reboot later the reboot must be performed from the Management page in order for the settings to take effect on the device.

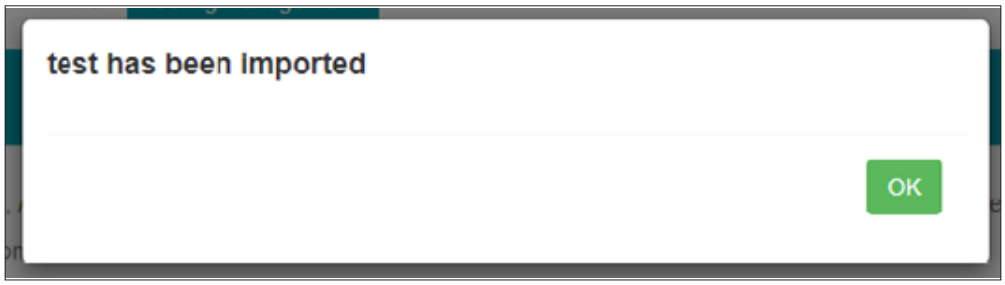


Users can remove any of the configuration files from the list at any time by selecting 'Delete'.

Import Configuration: Users can import a configuration file (.conf.an file format) to the PXM1K-ETHMULTI module.



Click on 'Import' to upload the configuration file to the PXM1K-ETHMULTI module.



The newly imported file will now appear in the List of Local Configurations.



List of Local Configurations

Title				
				Actions
AN12345678-v1.05-2019-06-03T16-49-25-0400.conf.an	Detail	Export	Apply	Delete
AN12345678-v1.05-2019-06-03T17-19-13-0400.conf.an	Detail	Export	Apply	Delete

Note: Users cannot import a file that already exists on the local configurations, when the list already contains 10 config files, and cannot import a config file that has been exported from a PXM1K-ETHMULTI module with a higher firmware version.



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